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ON THE GROUPS $H(\Pi, n)$, II

Methods of Computation

BY SAMUEL EILENBERG AND SAUNDERS MACLANE

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1. Introduction

The homology ring $H(\Pi, n)$ is defined, for any abelian group Π and any positive integer n , as the homology ring of the complex $K(\Pi, n)$. The main result of the first paper of this series¹ was the replacement of the complex $K(\Pi, n)$ by an equivalent complex $A(\Pi, n)$. The present investigation² uses the perspicuous algebraic structure of this complex $A(\Pi, n)$ to set up methods for the invariant computation of some of the integral homology groups $H_{n+k}(\Pi, n)$. This program is carried out in detail for $k \leq 5$ and $n \geq 2$, except for $H_7(\Pi, 2)$. Invariant cohomology results for the groups $H^{n+k}(\Pi, n; G)$ are also obtained for $k \leq n$, $k = 2, 3, 4$, and 5 .

To illustrate our methods, we cite the isomorphism

$$(1.1) \quad \theta: \Pi/2\Pi + \Pi/3\Pi \cong H_9(\Pi, 5),$$

obtained by assigning to the cosets $x + 2\Pi$ and $y + 3\Pi$ the homology classes of the cycles given in the complex $A(\Pi, 5)$ by the single cells denoted as $[x \mid_4 x]$ and $[y \mid_2 y \mid_2 y]$, respectively. The simplicity of this result is due to the fact that the homology group $H_9(\Pi, 5)$ is stable under suspension. Our main interest lies in a treatment which also includes the intrinsically more involved unstable cases. In these cases, simple functors like $\Pi/2\Pi$ and $\Pi/3\Pi$ do not suffice. We are therefore forced to treat a number of new and sometimes quite bizarre functors of Π .

Once the natural homomorphism θ above has been formulated for an arbitrary abelian group Π , the subsequent proof that it is an isomorphism can be restricted to finitely generated groups and thus to direct sums of cyclic groups.

For direct sums our first concern is the establishment of a chain equivalence

$$(1.2) \quad A(\Pi_1 + \Pi_2, n) \rightleftharpoons A(\Pi_1, n) \otimes A(\Pi_2, n).$$

Geometrically, this can be achieved rapidly by consideration of the cartesian product $X_1 \times X_2$ of two spaces X_1 and X_2 whose only nontrivial homotopy groups are $\pi_n(X_1) = \Pi_1$ and $\pi_n(X_2) = \Pi_2$. However, our subsequent use of the equivalence (1.2) requires that the chain mappings in both directions be explicit and natural. This need is met by the analysis of Chapter I. We begin by setting

¹ Samuel Eilenberg and Saunders MacLane. *On the groups $H(\Pi, n)$* , I, Ann. of Math. 58 (1953) 55-106. References to sections and theorems of this paper will be made by placing the prefix I on the section or theorem number. The bibliography below is a continuation of that of I; numbers [1] to [9] inclusive refer to the bibliography of I.

² A portion of this investigation was done under contracts Af-18(600)-562 and Nonr-218 (00).

up an explicit equivalence for the Eilenberg-Zilber theorem $K_1 \times K_2 \rightleftharpoons K_1 \otimes K_2$ on FD -complexes K_1 and K_2 , [8]. This leads up to the corresponding "tensor product theorem"

$$(1.3) \quad B(G_1 \otimes G_2) \rightleftharpoons B(G_1) \otimes B(G_2)$$

for the bar construction on graded boundary rings G_1 and G_2 . Since the complex $A(\Pi, n)$ is obtained by n -fold application of this bar construction to the integral group ring of Π , the direct sum theorem (1.2) is then obtained.

The direct sum theorem shows that the functor $H_q(\Pi, n)$ is additive in the stable cases ($q < 2n$); in particular

$$(1.4) \quad H_9(\Pi_1 + \Pi_2, 5) \cong H_9(\Pi_1, 5) + H_9(\Pi_2, 5).$$

This reduces the verification that θ is an isomorphism in (1.1) to the case of cyclic groups. This reduction is no longer possible for homology groups which are not stable; for instance, in the immediately preceding case the isomorphism (1.4) is replaced by

$$H_8(\Pi_1 + \Pi_2, 4) \cong H_8(\Pi_1, 4) + H_8(\Pi_2, 4) + \Pi_1 \otimes \Pi_2.$$

The extra term $\Pi_1 \otimes \Pi_2$ present here is called the "cross-effect" for the functor $H_8(\Pi, 4)$. Chapter II studies such cross-effects for an arbitrary covariant functor $T(\Pi)$, including the higher cross-effects which arise when the initial cross-effect is not additive in its arguments Π_1 and Π_2 . This theory of cross-effects is of independent interest, and our exposition of it is self-contained. With the direct sum theorem and the analysis of cross-effects, the verification in unstable cases that maps like θ are isomorphisms is reduced to the cases of cyclic groups and of cross-effects of several cyclic groups.

For a cyclic group Z_h of order h , Chapter III shows that the complex $A(Z_h, n)$ may be replaced by a much simpler complex $M(h, n)$ which is finite in each dimension. This guarantees the computability of the groups $H_q(\Pi, n)$ and their cross-effects in the category of all finitely generated abelian groups Π and of all homomorphisms of one such group into another.

Using the methods thus built up, Chapter IV derives the invariant results promised at the start of our introduction.

The following incidental results may be noted:

Since the well-known Künneth relations play an important part in our discussion of cross-effects on $H_q(\Pi, n)$, we find it necessary and useful to give a brief proof of these relations in §12. This is based on a discussion of cross-caps and a novel definition (§11), using generators and relations, of the torsion product of abelian groups.

The analysis of certain cycles present in $A(\Pi, n)$ suggests the introduction of two rings $\Gamma(\Pi)$ and $\Lambda(\Pi)$, (§§18, 19), defined for any abelian group Π . They are analogous to the polynomial algebra and the exterior algebra defined for a vector space.

As a by-product of the discussion of the cohomology group $H^4(\Pi, 2; G)$ we obtain a rapid proof (Theorem 26.2) of a theorem due to N. J. S. Hughes [12],

which in turn is connected with certain results on symmetric cohomology groups of $A(\Pi, 1)$. In this connection we also derive the useful isomorphism

$$\text{Extabel}(\Pi/p\Pi, G) \cong \text{Hom}(\Pi, G/pG),$$

valid for any prime p (Theorem 26.5).

CHAPTER I. TENSOR AND CARTESIAN PRODUCTS OF COMPLEXES

2. The Eilenberg-Zilber theorem

The bar construction was defined in I in stages, using FD -complexes and then R -complexes. The tensor product theorem (1.3) will be obtained in parallel stages and finally in an iterated form (§5). The transfer of cup products under the main map $k_n: A(\Pi, n) \rightarrow K(\Pi, n)$ is treated in §7.

For FD -complexes, the product ∇ defined in I, §5 yields a chain transformation

$$(2.1) \quad \nabla: K \otimes L \rightarrow K \times L$$

defined by the formula

$$(2.2) \quad \nabla(a \otimes b) = a\nabla b, \quad a \in K, b \in L.$$

THEOREM 2.1. *If K and L are FD -complexes, then the map ∇ and the map f defined in (2.9) below provide a chain equivalence*

$$K \times L \overset{f}{\underset{\nabla}{\rightleftarrows}} K \otimes L.$$

The statement that the pair f, ∇ is a chain equivalence means, as usual, that each of the composites $f\nabla$ and ∇f is chain homotopic to the appropriate identity map.

Each complex K, L , and $K \times L$ determines a normalized complex K_N, L_N and $K \times_N L = (K \times L)_N$; furthermore the product $K_N \otimes L_N$ may be regarded as the quotient of $K \otimes L$ by the subcomplex spanned by all norms $a \otimes Db$ and $Da \otimes b$, for $a \in K, b \in L$. The analogue of Theorem 2.1 for these normalized complexes is

THEOREM 2.1a. *If K and L are FD -complexes, then the maps f and ∇ above induce a chain equivalence*

$$K \times_N L \overset{f}{\underset{\nabla}{\rightleftarrows}} K_N \otimes L_N;$$

explicitly, there is a homotopy Φ in $K \times_N L$, and

$$(2.3) \quad f\nabla = i, \quad \partial\Phi + \Phi\partial = \nabla f - i,$$

where each i denotes the appropriate identity map. The homotopy Φ has the following annihilation properties (modulo norms):

$$(2.4) \quad \Phi\nabla = 0, \quad f\Phi = 0.$$

The maps f and ∇ and the homotopy Φ are natural.

PROOF. In view of the normalization Theorem I.4.1, it suffices to prove the second version of the theorem. The fact that ∇ induces a chain transformation for the normalized complexes was already established in Lemma I.5.3. We shall define f and Φ on the original (unnormalized) complexes, verify that they map norms into norms, and show that (2.3) and (2.4) hold modulo norms.

In any FD -complex K , denote the "last" face operator by $\tilde{F}$; thus if $a \in K_q$, then $\tilde{F}a = F_q a_q$. The powers of the first and last face operators are then

$$(2.5) \quad F_0^i a_q = F_0 F_1 \cdots F_{i-1} a_q, \quad \tilde{F}^{q-i} a_q = F_{i+1} \cdots F_q a_q, \quad 0 \leq i \leq q.$$

Note the following consequences of the basic FD -commutation rules of I, §2, for $b_{q-1} \in K_{q-1}$,

$$(2.6) \quad \begin{aligned} F_0^i D_j b_{q-1} &= D_{j-i} F_0^i b_{q-1}, & 0 \leq i \leq j \leq q-1, \\ &= F_0^{i-1} b_{q-1}, & 0 \leq j < i \leq q, \end{aligned}$$

$$(2.7) \quad \begin{aligned} \tilde{F}^{q-i} D_j b_{q-1} &= \tilde{F}^{q-1-i} b_{q-1}, & 0 \leq i \leq j \leq q-1, \\ &= D_j \tilde{F}^{q-i} b_{q-1}, & 0 \leq j < i \leq q. \end{aligned}$$

Now define $f_i: K_q \times L_q \rightarrow K_i \otimes L_{q-i}$ by

$$(2.8) \quad f_i(a_q \times b_q) = \tilde{F}^{q-i} a_q \otimes F_0^i b_q, \quad 0 \leq i \leq q,$$

and set $f = f_0 + \cdots + f_q$, so that

$$(2.9) \quad f(a_q \times b_q) = \sum_{i=0}^q \tilde{F}^{q-i} a_q \otimes F_0^i b_q, \quad a_q \in K_q, b_q \in L_q.$$

The proof that $f: K \times L \rightarrow K \otimes L$ is a chain transformation is straightforward and will be omitted; it is substantially identical with the familiar proof of the boundary formula for the Alexander-Whitney cup product in a simplicial complex.

To show that f maps norms into norms, compute, for $0 \leq j \leq q-1$,

$$f_i D_j (a_{q-1} \times b_{q-1}) = f_i (D_j a_{q-1} \times D_j b_{q-1}) = \tilde{F}^{q-i} D_j a_{p-1} \otimes F_0^i D_j b_{p-1}.$$

For $i \leq j$, the second factor is a norm by (2.6), while, for $i > j$, the first factor is a norm by (2.7).

To show that $f\nabla(a_p \otimes b_q) \equiv a_p \otimes b_q$, modulo norms, compound f_i with the explicit formula (I.5.3) for ∇ . The result is a sum of terms T of the form

$$T = \wp(\varepsilon(\mu)) \tilde{F}^{p+q-i} D_{\nu_q} \cdots D_{\nu_1} a_p \otimes F_0^i D_{\mu_p} \cdots D_{\mu_1} b_q, \quad (\wp(\varepsilon) = (-1)^{\varepsilon}),$$

with $0 \leq i \leq p+q$ and (μ, ν) a (p, q) -shuffle. Now apply the rules (2.6) and (2.7) to move the powers of F_0 and $\tilde{F}$ past the D 's. If $i < p$, the second factor of T will then have at least one D in front, and T is thus a norm. If $i > p$, the first factor has fewer F 's than D 's, hence will have at least one D in front, so that T is again a norm. If $i = p$, (2.6) shows that the second factor has a D in front except in the case when $\mu_p = p-1, \dots, \mu_1 = 0$, in which event the second factor of T is exactly b_q . In this one case $\nu_q = p+q-1, \dots, \nu_1 =$

p , and (2.7) shows that the first factor of T is exactly a_p . Furthermore the sign $\wp(\varepsilon(\mu))$ is positive in this case, so that this one term yields $a_p \otimes b_p$, as desired.

The notions of derived FD -operators, as developed in I, §3, may be applied in the present context. Thus any natural homomorphism,

$$M = M_{K,L}: K_q \otimes L_s \rightarrow K_p \otimes L_r,$$

as in Theorem I.3.1, can be written uniquely in the form

$$(2.10) \quad M(a_q \otimes b_s) = \sum m_i (\beta_i^* a_q \otimes \gamma_i^* b_s),$$

with integral coefficients m_i and monotonic maps $\beta_i: [p] \rightarrow [q]$ and $\gamma_i: [r] \rightarrow [s]$. The *derived* operator $M': K_{q+1} \otimes L_{s+1} \rightarrow K_{p+1} \otimes L_{r+1}$ is then defined by

$$M'(a_{q+1} \otimes b_{s+1}) = \sum m_i (\beta_i' a_q \otimes \gamma_i' b_s).$$

The operator M is *frontal* if each β_i and γ_i involved is frontal, and Lemma I. 3.3 applies to such operators on $K_q \times L_q \rightarrow K_p \times L_p$.

Now consider the composite chain transformation $h = \nabla f: K \times L \rightarrow K \times L$, and note that

$$(2.11) \quad \begin{aligned} h(a_0 \times b_0) &= a_0 \times b_0, \\ h(a_1 \times b_1) &= D_0 F_1 a_1 \times b_1 + a_1 \times D_0 F_0 b_1. \end{aligned}$$

The derived operator $h': (K \times L)_p \rightarrow (K \times L)_p$ is then defined for $p > 0$, and satisfies $\partial^* h' = h' \partial^*$ and $F_0 h' = h F_0$. Since f and ∇ separately map norms into norms, so does h ; in other words

$$h D_i (a_{q-1} \times b_{q-1}) \in D(K \times L), \quad 0 \leq i \leq q-1.$$

Therefore

$$(2.12) \quad h' D_i (a_q \times b_q) \in D(K \times L), \quad 1 \leq i \leq q.$$

Now define the homotopy Φ in $K \times L$ by induction as a natural homomorphism. Set $\Phi(a_0 \times b_0) = 0$. If $\Phi: (K \times L)_{q-1} \rightarrow (K \times L)_q$ is defined, for $q > 0$, so is the derived operator $\Phi': (K \times L)_q \rightarrow (K \times L)_{q+1}$. We define Φ in dimension $q > 0$ by

$$(2.13) \quad \Phi c_q = -\Phi' c_q + h' D_0 c_q, \quad c_q \in (K \times L)_q.$$

Since h' , like any derived operator, is frontal, we conclude, by induction, that Φ is frontal. To show that Φ maps norms into norms, calculate $\Phi D_i d_{q-1}$, for $0 \leq i \leq q-1$, $d_{q-1} \in (K \times L)_{q-1}$. If $0 < i$, $\Phi' D_i = \Phi' D_{i-1}'$ is a norm, by induction, while if $i = 0$, $\Phi' D_0 = D_0 \Phi$ is a norm, since Φ is frontal. As for the second term in the definition of Φ , we have $h' D_0 D_i d_{q-1} = h' D_{i+1} D_0 d_{q-1} \in D(K \times L)$ by (2.12).

We next prove that $(\partial \Phi + \Phi \partial) c_q \equiv h c_q - c_q$, modulo norms, for any $c_q \in (K \times L)_q$. For $q = 0$, this is immediate, while, for $q = 1$, it is proved by direct computation, using (2.11) and

$$\Phi(a_1 \times b_1) = D_1 D_0 F_1 a_1 \times D_0 b_1 + D_0 a_1 \times D_1 b_1.$$

If the result holds for dimension $q - 1 \geq 1$, then we have also

$$(\partial'\Phi' + \Phi'\partial')c_q = h'c_q - c_q, \quad q \geq 2.$$

Using (I.3.11) and the definition (2.13), we compute, for argument c_q ,

$$\begin{aligned} \partial'\Phi &= -\partial'\Phi' + \partial'h'D_0 = -\partial'\Phi' + h'\partial'D_0 \\ &= -\partial'\Phi' + h' - h'D_0\partial'. \end{aligned}$$

On the other hand,

$$\Phi\partial' = -\Phi'\partial' + h'D_0\partial',$$

so that

$$(\partial'\Phi + \Phi\partial')c_q = -c_q, \quad q \geq 2.$$

But, by the definition (2.13) and Lemma I.3.3,

$$F_0\Phi = -F_0\Phi' + F_0h'D_0 = -F_0\Phi' + h = -\Phi F_0 + h.$$

Since $\partial = F_0 - \partial'$, these results combine to yield $(\partial\Phi + \Phi\partial)c_q = hc_q - c_q$, as desired.

It remains to prove the annihilation properties (2.4). The first property, $\Phi\nabla = 0$, is proved by induction. It is trivial for dimension zero, as $\Phi = 0$ there. In higher dimensions, by (2.13),

$$(2.14) \quad \Phi\nabla(a_p \otimes b_q) = -\Phi'(a_p\nabla b_p) + h'D_0(a_p\nabla b_q).$$

For the first term we expand by the inductive definition (I.5.7) of ∇ to get

$$-\Phi'(a_p\nabla b_q) = -\Phi'(a_p\nabla'D_0b_q) + \wp(p+1)\Phi'(D_0a_p\nabla'b_q),$$

or only one of these two terms, in the event that p or q is zero, as in (I.5.7a). Both these terms are norms by the inductive assumption. For the second term of (2.14), observe that $f\nabla \equiv i$ implies $\nabla f\nabla \equiv \nabla$ and hence $h\nabla \equiv \nabla$ and $h'\nabla' \equiv \nabla'$. Thus, since ∇ is a frontal operator, $h'D_0\nabla = h'\nabla'D_0 \equiv \nabla'D_0 = D_0\nabla$ is a norm, q.e.d.

To establish the second annihilation property $f\Phi = 0$, we first claim that Φ can be written as a linear combination of operators P of the form

$$(2.15) \quad P(a_p \times b_q) = D_j\beta^*a_q \times D_k\gamma^*b_q, \quad j \leq k,$$

where $\beta, \gamma: [q] \rightarrow [q]$ are monotonic maps. Indeed, the definition of f (with the zeroth face F_0 always in the second factor) shows at once that h is a linear combination of operators $\beta^*a_q \times \gamma^*b_q$, with β frontal. As in Lemma I.3.3, this implies that $\beta'^*D_0 = D_0\beta^*$. Hence in the second term $h'D_0(a_q \times b_q) = h'(D_0a_q \times D_0b_q)$ of (2.13) we obtain operators of the form (2.15) with $j = 0$, and hence necessarily $j \leq k$. The assertion as to the form (2.15) of Φ then follows by induction.

It now suffices to show for each such P that

$$f_iP(a_q \times b_q) = \tilde{F}^{q+1-i}D_j\beta^*a_q \otimes F_0^iD_k\gamma^*b_q$$

is a norm. In case $j < i$, the first factor is a norm by (2.7). In case $j \geq i$, we have also $k \geq i$ by the stated form of P , and the second factor is a norm by (2.6).

An alternative homotopy Ψ may be obtained by setting

$$\Psi(a \times b) = -\Psi'(a \times b) + h'D_0(a \times b) - D_0^{q+1}\tilde{F}^qa \times D_0b,$$

for $a \in K_q$, $b \in L_q$. (This definition has the effect of removing from $h'D_0$ the term $D_0^{q+1}\tilde{F}^q \times D_0$ which is otherwise present.) It can be shown that $\partial\Psi + \Psi\partial = h - i$. (i.e., *not* modulo norms.)

A straightforward iteration of the theorem yields

COROLLARY 2.2. *If $K^{(i)}$, for $i = 1, \dots, t$, are FD-complexes, then there exist chain equivalences*

$$K^{(1)} \times \dots \times K^{(t)} \xrightleftharpoons[\nabla]{f} K^{(1)} \otimes \dots \otimes K^{(t)}$$

defined by the formulas

$$\nabla(a^{(1)} \otimes \dots \otimes a^{(t)}) = a^{(1)}\nabla \dots \nabla a^{(t)}, \quad a^{(i)} \in K^{(i)},$$

$$f(a^{(1)} \times \dots \times a^{(t)}) = \sum \tilde{F}^{q-j_1}a^{(1)} \otimes \dots \otimes \tilde{F}^{q-j_i}F_0^{j_i-1}a^{(i)} \otimes \dots \otimes F_0^{j_t-1}a^{(t)}$$

where in the second formula, each $a^{(i)} \in K_q^{(i)}$, and the summation is taken over all j_i with $0 \leq j_1 \leq j_2 \leq \dots \leq j_{t-1} \leq j_t$.

3. Cartesian products of R -complexes

If G and G' are graded ∂ -rings, as in I, §6, the tensor product $G \otimes G'$ of their additive groups becomes a graded ∂ -ring with the following grading, boundary, and multiplication:

$$(3.1) \quad \dim(a_p \otimes b_p) = p + q \quad a_p \in G_p, b_p \in G'_p,$$

$$(3.2) \quad \partial(a_p \otimes b_q) = \partial a_p \otimes b_q + \wp(p)a_p \otimes \partial b_q,$$

$$(3.3) \quad (a_p \otimes b_p) \circ (c_r \otimes d_s) = \wp(qr)(a_p \circ c_r) \otimes (b_q \circ d_s),$$

the latter for $a_p \in G_p$, $b_q \in G_q$, $c_r \in G'_r$, $d_s \in G'_s$, and with the sign denoted as $\wp(qr) = (-1)^{qr}$. The chain complex associated with $G \otimes G'$ is then just the tensor product, in the usual sense I, (§5), of the chain complex associated with G by that associated with G' . Note, however, that in the ring associated with $G \otimes G'$ the sign $\wp(qr)$ of (3.3) is different from that used in the tensor product of (ungraded) rings. If G and G' are skew-commutative, so is $G \otimes G'$. If $\rho: G \rightarrow H$ and $\rho': G' \rightarrow H'$ are homomorphisms of graded ∂ -rings, then $\rho \otimes \rho': G \otimes G' \rightarrow H \otimes H'$, defined by

$$(3.4) \quad (\rho \otimes \rho')(a \otimes a') = \rho a \otimes \rho' a', \quad a \in G, a' \in G',$$

is also such a homomorphism. In particular, if $\alpha: G \rightarrow J$ and $\alpha': G' \rightarrow J$ are augmentations of G and G' , respectively, then $\alpha \otimes \alpha'$ provides an augmentation of $G \otimes G'$.

By standard arguments on tensor products of complexes, one can establish

LEMMA 3.1. *Given two contractions*

$$(f, g, \Phi): (G, \alpha) \rightarrow (H, \beta), \quad (f', g', \Phi'): (G', \alpha') \rightarrow (H', \beta')$$

of augmented graded ∂ -rings, there is a contraction

$$(f \otimes f', g \otimes g', \Psi): (G \otimes G', \alpha \otimes \alpha') \rightarrow (H \otimes H', \beta \otimes \beta')$$

in which the homotopy Ψ is given, for homogeneous elements, by

$$\Psi(a \otimes b) = \Phi a \otimes b + \mathcal{P}(da)gfa \otimes \Phi'b, \quad a \in G, b \in G'.$$

In the sequel we omit the symbols for the augmentations in dealing with contractions (which are defined as in I, §12).

Now let R and S be two augmented R -complexes, as defined in I, §6. They may be regarded as graded ∂ -rings, relative to the product Δ , and the normalized complexes R_N and S_N are also graded ∂ -rings under the induced product. We may thus form the tensor products $R \odot S$ and $R_N \odot S_N$ of these graded ∂ -rings, where for subsequent clarity we use here the symbol $\odot$ for the tensor product. The contractions $R \rightarrow R_N$ and $S \rightarrow S_N$ of Theorem I.6.2 provide, by Lemma 3.1, a contraction

$$R \odot S \rightarrow R_N \odot S_N.$$

On the other hand, consider the cartesian product $R \times S$. This FD -complex may be regarded as an R -complex, if we define the inner product of two chains of dimension p in terms of the inner products in R and S as

$$(3.5) \quad (a_p \times b_p)(c_p \times d_p) = a_p c_p \times b_p d_p; \quad a_p, c_p \in R_p, b_p, d_p \in S_p.$$

In particular, if 1_p and $1'_p$ are the identity elements of the rings R_p and S_p , then $1_p \times 1'_p$ is the identity element of $(R \times S)_p$. The R -homomorphisms,

$$(3.6) \quad a_p \rightarrow a_p \times 1'_p, \quad b_p \rightarrow 1_p \times b_p, \quad a_p \in R_p, b_p \in S_p,$$

have kernel zero; indeed, if the first is followed by the ring homomorphism $R_p \times S_p \rightarrow R_p \otimes J \cong R_p$ given by $a_p \times b_p \rightarrow a_p \otimes \bar{\alpha} b_p$, with $\bar{\alpha}$ as in (I.6.13), the composite carries a_p into $a_p \otimes 1$, which is zero only if $a_p = 0$. Therefore we can and will use (3.6) to identify R and S with sub- R -complexes of $R \times S$.

The Eilenberg-Zilber Theorem 2.1, when applied to R -complexes, now yields

THEOREM 3.2. *If R and S are two augmented and commutative R -complexes, then there is a chain equivalence*

$$R \times S \underset{g}{\overset{f}{\rightleftarrows}} R \odot S$$

of the graded ∂ -rings associated with the cartesian product and the tensor product. The maps f and g are natural maps of graded ∂ -rings and are defined by

$$(3.7) \quad f(a_p \times b_p) = \sum_{i=0}^p \tilde{F}^{p-i} a_p \odot F^i b_p, \quad a_p \in R_p, b_p \in S_p,$$

where $\tilde{F}$ and F_0 are the last and initial faces, respectively, and

$$(3.8) \quad g(a \odot b) = a\Delta b, \quad a \in R, b \in S,$$

with the identifications (3.6).

THEOREM 3.2a. *The maps f and g above carry norms into norms and yield a contraction*

$$(f, g, \Phi): R \times_N S \rightarrow R_N \odot S_N.$$

The homotopy Φ also is natural.

PROOF. First, observe that the map g of (3.8) is in fact identical with the map ∇ defined in (2.2). For Δ is defined in terms of ∇ by (I.6.6), using the inner product in $R \times S$ to define the map $\Pi: (R \times S) \times (R \times S) \rightarrow R \times S$. Now $a\Delta b = \Pi(a\nabla b)$, and $a\nabla b$ is a sum of terms $c \times d$ with $c \in R \subset R \times S$, $d \in S \subset R \times S$. For each such term, by (3.6) and (3.5)

$$\Pi(c \times d) = \Pi((c \times 1) \times (1 \times d)) = c1 \times 1d = c \times d.$$

Hence $a\Delta b = \Pi(a\nabla b) = a\nabla b$, as desired.

In view of the results of Theorems 2.1 and 2.1.a, which include in particular all the annihilation properties required for a contraction, we need only show that the maps f and g preserve the products in the respective graded ∂ -rings. As for g , take $a \in R_p$, $b \in S_q$, $c \in R_r$, and $d \in S_s$; then

$$\begin{aligned} g[(a \odot b) \circ (c \odot d)] &= \mathcal{P}(qr)g[(a\Delta c) \odot (b\Delta d)] \\ &= \mathcal{P}(qr)(a\Delta c)\Delta(b\Delta d). \end{aligned}$$

On the other hand,

$$g(a \odot b)\Delta g(c \odot d) = (a\Delta b)\Delta(c\Delta d).$$

Since $R \times S$ is commutative as an R -complex and hence skew-commutative for the product Δ , the two results agree.

As the map f is the sum of the maps f_i of (2.8), the assertion that f preserves the products will follow from the following more explicit property of the maps f_i , for $0 \leq i \leq p + q$, $a_p \in R_p$, $b_p \in S_p$, $c_q \in R_q$, and $d_q \in S_q$:

$$(3.9) \quad f_i[(a_p \times b_p)\Delta(c_q \times d_q)] = \sum_{r+s=i} f_r(a_p \times b_p) \circ f_s(c_q \times d_q),$$

when the sum on the right is taken over all pairs of integers r, s with

$$(3.10) \quad 0 \leq r \leq p, \quad 0 \leq s \leq q, \quad r + s = i,$$

where the product Δ on the left of (3.9) is the standard Δ product in $R \times S$, while the product $\circ$ on the right is determined by the products in R and S , as in (3.3).

As a preliminary to the proof of (3.9), let (μ, ν) be any (p, q) -shuffle and i any integer with $0 \leq i \leq p + q$. The conditions

$$\mu_r < i, \quad \mu_{r+1} \geq i; \quad \nu_s < i, \quad \nu_{s+1} \geq i$$

(or, $r = 0$, if $\mu_1 \geq 1$; $r = p$, if $\mu_p < i$, and similarly for s) uniquely determine a pair of integers (r, s) satisfying (3.10), and yield an (r, s) shuffle

$$(\mu', \nu') = (\mu_1, \dots, \mu_r; \nu_1, \dots, \nu_s)$$

and a $(p - r, q - s)$ shuffle

$$(\mu'', \nu'') = (\mu_{r+1} - i, \dots, \mu_p - i; \nu_{s+1} - i, \dots, \nu_q - i).$$

We then say that (μ, ν) has been *split* at i into the two shuffles (μ', ν') and (μ'', ν'') . Conversely, given integers r, s , as in (3.10), an (r, s) shuffle (μ', ν') , and a $(p - r, q - s)$ shuffle (μ'', ν'') , there exists a unique (p, q) -shuffle (μ, ν) which splits at $i = r + s$ into (μ', ν') and (μ'', ν'') . The signatures of these shuffles, as defined in I, §5, are related by the equation

$$\varepsilon(\mu') + \varepsilon(\mu'') = \varepsilon(\mu) - s(p - r).$$

It is convenient to write

$$(3.11) \quad D_\mu = D_{\mu_p} \cdots D_{\mu_1} \quad D_\nu = D_{\nu_q} \cdots D_{\nu_1},$$

for the operators appearing in the definition of the product Δ . Then when (μ, ν) is split at i , the rules (2.6) and (2.7) for transfer of first and last faces yield

$$\begin{aligned} F_0^i D_\nu b_p &= D_{\nu'} F_0^r b_p, & F_0^i D_\mu d_q &= D_{\mu'} F_0^s d_q, \\ \tilde{F}^{p+q-i} D_\nu a_p &= D_{\nu'} \tilde{F}^{p-r} a_p, & \tilde{F}^{p+q-i} D_\mu c_q &= D_{\mu'} \tilde{F}^{q-s} c_q. \end{aligned}$$

The left side of (3.9) is, by the definitions of Δ and f_i ,

$$\sum_{(\mu, \nu)} \wp(\varepsilon(\mu)) \tilde{F}^{p+q-i} (D_\nu a D_\mu c) \otimes F_0^i (D_\nu b D_\mu d),$$

with the sum taken over all (p, q) -shuffles. Upon splitting each shuffle at i and applying the rules just above, this becomes

$$\sum_{(\mu, \nu)} \wp(\varepsilon(\mu)) (D_{\nu'} \tilde{F}^{p-r} a) (D_{\mu'} \tilde{F}^{q-s} c) \otimes (D_{\nu''} F_0^r b) (D_{\mu''} F_0^s d).$$

On the other hand, the right side of (3.9) is

$$\begin{aligned} \sum_{r+s=i} (\tilde{F}^{p-r} a \otimes F_0^r b) \circ (\tilde{F}^{q-s} c \otimes F_0^s d) \\ = \sum_{r+s=i} \wp(s(p-r)) (\tilde{F}^{p-r} a \Delta \tilde{F}^{q-s} c) \otimes (F_0^r b \Delta F_0^s d) \\ = \cdot \sum_{r+s=i} \sum \wp(\eta) (D_{\nu'} \tilde{F}^{p-r} a) (D_{\mu'} \tilde{F}^{q-s} c) \otimes (D_{\nu''} F_0^r b) (D_{\mu''} F_0^s d), \end{aligned}$$

with sign given by $\eta = s(p - r) + \varepsilon(\mu') + \varepsilon(\mu'')$, and with the second sum taken over all (r, s) shuffles (μ', ν') and all $(p - r, q - s)$ shuffles (μ'', ν'') . This agrees with the previous result, hence proves (3.9), and with it completes the proof of the theorems.

The theorems extend easily to products of t different R -complexes.

REMARK. There is an analogous result for the half-product $\downarrow$ of I, §18. The

appropriate definition of the half-product $\downarrow$ in $R \odot S$ proves to be

$$(3.12) \quad \begin{aligned} (a_p \odot b_p) \downarrow (c_r \odot d_s) &= \wp(qr)(a_p \downarrow c_r) \odot (b_q \Delta d_s), & p > 0, \\ &= 0 & p = 0, r > 0, \\ &= a_p c_r \odot (b_q \downarrow d_s), & p = r = 0; \end{aligned}$$

this for $a, c \in R$, and $b, d \in S$. By way of motivation, note the readily proven fact that for $u \in (R \odot S)_m$, $v \in (R \odot S)_n$, this definition yields

$$(3.13) \quad u \circ v = u \downarrow v + \wp(mn)v \downarrow u, \quad (m, n) \neq (0, 0),$$

showing that the "full" product can be constructed from two half-products, exactly as in (I.18.6).

COROLLARY 3.3. *For the map f of (3.7) and the half-product of (3.12)*

$$(3.14) \quad f(u \downarrow v) = fu \downarrow fv, \quad u, v \in R \times S.$$

The proof is exactly analogous to that given above, with due account of the cases present in the definition (3.12), noting that the shuffles (μ, ν) used for the half-product $\downarrow$ in R are restricted by $\mu_0 = 0$ and that (3.9) holds also for the half-products, with the restriction $i > 0$, $r > 0$, and also in the special case $i = 0$, $r = s = 0$.

4. The tensor product theorem

The Eilenberg-Zilber theorem can be used to contract the bar construction on the tensor product $G \otimes G'$ of two augmented graded ∂ -rings. We define a map

$$(4.1) \quad f_B: B(G \otimes G') \rightarrow B(G) \otimes B(G')$$

for homogeneous elements a_i in G and x_i in G' by the formula

$$(4.2) \quad \begin{aligned} f_B[a_1 \otimes x_1 \mid \cdots \mid a_m \otimes x_m] \\ = \sum_{i=0}^m \alpha(x_1 \circ \cdots \circ x_i) \alpha'(a_{i+1} \circ \cdots \circ a_m) [a_1 \mid \cdots \mid a_i] \\ \otimes [x_{i+1} \mid \cdots \mid x_m]; \end{aligned}$$

in the summand for $i = 0$, the term $[a_1 \mid \cdots \mid a_i]$ is understood to be the identity 1 of $B(G)$, while, for $i = m$, $[x_{i+1} \mid \cdots \mid x_m]$ similarly designates the identity 1' of $B(G')$.

The augmentation $\alpha: G \rightarrow J$ together with the injection $\beta: J \rightarrow G$ with $\beta(1) = \theta$, where θ is the identity of G , show that the group of multiples of θ is a direct summand of the additive group of G . We may therefore identify G and G' with subrings of $G \otimes G'$ by the graded ∂ -ring homomorphisms

$$(4.3) \quad a \rightarrow a \otimes \theta', \quad x \rightarrow \theta \otimes x, \quad a \in G, x \in G'.$$

Since the additive group of G or G' is a direct summand of the group of $G \otimes G'$, this yields induced identifications of $B(G)$ and $B(G')$ with subrings of $B(G \otimes G')$.

as in

$$(4.4) \quad [a_1 \mid \cdots \mid a_m] = [a_1 \otimes \theta' \mid \cdots \mid a_m \otimes \theta'],$$

$$[x_1 \mid \cdots \mid x_m] = [\theta \otimes x_1 \mid \cdots \mid \theta \otimes x_m].$$

With this identification, we define a map

$$(4.5) \quad g_B: B(G) \otimes B(G') \rightarrow B(G \otimes G')$$

by the formula

$$(4.6) \quad g_B(u \otimes v') = u * v', \quad u \in B(G), v' \in B(G').$$

THEOREM 4.1. (Tensor product theorem) *If G and G' are two augmented, skew-commutative graded ∂ -rings, then the maps f_B and g_B of (4.2) and (4.6) are natural homomorphisms of graded ∂ -rings and yield a chain equivalence*

$$(4.7) \quad B(G \otimes G') \xrightleftharpoons[g_B]{f_B} B(G) \otimes B(G').$$

THEOREM 4.1a. *The maps f_B and g_B above carry norms into norms and yield a contraction*

$$(f_B, g_B, \Phi_B): B_N(G \otimes G') \rightarrow B_N(G) \otimes B_N(G').$$

The homotopy Φ_B also is natural.

The rest of the section is devoted to the proof. In order to exploit the Eilenberg-Zilber maps f and g we regard $B(G)$ and $B(G')$ as R -complexes and take their cartesian product $B(G) \times B(G')$. Also, $B(G)$ and $B(G')$ are graded ∂ -rings relative to the grading d_* , the boundary ∂_* , and the product Δ (Lemma I.8.2); their tensor product in this sense is denoted by $B(G) \odot B(G')$. They are also skew-commutative graded ∂ -rings with the grading d_B , boundary ∂_B , and product $*$ (Theorem I.7.1); their tensor product in this sense is denoted, as in the statement of the theorem, by $B(G) \otimes B(G')$. By introducing suitable group isomorphisms λ and κ , which amount largely to sign changes, we shall obtain a diagram

$$(4.8) \quad B(G \otimes G') \xrightarrow{\lambda} B(G) \times B(G') \xrightleftharpoons[g]{f} B(G) \odot B(G') \xrightarrow{\kappa} B(G) \otimes B(G').$$

The maps f_B and g_B and their properties are then obtained by composition from this diagram.

The map λ is defined for homogeneous elements of $B(G \otimes G')$ of the form

$$U = [a_1 \otimes x_1 \mid \cdots \mid a_m \otimes x_m],$$

with a_i and x_i homogeneous elements of G and G' , respectively, by the formula

$$\lambda[a_1 \otimes x_1 \mid \cdots \mid a_m \otimes x_m] = \wp(\sum_{j>k} da_j dx_k)([a_1 \mid \cdots \mid a_m] \times [x_1 \mid \cdots \mid x_m]).$$

Alternatively, the element U is determined by its two "components"

$$(4.9) \quad u = [a_1 \mid \cdots \mid a_m] \in B(G), \quad u' = [x_1 \mid \cdots \mid x_m] \in B(G').$$

Indeed, in view of the identification (4.4) and the definition (I.8.3) of the inner product uu' in $B(G \otimes G')$, we have

$$(4.10) \quad U = \wp(\sum_{j>k} da_j dx_k)uu'.$$

The map λ above thus has the form

$$(4.11) \quad \lambda(uu') = u \times u', \quad u \in B(G), u' \in B(G'), d_s u = d_s u'.$$

The R -complex $B(G) \times B(G')$ is a graded ∂ -ring with dimension d_s , (simplicial) boundary ∂_s , and product Δ . It will appear that the map λ can be used to transfer to this complex the tensor dimension d_t , the residual boundary ∂_r , and the product $*$, defined in $B(G) \times B(G')$ according to the formulas

$$(4.12) \quad \begin{aligned} d_t(u \times u') &= d_t u + d_t u', \\ \partial_r(u \times u') &= \partial_r u \times u' + \wp(d_t u)u \times \partial_r u', \end{aligned}$$

$$(4.13) \quad \begin{aligned} (u \times u') * (v \times v') \\ = \wp(d_t(u \times u')d_s v + d_t u' d_s v)(u \times u')\Delta(v \times v'), \end{aligned}$$

where in the last formula u, u', v, v' are homogeneous elements with $d_s u = d_s u', d_s v = d_s v'$.

LEMMA 4.2. *The map λ of (4.11) is an FD-isomorphism of $B(G \otimes G')$ to $B(G) \times B(G')$, and hence induces a chain isomorphism*

$$(4.14) \quad \lambda: B_N(G \otimes G') \rightarrow B(G) \times_N B(G')$$

for the simplicial boundary operator. For elements $u \in B(G), v' \in B(G')$,

$$(4.15) \quad \lambda u = u, \quad \lambda v' = v'.$$

Furthermore, under the definitions (4.12) and (4.13) in $B(G) \times B(G')$

$$(4.16) \quad \lambda d_t = d_t \lambda, \quad \lambda \partial_r = \partial_r \lambda, \quad \lambda(U * V) = \lambda U * \lambda V.$$

PROOF. By the definition (4.11), λ is clearly an FD-isomorphism, and hence $\lambda \partial_s = \partial_s \lambda$. To prove (4.15), take u homogeneous with $d_s u = m$, and let $1'_m$ be the identity in dimension m for the inner product in $B(G')$. Then, according to the identification (4.4), $u = u1'_m$, and hence by (4.11), $\lambda u = u \times 1'_m = u \in B(G) \times B(G')$, according to the identification (3.6). The first two parts of (4.16) follow by direct computation.

The inner product of the R -structure of $B(G)$ is not commutative; indeed, the definition (I.8.3) and the skew-commutativity of the product $\circ$ in G show that

$$(4.17) \quad uv = \wp(d_v d_t u)vu, \quad d_s u = d_s v.$$

With the definition (3.5) of the inner product in $R \times S$, this gives at once, for elements u, v, u', v' of the same simplicial dimension,

$$(4.18) \quad \lambda[(uu')(vv')] = \wp(d_i u' d_i v) \lambda(uu') \lambda(vv').$$

This shows that λ is not an R -map (not a homomorphism for the inner product). Two homogeneous elements $u \in B(G)$ and $v' \in B(G')$ with the same simplicial dimension m have an inner product $uv' \in B(G \otimes G')$ and an inner product $uv' \in B(G) \times B(G')$ under our identifications; however, by (4.18), $\lambda(uv') = \lambda((u1_m')(1_mv')) = \lambda u \lambda v' = uv'$, so that no ambiguity arises.

To prove the last result of (4.16), take homogeneous elements $u, v \in B(G)$ and $u', v' \in B(G')$ with $d_s u = d_s u', d_s v = d_s v'$. The definitions (I.6.6) of the product Δ in $B(G \otimes G')$ and in $B(G) \times B(G')$, written with the notation (3.11), are:

$$\begin{aligned} (uu')\Delta(vv') &= \sum \wp(\varepsilon(\mu))(D_\nu u D_\nu u')(D_\mu v D_\mu v') \\ (u \times u')\Delta(v \times v') &= \sum \wp(\varepsilon(\mu))(D_\nu u D_\mu v) \times (D_\nu u' D_\mu v'), \end{aligned}$$

with summations over all $(d_s u, d_s v)$ shuffles (μ, ν) . Now in the bar construction, $d_i D_i u = d_i u$ for every i , hence $d_i D_i u = d_i u$. In the first formula above, the terms $D_\nu u'$ and $D_\mu v$ may then be commuted by (4.17) with the sign $\wp(d_i u' d_i v)$. This shows that

$$(4.19) \quad \lambda((uu')\Delta(vv')) = \wp(d_i u' d_i v) \lambda(uu') \Delta \lambda(vv').$$

Using the sign change (I.8.8) from Δ to $*$ in $B(G \otimes G')$ yields an additional sign $\wp(d_i(uu')d_s v)$. The total sign is then exactly that used in the definition (4.13) of the $*$ -product in $B(G) \times B(G')$. This proves the last of (4.16).

The boundary operators of $B(G) \odot B(G')$ and $B(G) \otimes B(G')$ are, by (3.2),

$$\begin{aligned} \partial_s(u \odot v') &= \partial_s u \odot v' + \wp(d_s u) u \odot \partial_s v', \\ \partial_B(u \otimes v') &= \partial_B u \otimes v' + \wp(d_B u) u \otimes \partial_B v'. \end{aligned}$$

Since the total boundary in $B(G)$ is the sum $\partial_B = \partial_s + \partial_r$, we can similarly split the boundary ∂_B of $B(G) \otimes B(G')$ as $\partial_B = \partial_s + \partial_r$, where

$$(4.20) \quad \partial_s(u \otimes v') = \partial_s u \otimes v' + \wp(d_B u) u \otimes \partial_s v',$$

$$(4.21) \quad \partial_r(u \otimes v') = \partial_r u \otimes v' + \wp(d_B u) u \otimes \partial_r v'.$$

We now define a dimension d_i and an operator ∂_r in $B(G) \odot B(G')$ by

$$(4.22) \quad d_i(u \odot v') = d_i u + d_i v',$$

$$(4.23) \quad \partial_r(u \odot v') = \wp(d_s v') \partial_r u \odot v' + \wp(d_B u) u \odot \partial_r v',$$

and a $*$ -product for homogeneous elements, $w, z \in B(G)$, $w', z' \in B(G')$ by

$$(4.24) \quad (w \odot w') * (z \odot z') = \wp(\eta)(w \odot w') \Delta(z \odot z'),$$

with sign determined by

$$(4.25) \quad \eta = d_s(z \odot z') d_i(w \odot w') + d_i w' d_i z.$$

The map κ of the main diagram (4.8) is defined for homogeneous elements $u \in B(G)$, $v' \in B(G')$ as

$$(4.26) \quad \kappa(u \odot v') = \wp(d_i u d_s v') u \otimes v'.$$

LEMMA 4.3. *The map κ defined by (4.26) is a group isomorphism of $B(G) \odot B(G')$ to $B(G) \otimes B(G')$, carries norms into norms, and satisfies*

$$(4.27) \quad \partial_s \kappa = \kappa \partial_s, \quad \partial_r \kappa = \kappa \partial_r, \quad d_i \kappa = d_i, \quad d_s \kappa = d_s,$$

$$(4.28) \quad \kappa[(w \odot w') * (z \odot z')] = \kappa(w \odot w') * \kappa(z \odot z'),$$

with ∂_r , d_i , and $*$ defined in $B(G) \odot B(G')$ by (4.23), (4.22), and (4.24).

PROOF. That κ is a group isomorphism mapping norms to norms is clear from the definition; the preservation of the dimensions is also clear. The commutation of κ with ∂_s and ∂_r follows from the definitions by direct computation. In the definition (4.24) of the $*$ -product, the product Δ is that given by (3.3) for the tensor product; hence

$$(w \odot w') * (z \odot z') = \wp(\eta + d_s w' d_s z)(w \Delta z) \odot (w' \Delta z').$$

Application of κ yields $(w \Delta z) \otimes (w' \Delta z')$ with the sign $\wp(\varepsilon_1)$, where

$$\varepsilon_1 = \eta + d_s w' d_s z + (d_i w + d_i z)(d_s w' + d_s z').$$

A similar computation for the right side of (4.28), using the sign shift from $*$ to Δ , yields the same result with sign $\wp(\varepsilon_2)$, where

$$\varepsilon_2 = d_i w d_s w' + d_i z d_s z' + d_s w' d_s z + d_i w d_s z + d_i w' d_s z'.$$

Since $\varepsilon_1 + \varepsilon_2 \equiv 0 \pmod{2}$, the result (4.28) follows.

LEMMA 4.4. *The map $f: B(G) \times B(G') \rightarrow B(G) \odot B(G')$ of the Eilenberg-Zilber theorem commutes with ∂_r and carries $*$ -products to $*$ -products.*

PROOF. For an element of simplicial dimension m the map f is a sum $f = f_0 + \dots + f_m$, as in (2.8). Take u, u' as in (4.9); then

$$f_i(u \times u') = \tilde{F}^{m-i}[a_1 | \dots | a_m] \odot F_0^i[x_1 | \dots | x_m].$$

The definition (I.8.1) of the last face $\tilde{F}$ is

$$\tilde{F}[a_1 | \dots | a_m] = \wp(d_i[a_1 | \dots | a_m])\alpha(a_m)[a_1 | \dots | a_{m-1}].$$

However, $\alpha(a_m) = 0$ unless $da_m = 0$, so that the sign can be rewritten as $\wp(d_i[a_1 | \dots | a_{m-1}])$. Hence, by iteration,

$$\tilde{F}^{m-i}[a_1 | \dots | a_m] = \wp((m-i)d_i[a_1 | \dots | a_i])\alpha(a_{i+1} \circ \dots \circ a_m)[a_1 | \dots | a_i].$$

Similarly, but without sign trouble

$$F_0^i[x_1 | \dots | x_m] = \alpha'(x_1 \circ \dots \circ x_i)[x_{i+1} | \dots | x_m].$$

Therefore $f_i(u \times u')$ vanishes except when

$$(4.29) \quad dx_1 = \dots = dx_i = 0, \quad da_{i+1} = \dots = da_m = 0;$$

in this event

$$(4.30) \quad f_i(u \times u') = \wp((m-i)d_i u) \alpha_i[a_1 | \cdots | a_i] \odot [x_{i+1} | \cdots | x_m],$$

where $\alpha_i = \alpha(a_{i+1} \circ \cdots \circ a_m) \alpha'(x_1 \circ \cdots \circ x_i)$. By (4.29) the sign can also be written as $\wp(d_s[x_{i+1} | \cdots | x_m] d_i[a_1 | \cdots | a_i])$, so that application of κ yields

$$(4.31) \quad \kappa f_i(u \times u') = \alpha_i[a_1 | \cdots | a_i] \otimes [x_{i+1} | \cdots | x_m].$$

By the definitions (I.7.4) of ∂_r , $\partial_r u$ is a sum of signed terms $[a_1 | \cdots | \partial a_j | \cdots | a_m]$. Since $\alpha(\partial a_j) = 0$, the terms with $j > i$ will vanish upon application of f_i to $\partial_r u \times u'$, and we have

$$\kappa f_i(\partial_r u \times u') = \alpha_i \partial_r [a_1 | \cdots | a_i] \otimes [x_{i+1} | \cdots | x_m].$$

Similarly,

$$\kappa f_i(u \times \partial_r u') = \wp(d_B[x_1 | \cdots | x_i]) \alpha_i[a_1 | \cdots | a_i] \otimes \partial_r [x_{i+1} | \cdots | x_m].$$

But, in view of (4.29), $d_B[x_1 | \cdots | x_i] = i$. By the definition (4.12) of ∂_r in $B(G) \times B(G')$ and the definition (4.21) of ∂_r in $B(G) \otimes B(G')$, the last three equations show that $\kappa f_i \partial_r = \partial_r \kappa f_i$. Therefore κf and hence f commutes with ∂_r .

Next, consider the $\ast$ -product of $u \times u'$ and $v \times v'$, where

$$v = [b_1 | \cdots | b_n], \quad v' = [y_1 | \cdots | y_n].$$

From §3 we already know that

$$f[(u \times u') \Delta (v \times v')] = f(u \times u') \Delta f(v \times v').$$

The shift from Δ to $\ast$ on the left by (4.13) introduces the sign

$$(4.32) \quad \wp(d_i(u \times u') d_s v + d_i u' d_i v).$$

On the right the product may be written as a double sum $\sum f_i(u \times u') \Delta f_j(v \times v')$ with $i = 1, \dots, m; j = 1, \dots, n$. A typical term is

$$([a_1 | \cdots | a_i] \odot [x_{i+1} | \cdots | x_m]) \Delta ([b_1 | \cdots | b_j] \odot [y_{j+1} | \cdots | y_n])$$

with a sign and coefficients involving the augmentations which vanish except when (4.29) and

$$(4.29') \quad dy_1 = \cdots = dy_j = 0, \quad db_{j+1} = \cdots = db_m = 0$$

hold. The shift from Δ to $\ast$ by (4.24) in this term introduces a sign

$$\wp(n(d_i[a_1 | \cdots | a_i] + d_i[x_{i+1} | \cdots | x_m]) + d_i[x_{i+1} | \cdots | x_m] d_i[b_1 | \cdots | b_j]).$$

In view of (4.29) and (4.29') this is exactly the sign (4.32) involved in the first shift and is independent of i and j . This shows that f preserves the $\ast$ -products and completes the proof of the lemma.

LEMMA 4.5. *The maps (4.2) and (4.6) are given by composition in the diagram (4.8) as*

$$(4.33) \quad f_B = \kappa f \lambda, \quad g_B = \lambda^{-1} g \kappa^{-1}.$$

PROOF. Take a homogeneous argument $U = [a_1 \otimes x_1 \mid \cdots \mid a_m \otimes x_m]$ in $B(G \otimes G')$. Then, by (4.10), $\lambda U = \pm u \times u'$, with a sign $\wp(\sum_{j>k} da_j dx_k)$ which will be $+1$ in the presence of any one of the conditions (4.29). Therefore, by (4.31), $\kappa f \lambda U$ is the i^{th} term in the definition (4.2) of f_B , as desired.

As for g_B , we have $\lambda g_B(u \otimes v') = u * v'$ by (4.6), (4.15), and (4.16), while by the definitions of g and κ ,

$$g \kappa^{-1}(u \otimes v') = \wp(d_i u d_s v') u \Delta v'.$$

The sign here is exactly that required for the shift (4.13) from Δ to $*$ in $B(G) \times B(G')$. Hence, $\lambda g_B = g \kappa^{-1}$, as desired.

Now return to the proof of the Theorems 4.1 and 4.1a. Since κ , f , and λ are known to commute with ∂_s , ∂_r , and $*$, it follows from (4.33) that f_B is a homomorphism of graded ∂ -rings. On the other hand the definition (4.6) of g_B shows directly that it is such a homomorphism. It also follows that f_B and g_B map norms into norms. The composite is, by Lemma 4.5,

$$f_B g_B = \kappa f \lambda \lambda^{-1} g \kappa^{-1} = \kappa f g \kappa^{-1} = \kappa \kappa^{-1} = i,$$

in view of the corresponding property of fg .

It remains to define the homotopy Φ_B ; to do this, we use the homotopy $\Phi: B(G) \times_N B(G') \rightarrow B(G) \times_N B(G')$ of the Eilenberg-Zilber theorem, and set

$$(4.34) \quad \Phi_B = \lambda^{-1} \Phi \lambda: B_N(G \otimes G') \rightarrow B_N(G \otimes G'),$$

which is justified because λ preserves norms.

We first verify that Φ_B raises the total dimension by 1. Since Φ is natural, we may write, as in (2.10),

$$(4.35) \quad \Phi(u \times u') = \sum_k l_k \beta_k^* u \times \gamma_k^* u', \quad d_s u = m = d_s u',$$

where the l_k are integers and β_k and γ_k monotonic maps of $[m+1]$ to $[m]$. Now inspection of (I.8.1.) and (I.8.2.) show that $d_i D_i u = d_i u$ and $d_i F_i u = d_i u$, the latter provided $F_i u \neq 0$. Hence for any monotonic β we have $d_i \beta^* u = d_i u$. Therefore, each non-zero term in $\Phi(u \times u')$ has tensor dimension $d_i u + d_i u' = d_i(u \times u')$. Since Φ by definition increases the simplicial dimension by 1, it therefore increases the dimension $d_s + d_i$ by 1. However, λ preserves both d_s and (by (4.16)) d_i ; therefore Φ_B , as defined by (4.34), raises the total dimension $d_B = d_i + d_s$ by 1.

We now show that Φ_B is the requisite homotopy, as in

$$\partial_B \Phi_B + \Phi_B \partial_B \equiv g_B f_B - i, \quad (\text{mod norms}).$$

Inserting the representations (4.33) and (4.34) of the B -operations as composites, and noting that $\partial_B = \partial_s + \partial_r$ commutes with λ in virtue of Lemma 4.2, this

congruence reduces to

$$\partial_B \Phi + \Phi \partial_B \equiv gf - i.$$

By the Eilenberg-Zilber theorem, this is known to hold with ∂_B replaced by ∂_* ; hence we need only prove that

$$(4.36) \quad \partial_r \Phi = -\Phi \partial_r,$$

where ∂_r is defined as in (4.12).

To this end use the "qualitative" formula (4.35). Since each β_k^* raises the simplicial dimension by 1 and can be written as a composite of F_i 's and D_j 's, while by (I.9.5) and (I.9.6) $\partial_r F_i u = -F_i \partial_r u$, $\partial_r D_j u = -D_j \partial_r u$, we have

$$\partial_r \beta_k^* u = -\beta_k^* \partial_r u, \quad \partial_r \gamma_k^* u = -\gamma_k^* \partial_r u,$$

and hence by (4.12)

$$\partial_r \Phi(u \times u') = -\sum_k l_k(\beta_k^* \partial_r u) \times (\gamma_k^* u') - \sum_k \wp(d_i \beta_k^* u) l_k(\beta_k^* u) \times (\gamma_k^* \partial_r u').$$

On the other hand, each term in (4.12) has the same simplicial dimension m as does $u \times u'$; hence $\Phi \partial_r(u \times u')$ may be calculated by the same qualitative formula (4.35). Since $d_i \beta_k^* u = d_i u$, the result differs from that above in sign only, thus proving (4.36).

This type of qualitative proof, independent of the explicit formula for Φ , is analogous to that used in proving Theorem I.11.2, and could also have been used above (but without advantage) to prove that $\partial_r f_B = f_B \partial_r$.

Finally, the requisite annihilation properties of the homotopy Φ_B follow trivially from the definition (4.34) and the corresponding properties of Φ .

5. Iterated tensor product theorem

We now consider the tensor product theorem in which the simple bar construction B is replaced by its n -fold iterate B^n . For any two augmented skew-commutative graded ∂ -rings G and G' there is a chain equivalence

$$(5.1) \quad B^n(G \otimes G') \underset{g_n}{\overset{f_n}{\rightleftarrows}} B^n(G) \otimes B^n(G').$$

f_0 and g_0 are the identity maps, and f_n and g_n are defined by iteration as

$$(5.2) \quad f_n = f(Bf_{n-1}), \quad g_n = (Bg_{n-1})g, \quad n > 0,$$

using the diagram

$$(5.3) \quad B^n(G \otimes G') \underset{Bg_{n-1}}{\overset{Bf_{n-1}}{\rightleftarrows}} B(B^{n-1}G \otimes B^{n-1}G') \underset{g}{\overset{f}{\rightleftarrows}} B^n G \otimes B^n G',$$

in which f and g are the maps f_B and g_B of §4.

To verify that (5.1) is a chain equivalence we replace the iterated bar construction B^n by the normalized bar construction B_N^n . Then, instead of (5.1)–

(5.3) we consider

$$(5.1)_N \quad B_N^n(G \otimes G') \xrightleftharpoons[g_n]{f_n} B_N^n(G) \otimes B_N^n(G'),$$

$$(5.2)_N \quad f_n = f(B_N f_{n-1}), \quad g_{n-1} = (B_N g_{n-1})g,$$

$$(5.3)_N \quad B_N^n(G \otimes G') \xrightleftharpoons[B_N g_{n-1}]{B_N f_{n-1}} B_N(B_N^{n-1}G \otimes B_N^{n-1}G') \xrightleftharpoons[g]{f} B_N^n G \otimes B_N^n G'.$$

We prove by induction that f_n and g_n yield a contraction

$$(5.4) \quad (f_n, g_n, \Phi_n): B_N^n(G \otimes G') \rightleftharpoons B_N^n G \otimes B_N^n G'$$

with a natural homotopy Φ_n . For $n = 1$ this is assured by Theorem 4.1a. Assuming the conclusion for $n - 1$ it follows from Theorem I.12.1 that $(B_N f_{n-1}, B_N g_{n-1}, \Psi)$ is a contraction with some natural homotopy Ψ . Thus (5.4) is the composition of the two contractions displayed in (5.3)_N and consequently is a contraction by I, §12.

Given that (5.4) is a contraction, the fact that (5.1) is a chain equivalence follows from the commutativity in the diagram

$$\begin{array}{ccccc} B^n(G \otimes G') & \xrightarrow{f_n} & B^n(G) \otimes B^n(G') & \xrightarrow{g_n} & B^n(G \otimes G') \\ \downarrow & & \downarrow & & \downarrow \\ B_N^n(G \otimes G') & \xrightarrow{f_n} & B_N^n(G) \otimes B_N^n(G') & \xrightarrow{g_n} & B_N^n(G \otimes G'), \end{array}$$

where the vertical maps are the canonical maps associated with the normalization. These commutativity relations are proved again by induction.

We shall now give explicit formulas for the maps f_n, g_n defined above by iteration.

To express g_n we first observe that g_n commutes with the products. Further, if we identify $B^n G$ and $B^n G'$ with subrings of $B^n(G \otimes G')$ under the natural maps $a \rightarrow a \otimes \theta'$ and $x \rightarrow \theta \otimes x$, as in (4.3), then it is easy to see (by induction on n) that

$$g_n(u \otimes 1) = u, \quad g_n(1 \otimes u') = u'$$

for $u \in B^n(G)$, $u' \in B^n(G')$. Since $u \otimes u'$ is the product $(u \otimes 1) \circ (1 \otimes u')$ in $B^n(G) \otimes B^n(G')$, it follows that

$$(5.5) \quad g_n(u \otimes u') = u *_{\mathbf{n}} u',$$

where $*_{\mathbf{n}}$ is the product in $B^n(G \otimes G')$.

The computation of the explicit form of f_n requires some preliminaries. For each $n \geq 0$, introduce the maps

$$\mu_n: B^n G \otimes B^n G' \rightarrow B^n G, \quad \mu'_n: B^n G \otimes B^n G' \rightarrow B^n G'$$

given by

$$(5.6) \quad \mu_n(u \otimes u') = \alpha_n(u')u, \quad \mu'_n(u \otimes u') = \alpha_n(u)u',$$

where α_n is the augmentation map belonging to B^n .

LEMMA 5.1. *The map f_n defined by the iteration (5.2) satisfies*

$$\mu_n f_n = B^n \mu_0, \quad \mu'_n f_n = B^n \mu'_0.$$

PROOF. For $n = 1$, the proof is direct; indeed, since $\alpha_1[x_{i+1} | \cdots | x_m] = 0$ if $i < m$, the formula (4.2) for $f_1 = f_B$ applied to an argument $U = [a_1 \otimes x_1 | \cdots | a_m \otimes x_m]$ yields

$$\mu_1 f_1 U = \alpha(x_1 \circ \cdots \circ x_m)[a_1 | \cdots | a_m] = (B\mu_0)U.$$

For $n > 1$ we have

$$\mu_n f_n = \mu_n f B(f_{n-1}) = B(\mu_{n-1})B(f_{n-1}) = B(\mu_{n-1} f_{n-1}),$$

which proves the result by induction.

LEMMA 5.2. *The map f_n defined by the iteration (5.2) coincides with the composition*

$$B^n(G \otimes G') \xrightarrow{e} B^n(G \otimes G') \otimes B^n(G \otimes G') \xrightarrow{B^n(\mu_0) \otimes B^n(\mu'_0)} B^n(G) \otimes B^n(G'),$$

where e is a natural group homomorphism $e: B(H) \rightarrow B(H) \otimes B(H)$ defined for any augmented graded ∂ -ring H by the formulas

$$e1 = 1 \otimes 1,$$

$$e[a_1 | \cdots | a_m] = \sum_{i=0}^m [a_1 | \cdots | a_i] \otimes [a_{i+1} | \cdots | a_m].$$

In this definition of e we use again the convention that $[a_1 | \cdots | a_i]$ for $i = 0$ or $[a_{i+1} | \cdots | a_m]$ for $i = m$ denotes the identity 1 of $B(H)$. This map e actually commutes with boundary and product, but no property except naturality will be used in the following proof.

PROOF. For $n = 1$, the proof is direct. For $n > 1$, we consider the following diagram, with $H = G \otimes G'$,

$$\begin{array}{ccc} B^n H & \xrightarrow{Bf_{n-1}} & B(B^{n-1}G \otimes B^{n-1}G') \\ \downarrow e & & \downarrow e \\ B^n H \otimes B^n H & \xrightarrow{Bf_{n-1} \otimes Bf_{n-1}} & B(B^{n-1}G \otimes B^{n-1}G') \otimes B(B^{n-1}G \otimes B^{n-1}G') \\ \downarrow i & & \downarrow B\mu_{n-1} \otimes B\mu_{n-1}^1 \\ B^n H \otimes B^n H & \xrightarrow{B^n \mu_0 \otimes B^n \mu'_0} & B^n G \otimes B^n G', \end{array}$$

where i is the identity map. The upper square is commutative by the naturality of e ; the lower square is commutative because of lemma 5.1. Since the right column is a factorization of f for the case $n = 1$, the lemma follows.

Summarizing the results of this section, we have

THEOREM 5.3. *If G and G' are two augmented and skew-commutative graded ∂ -rings, then there is a chain equivalence*

$$B^n(G \otimes G') \underset{g}{\overset{f}{\rightleftarrows}} B^n(G) \otimes B^n(G').$$

The maps f and g are natural maps of graded ∂ -rings. The map g is defined by the formula

$$(5.7) \quad g(u \otimes u') = u *_{\mathbf{n}} u', \quad u \in B^n(G), u' \in B^n(G'),$$

where $B^n(G)$ and $B^n(G')$ are identified with subrings of $B^n(G \times G')$ by the homomorphisms induced by $a \rightarrow a \otimes \theta'$, $x \rightarrow \theta \otimes x$, as in (4.3). The map f is defined by the formula.

$$(5.8) \quad f[c_1 |_{\mathbf{n}} \cdots |_{\mathbf{n}} c_m] = \sum_{i=0}^m B^n(\mu)[c_1 |_{\mathbf{n}} \cdots |_{\mathbf{n}} c_i] \otimes B^n(\mu')[c_{i+1} |_{\mathbf{n}} \cdots |_{\mathbf{n}} c_m]$$

for $c_i \in B^{n-1}(G \otimes G')$, where $\mu = \mu_0$ and $\mu' = \mu'_0$ are as in (5.6).

THEOREM 5.3a. *The maps f and g above carry norms into norms and yield a contraction*

$$(f, g, \Phi): B_N^n(G \otimes G') \rightarrow B_N^n(G) \otimes B_N^n(G').$$

The homotopy Φ also is natural.

This pair of theorems can be used to prove similar theorems for a t -fold, tensor product $G^{(1)} \otimes \cdots \otimes G^{(t)}$.

6. The direct product theorem

Let $\Pi_1, \dots, \Pi_t$ be associative and commutative multiplicative systems, each with an identity, and

$$(6.1) \quad \Pi = \Pi_1 \times \cdots \times \Pi_t$$

their direct product. Introduce the projections

$$\nu^{(i)}: \Pi \rightarrow \Pi_i, \quad \nu^{(i)}(x_1 \cdots x_t) = x_i, \quad i = 1, \dots, t,$$

which induce maps

$$A(\nu^{(i)}, n): A(\Pi, n) \rightarrow A(\Pi_i, n), \quad i = 1, \dots, t.$$

THEOREM 6.1. *For the direct product (6.1) there is a chain equivalence*

$$(6.2) \quad A(\Pi, n) \underset{g_A}{\overset{f_A}{\rightleftarrows}} A(\Pi_1, n) \otimes \cdots \otimes A(\Pi_t, n).$$

The maps f_A and g_A are natural maps of graded ∂ -rings. The map g_A is defined by the formula

$$(6.3) \quad g_A(a_1 \otimes \cdots \otimes a_t) = a_1 *_{\mathbf{n}} \cdots *_{\mathbf{n}} a_t, \quad a_i \in A(\Pi_i, n),$$

where $\ast_n$ denotes the product in the ring $A(\Pi_i, n)$ and each $A(\Pi_i, n)$ is identified with a subring of $A(\Pi, n)$ by the injection $\Pi_i \rightarrow \Pi$. The map f_A is given, for a homogeneous element $u = [a_1 |_n \cdots |_n a_m]$ of $A(\Pi, n)$, with $n \geq 1$ and $a_i \in A(\Pi, n-1)$, by the formula

$$(6.4) \quad f_A u = \sum_P A(\nu^{(1)}, n) [a_{p_0+1} |_n \cdots |_n a_{p_1}] \otimes \cdots \otimes A(\nu^{(t)}, n) [a_{p_{t-1}+1} |_n \cdots |_n a_{p_t}],$$

where the sum is taken over all sets P of $t+1$ integers $p_0, \dots, p_t$ with

$$0 = p_0 \leq \cdots \leq p_t = m.$$

In case $t = 2$, the summation is in effect over $p_1 = 0, \dots, m$. In the case of general t it is convenient to introduce "portions" $P_p^q u$ for $0 \leq p \leq q \leq m$, where

$$(6.5) \quad P_p^p u = 1, \quad P_p^q u = [a_{p+1} |_n \cdots |_n a_q], \quad 0 \leq p < q \leq m.$$

Then the formula (6.4) takes the form

$$(6.4') \quad f_A u = \sum_P A(\nu^{(1)}, n) P_{p_0}^{p_1} u \otimes \cdots \otimes A(\nu^{(t)}, n) P_{p_{t-1}}^{p_t} u.$$

THEOREM 6.1a. *The maps f_A and g_A above carry norms into norms and yield a contraction*

$$(f_A, g_A, \Phi_A): A_N(\Pi, n) \rightarrow A_N(\Pi_1, n) \otimes \cdots \otimes A_N(\Pi_t, n).$$

The homotopy Φ_A is also natural.

PROOF. Suppose first that $n = 0$. Then $A(\Pi, 0)$ is the integral "group ring" of the multiplicative system Π , and the map $g_A = g_0$ is given by

$$g_0(x_1 \otimes \cdots \otimes x_t) = x_1 \cdots x_t, \quad x_i \in \Pi_i,$$

This clearly induces an isomorphism

$$A(\Pi_1, 0) \otimes \cdots \otimes A(\Pi_t, 0) \cong A(\Pi, 0),$$

and in this case f_A is defined to be the inverse of g_0 . If we regard these maps as identifications, then the chain equivalence (6.2) is the one obtained by applying the iterated tensor product theorem for t factors.

First assume $t = 2$. An application of Theorem 5.3 then gives chain equivalences

$$B^n(A(\Pi, 0)) \xrightleftharpoons[g]{f} B^n(A(\Pi_1, 0)) \otimes B^n(A(\Pi_2, 0)).$$

Since f and g are defined by iteration and preserve norms at each stage of this iteration, passage to $A(\Pi, n) = B_N^{n-1} B(A(\Pi, 0))$ yields chain equivalences

$$A(\Pi, n) \xrightleftharpoons[g_A]{f_A} A(\Pi_1, n) \otimes A(\Pi_2, n).$$

The formula (5.7) for g in the tensor product theorem clearly gives the above formula (6.3) for g_A . As for f , recall that the augmentation $\alpha^{(i)}(x_i)$ of an element $x_i \in \Pi_i$ in the augmented ring $A(\Pi_i, 0)$ is defined as $\alpha^{(i)}(x_i) = 1$. Hence, for any $x_1 x_2 = x_1 \otimes x_2$ in $A(\Pi, 0)$,

$$A(\nu^{(1)}, 0)(x_1 x_2) = x_1 = \alpha(x_2) x_1,$$

and therefore the map

$$A(\nu^{(1)}, 0): A(\Pi_1, 0) \otimes A(\Pi_2, 0) \rightarrow A(\Pi_1, 0)$$

is exactly the map $\mu = \mu_0$ of (5.6) used in the definition of f , and similarly $A(\nu^{(2)}, 0)$ is the map $\mu' = \mu'_0$. The formula (5.8) for f then becomes (6.4), in the case $t = 2$.

This proves the case $t = 2$. The general case now follows by straightforward iteration.

With this theorem, the homology groups $H(\Pi, n)$ may be computed from the homology groups $H(\Pi_i, n)$ of the direct factors Π_i of Π , using the Künneth formulas (§12) for the homology groups of the tensor product of complexes.

A similar discussion can be applied directly to the complexes $K(\Pi, n)$, where Π is now a group. Observe that if σ is an n -simplex of $K(\Pi, n)$ with $\Pi = \Pi_1 \times \cdots \times \Pi_t$, as in (6.1), then the projection $\nu^{(i)}$ of Π upon a component Π_i carries σ into a simplex σ_i of $K(\Pi_i, n)$. The correspondence $\sigma \rightarrow (\sigma_1 \times \cdots \times \sigma_t)$ then provides an identification of the R -complex $K(\Pi, n)$ with the cartesian product $K(\Pi_1, n) \times \cdots \times K(\Pi_t, n)$. By Corollary 2.2, we obtain a chain equivalence

$$(6.6) \quad K(\Pi, n) \xrightleftharpoons[\nabla]{f} K(\Pi_1, n) \otimes \cdots \otimes K(\Pi_t, n).$$

If we compare this with (6.2) using the main map $k_n: A(\Pi, n) \rightarrow K(\Pi, n)$ of (I.20.3) we obtain the diagram

$$(6.7) \quad \begin{array}{ccc} A(\Pi, n) & \xrightleftharpoons[g_A]{f_A} & A(\Pi_1, n) \otimes \cdots \otimes A(\Pi_t, n) \\ \downarrow k_n & & \downarrow k'_n \\ K(\Pi, n) & \xrightleftharpoons[\nabla]{f} & K(\Pi_1, n) \otimes \cdots \otimes K(\Pi_t, n), \end{array}$$

where $k'_n = k_n \otimes \cdots \otimes k_n$.

THEOREM 6.2. *In the diagram (6.7), the commutativity*

$$(6.8) \quad k_n g_A = \nabla k'_n.$$

holds, while for the maps f there is a chain homotopy

$$(6.9) \quad k'_n f_A \sim f k_n.$$

PROOF. The commutativity relation (6.8) follows from the fact that g and ∇ are defined using products, while k_n and k'_n commute with products. This implies

$$k'_n f_A \sim f \nabla k'_n f_A = f k_n g_A f_A \sim f k_n.$$

REMARK. If all the complexes K and A are normalized, it can be proved that the homotopy in (6.9) can be replaced by an equality.

7. Diagonal maps and cup products

If K and K' are two chain complexes, while E and E' are two coefficient groups, the well-known "external" cohomology product is the homomorphism

$$(7.1) \quad H^p(K, E) \otimes H^q(K', E') \rightarrow H^{p+q}(K \otimes K'; E \otimes E'),$$

which is obtained by associating with each pair of cochains $f \in C^p(K, E)$, $f' \in C^q(K', E')$ the cochain $g \in C^{p+q}(K \otimes K', E \otimes E')$ defined by

$$g(a \otimes a') = f(a) \otimes f'(a'), \quad a \in K_p, a' \in K'_q,$$

and otherwise zero.

For any chain transformation

$$e: K \rightarrow K \otimes K,$$

called in the sequel a "diagonal map," one defines a cup product

$$\smile: H^p(K, E) \otimes H^q(K, E') \rightarrow H^{p+q}(K, E \otimes E')$$

in K by the composition of the map (7.1) with the map $H^{p+q}(K \otimes K; E \otimes E') \rightarrow H^{p+q}(K; E \otimes E')$ induced by e . Clearly two homotopic diagonal maps induce the same cup product.

We shall define such diagonal maps for the complexes $K(\Pi, n)$ and $A(\Pi, n)$. We begin by defining

$$(7.2) \quad e_\Pi: \Pi \rightarrow \Pi \times \Pi$$

by setting $e_\Pi x = (x, x)$ for $x \in \Pi$. Then we define

$$(7.3) \quad e_K: K(\Pi, n) \rightarrow K(\Pi, n) \otimes K(\Pi, n)$$

as the composition

$$(7.4) \quad K(\Pi, n) \xrightarrow{K(e_\Pi, n)} K(\Pi \times \Pi, n) \xrightarrow{f} K(\Pi, n) \otimes K(\Pi, n),$$

where f is the map of (6.6). Alternatively, if we identify $K(\Pi \times \Pi, n)$ with $K(\Pi, n) \times K(\Pi, n)$, f becomes the map in the Eilenberg-Zilber theorem. If σ is a q -simplex of $K(\Pi, n)$, then $K(e_\Pi, n)$ maps σ into the simplex $\sigma \times \sigma$ of $K(\Pi, n) \times K(\Pi, n)$. Composition with the map f then yields the following formula for e_K :

$$(7.5) \quad e_K(\sigma) = \sum_{i=0}^q \tilde{F}^{q-i} \sigma \otimes F_0^i \sigma.$$

This is the standard formula for the diagonal map in complexes with a simplicial structure. This insures that in any topological applications the cup-products obtained from the diagonal map e_K will correspond to the Alexander-Whitney cup-products in topology.

For the complexes $A(\Pi, n)$ we define the diagonal map

$$(7.6) \quad e_A: A(\Pi, n) \rightarrow A(\Pi, n) \otimes A(\Pi, n)$$

as the composition

$$(7.7) \quad A(\Pi, n) \xrightarrow{A(e_\Pi, n)} A(\Pi \times \Pi, n) \xrightarrow{f_A} A(\Pi, n) \otimes A(\Pi, n)$$

where f_A is the map of Theorem 6.1. For a homogeneous element

$$u = [a_1 |_n \cdots |_n a_m]$$

of $A(\Pi, n)$ with $n \geq 1$ and $a_i \in A(\Pi, n-1)$ we have

$$A(e_\Pi, n)u = [A(e_\Pi, n-1)a_1 |_n \cdots |_n A(e_\Pi, n-1)a_m].$$

This implies

$$P_p^q A(e_\Pi, n)u = A(e_\Pi, n)P_p^q u$$

for the "portions" defined in (6.5). Since for the maps $\nu^{(i)}: \Pi \times \Pi \rightarrow \Pi$ of §6 we have $\nu^{(i)}e_\Pi = \text{identity}$ for $i = 1, 2$, we obtain the following closed formula for e_A :

$$(7.8) \quad e_A u = \sum_{i=0}^m [a_1 |_n \cdots |_n a_i] \otimes [a_{i+1} |_n \cdots |_n a_m].$$

When $n = 1$, $A(\Pi, 1)$ coincides with $K(\Pi, 1)$ and $e_K = e_A$.

THEOREM 7.1. *In the diagram*

$$\begin{array}{ccc} A(\Pi, n) & \xrightarrow{e_A} & A(\Pi, n) \otimes A(\Pi, n) \\ \downarrow k_n & & \downarrow k_n \otimes k_n \\ K(\Pi, n) & \xrightarrow{e_K} & K(\Pi, n) \otimes K(\Pi, n), \end{array}$$

where k_n is the main map, we have the homotopy

$$(k_n \otimes k_n)e_A \sim e_K k_n.$$

This theorem insures that the diagonal map e_A yields the "correct" cup-products in $A(\Pi, n)$. The proof follows directly from the definitions (7.4) and (7.7) and from the homotopy (6.9) of Theorem 6.2. Both diagonal maps e_K and e_A carry norms into norms, and therefore apply also to the normalized complexes. In view of the remark at the end of §6, the homotopy of Theorem 7.1 may be replaced by an equality provided that we pass to the normalized complexes.

We shall now make some general remarks about diagonal maps that will not be used in the sequel.

Let G be an augmented skew-commutative graded ∂ -ring. A map

$$e: G \rightarrow G \otimes G$$

of augmented graded ∂ -rings will be called a *diagonal map* if the compositions

$$G \xrightarrow{e} G \otimes G \xrightarrow{\eta^{(1)}} G$$

$$G \xrightarrow{e} G \otimes G \xrightarrow{\eta^{(2)}} G$$

are the identity maps, where

$$\eta^{(1)}(g_1 \otimes g_2) = (\alpha g_2)g_1, \quad \eta^{(2)}(g_1 \otimes g_2) = (\alpha g_1)g_2.$$

In general diagonal maps need not exist.

THEOREM 7.2. *If G is an augmented skew-commutative graded ∂ -ring with a diagonal map $e: G \rightarrow G \otimes G$, then the composition*

$$B(G) \xrightarrow{B(e)} B(G \otimes G) \xrightarrow{f} B(G) \otimes B(G)$$

is a diagonal map

$$e_B: B(G) \rightarrow B(G) \otimes B(G)$$

given for a homogeneous element $[a_1 | \cdots | a_m]$, $a_i \in G$, by the formula

$$(7.9) \quad e_B[a_1 | \cdots | a_m] = \sum_{i=0}^m [a_1 | \cdots | a_i] \otimes [a_{i+1} | \cdots | a_m].$$

We recall that the map e_B has already made an appearance in Lemma 5.2, but none of its properties, except for naturality, was used there. Actually by fairly tedious and straight-forward calculations it is possible to see that e_B is a diagonal map for any ∂ -ring $B(G)$. For those ∂ -rings $B(G)$ where G has a diagonal map this fact follows from Theorem 7.2 without calculations. We further note the curious fact that the composition $fB(e)$ is independent of the diagonal map e in G .

The proof of Theorem 7.2 is based on the diagram

$$\begin{array}{ccccc} BG & \xrightarrow{e_B} & BG \otimes BG & \xrightarrow{i} & BG \otimes BG \\ \downarrow B(e) & & \downarrow B(e) \otimes B(e) & & \downarrow i \\ B(G \otimes G) & \xrightarrow{e_B} & B(G \otimes G) \otimes B(G \otimes G) & \xrightarrow{B(\eta^{(1)}) \otimes B(\eta^{(2)})} & BG \otimes BG \end{array},$$

where i is the identity map. The commutativity in the first square expresses the naturality of e_B , while that in the second square follows from $\eta^{(1)}e = \eta^{(2)}e = \text{identity}$. Since, by Lemma 5.2, the composition in the lower row is f , the conclusion follows.

A similar discussion applies to the W -construction (I, §17). Given an augmented commutative R -complex R , a diagonal map

$$e_W: W(R) \rightarrow W(R) \otimes W(R)$$

is defined by

$$(7.10) \quad e_W \langle a_{q-1}, \dots, a_0 \rangle = \sum_{i=0}^q \langle \tilde{F}^{q-i} a_{q-1}, \dots, \tilde{F}^{q-i} a_{q-i} \rangle \otimes \langle a_{q-i-1}, \dots, a_0 \rangle$$

with appropriate conventions for the terms $i = 0, i = q$. The map e_W coincides with e_K in case $R = K(\Pi, n-1)$, $W(R) = K(\Pi, n)$. It can then be proved that the appropriate diagram

$$\begin{array}{ccc} B(R) & \xrightarrow{e_B} & B(R) \otimes B(R) \\ \downarrow k & & \downarrow k \otimes k \\ W(R) & \xrightarrow{e_W} & W(R) \otimes W(R) \end{array}$$

for the main transfer k is commutative modulo norms.

CHAPTER II. DEVIATIONS AND CROSS-EFFECTS

8. The algorithm for deviations

Let C be an associative multiplicative system, and let $J(C)$ be its "integral group ring," whose elements are all finite formal sums $\sum m_i c_i$ with integral coefficients m_i and $c_i \in C$, with the obvious addition and multiplication. We define a new composition in $J(C)$ by

$$(8.1) \quad a \tau b = ab - a - b, \quad a, b \in C$$

and distributivity. This composition is readily shown to be associative.

Let G be an additive abelian group and f a function on C to G . Extend f to $J(C)$ by linearity; i.e., set

$$f(\sum_{i=1}^t m_i c_i) = \sum_{i=1}^t m_i f(c_i), \quad c_i \in C, m_i \text{ integers.}$$

Then, according to the definition (8.1),

$$(8.2) \quad f(a \tau b) = f(ab) - f(a) - f(b),$$

and we may regard $f(a \tau b)$ as a new function on $C \times C$ to G , which we call the *deviation* of f . It is zero precisely when f is a homomorphism. In view of the associativity of the composition τ , we have without ambiguity a *second deviation*

$$f(a \tau b \tau c) = f(abc) - f(ab) - f(ac) - f(bc) + f(a) + f(b) + f(c).$$

By induction on n one obtains an analogous explicit formula for the n^{th} deviation of the function f as

$$(8.3) \quad f(c_1 \tau \dots \tau c_n) = \sum_{k=1}^n \varphi(n-k) \sum_{i_1 < \dots < i_k} f(c_{i_1} \dots c_{i_k}), \quad c_i \in C.$$

We prove at once the following additivity property.

LEMMA 8.1. *If $f = f_1 + f_2$ is the sum of two functions f_1 and f_2 on C to G , then the n^{th} deviation of f is the sum of the n^{th} deviations of f_1 and f_2 .*

LEMMA 8.2. *If C has an identity element 1 and $f(1) = 0$, then*

$$f(c_1 \tau \cdots \tau c_n) = 0$$

whenever any one $c_i = 1$.

PROOF. In (8.3) separate the terms in which c_i is present from those in which c_i is absent. When $c_i = 1$, these terms cancel in pairs, except for the one term in which only c_i is present; this term is zero by hypothesis.

By (8.2) we can express the value of f for a product in terms of f and the first deviation of f as

$$(8.4) \quad f(ab) = f(a \tau b) + f(a) + f(b), \quad a, b \in C.$$

The corresponding formula for an n -fold product

$$(8.5) \quad f(c_1 \cdots c_n) = \sum_{k=1}^n \sum_{i_1 < \cdots < i_k} f(c_{i_1} \tau \cdots \tau c_{i_k}), \quad c_i \in C,$$

is readily established by induction.

If C is commutative, so is the composition τ , and in this case

$$(8.6) \quad f(c_{\sigma(1)} \tau \cdots \tau c_{\sigma(n)}) = f(c_1 \tau \cdots \tau c_n), \quad c_i \in C,$$

for any permutation σ of the subscripts $1, \cdots, n$.

If C and $D_1, \cdots, D_m$ are associative multiplicative systems and f is a function on $C \times D_1 \times \cdots \times D_m$ to G , we define the deviation of f in its first argument by the formula

$$(8.7) \quad f(a \tau b, d_1 \cdots, d_m) = f(ab, d_1, \cdots, d_m) - f(a, d_1, \cdots, d_m) \\ - f(b, d_1, \cdots, d_m),$$

for $a, b \in C, d_i \in D_i$. All the previous results hold, *mutatis mutandis*, for such deviations. Furthermore, the first deviation on C of the first deviation on D_1 of the function f is identical with the first deviation on D_1 of the first deviation on C ; for $m = 1$, it is given by the formula

$$f(a \tau b, r \tau s) = f(ab, rs) - f(ab, r) - f(ab, s) - f(a, rs) - f(b, rs) \\ + f(a, r) + f(a, s) + f(b, r) + f(b, s),$$

for $a, b \in C$ and $r, s \in D$. More generally, this fact and (8.3) yield the formula

$$(8.8) \quad f(c_1 \tau \cdots \tau c_n, d_1 \tau \cdots \tau d_m) = \sum_{k=1}^n \sum_{l=1}^m \wp(n-k+m-l) \\ \cdot \sum_{i_1 < \cdots < i_k} \sum_{j_1 < \cdots < j_l} f(c_{i_1} \cdots c_{i_k}, d_{j_1} \cdots d_{j_l}).$$

9. Cross-effect functors

This section is concerned with properties of a covariant functor T on abelian groups to abelian groups or, more generally, on modules over a commutative

ring to modules over another commutative ring. Throughout we assume that $T(0) = 0$, where 0 designates the abelian group (or module) containing only the zero element. If $O_{BA}:B \rightarrow A$ designates the zero homomorphism of the group B into the group A , our assumption implies that $T(O_{BA}) = O_{T(B)T(A)}$.

The functor T is said to be *additive* if, for any two maps $f, g:A \rightarrow B$,

$$T(f + g) = T(f) + T(g):T(A) \rightarrow T(B).$$

This condition is equivalent (cf. Theorem 9.11 below) to the requirement that $T(A + B) \cong T(A) + T(B)$ under the canonical map.

When T is not additive, then the deviation

$$T(f \tau g) = T(f + g) - T(f) - T(g)$$

is not zero; this is indeed an application of the notion of deviations of the previous section, regarding T as a function on the (additive) system $C = \text{Hom}(A, B)$ to the system $G = \text{Hom}(T(A), T(B))$. More generally, the k -fold deviation

$$T(f_1 \tau \cdots \tau f_k):T(A) \rightarrow T(B)$$

for $f_1, \dots, f_k:A \rightarrow B$ is defined as in (8.3) by

$$(9.1) \quad T(f_1 \tau \cdots \tau f_k) = \sum_{s=1}^k \wp(k-s) \sum_{i_1 < \dots < i_s} T(f_{i_1} + \cdots + f_{i_s}).$$

Since T is a functor, we have $T(f_i g) = T(f_i)T(g)$ for any $g:B \rightarrow B'$ and hence

$$(9.2) \quad T(f_1 \tau \cdots \tau f_k)T(g) = T(f_1 g \tau \cdots \tau f_k g).$$

Similarly, for any $h:A' \rightarrow A$,

$$(9.2') \quad T(h)T(f_1 \tau \cdots \tau f_k) = T(hf_1 \tau \cdots \tau hf_k).$$

DEFINITION. Let $A = A_1 + \cdots + A_k$ be the direct sum of k abelian groups with the endomorphisms $\psi_i:A \rightarrow A$ defined for $a_i \in A_i$ by

$$(9.3) \quad \psi_i(a_1, \dots, a_k) = (0, \dots, 0, a_i, 0, \dots, 0), \quad i = 1, \dots, k.$$

The k^{th} cross-effect functor for T is a covariant functor of k abelian groups $A_1, \dots, A_k$. The object function $T(A_1 | \cdots | A_k)$ is defined as the image of a deviation,

$$(9.4) \quad T(A_1 | \cdots | A_k) = T(\psi_1 \tau \cdots \tau \psi_k)T(A).$$

The mapping function $T(f_1 | \cdots | f_k)$ for k homomorphisms $f_i:A_i \rightarrow B_i$ is defined to be the homomorphism

$$(9.5) \quad T(f_1 | \cdots | f_k):T(A_1 | \cdots | A_k) \rightarrow T(B_1 | \cdots | B_k)$$

induced by $f = f_1 + \cdots + f_k:A_1 + \cdots + A_k \rightarrow B_1 + \cdots + B_k$.

Since A is a direct sum, the endomorphisms ψ_i satisfy

$$(9.6) \quad \psi_i \psi_i = \psi_i; \quad \psi_i \psi_j = 0, i \neq j; \quad \psi_1 + \cdots + \psi_k = I_A,$$

where I_A denotes the identity map of A . Furthermore, the deviation

$$(9.7) \quad D_\sigma = T(\psi_{\sigma_1} \tau \cdots \tau \psi_{\sigma_s}):T(A) \rightarrow T(A)$$

is defined for each non-void subset $\sigma = \{\sigma_1, \dots, \sigma_s\}$ of $\{1, \dots, k\}$. The group

$$A^\sigma = A_{\sigma_1} + \dots + A_{\sigma_s}, \quad \sigma = \{\sigma_1, \dots, \sigma_s\}$$

may be regarded as a direct summand of A , with injection and projection denoted by

$$(9.8) \quad \lambda_\sigma: A^\sigma \rightarrow A, \quad q_\sigma: A \rightarrow A^\sigma.$$

The basic property of the cross-effects is the representation of

$$T(A_1 + \dots + A_k)$$

as a direct sum of s -fold cross-effects, for $1 \leq s \leq k$. This may be stated as follows:

THEOREM 9.1. *There is a natural isomorphism*

$$T(A_1 + \dots + A_k) \cong \sum_\sigma T(A_{\sigma_1} | \dots | A_{\sigma_s}),$$

in which the sum is taken over all non-void subsets σ of $\{1, \dots, k\}$. More precisely, $T(A)$ is the direct sum

$$T(A_1 + \dots + A_k) = \sum_\sigma T(\lambda_\sigma)T(A_{\sigma_1} | \dots | A_{\sigma_s}),$$

where λ_σ is the injection of (9.8). The induced map

$$T(\lambda_\sigma): T(A_{\sigma_1} | \dots | A_{\sigma_s}) \rightarrow T(A_1 + \dots + A_k)$$

has kernel zero and image $T(\psi_{\sigma_1} \tau \dots \tau \psi_{\sigma_s})T(A)$; on this image the inverse $T(\lambda_\sigma)^{-1}$ is induced by $T(q_\sigma)$.

PROOF. In the notation (9.7) we shall first establish a direct sum decomposition

$$(9.9) \quad T(A_1 + \dots + A_k) = \sum_\sigma D_\sigma T(A_1 + \dots + A_k),$$

then show that

$$(9.10) \quad T(\lambda_\sigma)[T(A_{\sigma_1} | \dots | A_{\sigma_s})] \subset D_\sigma T(A), \quad T(q_\sigma)[D_\sigma T(A)] \subset T(A_{\sigma_1} | \dots | A_{\sigma_s}),$$

and finally prove that both composites in the diagram

$$(9.11) \quad T(A_{\sigma_1} | \dots | A_{\sigma_s}) \xrightleftharpoons[T(q_\sigma)]{T(\lambda_\sigma)} D_\sigma T(A)$$

are the identities. These facts then give the theorem.

To obtain (9.9) it suffices, according to a familiar description of direct sums, to show that

$$\sum_\sigma D_\sigma = I_{T(A)}, \quad D_\sigma D_\tau = 0, \quad \sigma \neq \tau,$$

where $I_{T(A)}$ denotes the identity map of $T(A)$. The first of these follows by (8.5), for

$$\sum_\sigma D_\sigma = \sum_\sigma T(\psi_{\sigma_1} \tau \dots \tau \psi_{\sigma_s}) = T(\psi_1 + \dots + \psi_k) = T(I_A) = I_{T(A)}.$$

To obtain the second, set

$$\psi^i = \psi_1 + \cdots + \psi_{i-1} + \psi_{i+1} + \cdots + \psi_k, \quad i = 1, \dots, k.$$

Then, by (9.6)

$$\psi^i \psi_i = 0 = \psi_i \psi^i, \quad \psi^i \psi_j = \psi_j = \psi_j \psi^i, \quad i \neq j.$$

If σ and τ are distinct subsets of $\{1, \dots, k\}$, choose an index i which is, say, in σ and not in τ . Then by (9.2) and (9.2')

$$\begin{aligned} D_\sigma D_\tau &= T(\psi_{\sigma_1} \tau \cdots \tau \psi_{\sigma_s}) T(\psi_{\tau_1} \tau \cdots \tau \psi_{\tau_t}) \\ &= T(\psi_{\sigma_1} \tau \cdots \tau \psi_{\sigma_s}) T(\psi^i \psi_{\tau_1} \tau \cdots \tau \psi^i \psi_{\tau_t}) \\ &= T(\psi_{\sigma_1} \tau \cdots \tau \psi_{\sigma_s}) T(\psi^i) T(\psi_{\tau_1} \tau \cdots \tau \psi_{\tau_t}) \\ &= T(\psi_{\sigma_1} \psi^i \tau \cdots \tau \psi_{\sigma_s} \psi^i) T(\psi_{\tau_1} \tau \cdots \tau \psi_{\tau_t}). \end{aligned}$$

Since $i \in \sigma$, one argument $\psi_{\sigma_i} \psi^i$ is zero, hence the result is zero by Lemma 8.2.

Next, to prove (9.10), let $\psi_i^\sigma: A^\sigma \rightarrow A^\sigma$ for $i \in \sigma$ be the endomorphisms of A^σ defined exactly as in (9.3). They are related to the endomorphisms ψ_i of A by

$$\psi_i^\sigma = q_\sigma \psi_i \lambda_\sigma, \quad \psi_i = \lambda_\sigma \psi_i^\sigma q_\sigma, \quad i \in \sigma.$$

From these we deduce that

$$q_\sigma \psi_i = \psi_i^\sigma q_\sigma, \quad \lambda_\sigma \psi_i^\sigma = \psi_i \lambda_\sigma, \quad i \in \sigma.$$

Now, by (9.2 and (9.2'),

$$\begin{aligned} T(q_\sigma) D_\sigma &= T(q_\sigma) T(\psi_{\sigma_1} \tau \cdots \tau \psi_{\sigma_s}) = T(q_\sigma \psi_{\sigma_1} \tau \cdots \tau q_\sigma \psi_{\sigma_s}) \\ &= T(\psi_{\sigma_1}^\sigma q_\sigma \tau \cdots \tau \psi_{\sigma_s}^\sigma q_\sigma) = T(\psi_{\sigma_1}^\sigma \tau \cdots \tau \psi_{\sigma_s}^\sigma) T(q_\sigma), \\ T(\lambda_\sigma) T(\psi_{\sigma_1}^\sigma \tau \cdots \tau \psi_{\sigma_s}^\sigma) &= T(\psi_{\sigma_1} \tau \cdots \tau \psi_{\sigma_s}) T(\lambda_\sigma). \end{aligned}$$

These prove (9.10) and show that $T(q_\sigma) T(\lambda_\sigma)$ is the identity on

$$T(\psi_{\sigma_1}^\sigma \tau \cdots \tau \psi_{\sigma_s}^\sigma) T(A^\sigma).$$

On the other hand,

$$\lambda_\sigma q_\sigma \psi_i = \lambda_\sigma q_\sigma \lambda_\sigma \psi_i^\sigma q_\sigma = \lambda_\sigma \psi_i^\sigma q_\sigma = \psi_i, \quad i \in \sigma,$$

hence

$$T(\lambda_\sigma) T(q_\sigma) D_\sigma = T(\lambda_\sigma q_\sigma \psi_{\sigma_1} \tau \cdots \tau \lambda_\sigma q_\sigma \psi_{\sigma_s}) = D_\sigma,$$

so that $T(\lambda_\sigma) T(q_\sigma)$ is the identity on $D_\sigma T(A)$. This gives (9.11) and completes the proof of the theorem.

More elementary properties of the cross-effects are given by the next three theorems.

THEOREM 9.2. $T(A_1 | \cdots | A_k) = 0$, whenever one argument $A_i = 0$.

PROOF. If $A_i = 0$, then $\psi_i = 0$, hence $D_\sigma = 0$, by Lemma 8.2.

THEOREM 9.3. *The k^{th} cross-effect functor is symmetric in its arguments; that is, any permutation π of the subscripts $1, \dots, k$ induces a natural isomorphism $T(A_1 | \dots | A_k) \cong T(A_{\pi(1)} | \dots | A_{\pi(k)})$.*

PROOF. Let π also denote the endomorphism $\pi: A \rightarrow A$ induced by the permutation π . Then

$$T(\pi)T(\psi_1 \tau \dots \tau \psi_k) = T(\pi\psi_1 \tau \dots \tau \pi\psi_k) = T(\psi_{\pi(1)} \tau \dots \tau \psi_{\pi(k)});$$

hence $T(\pi)$ induces a homomorphism of $T(A_1 | \dots | A_k)$ into $T(A_{\pi(1)} | \dots | A_{\pi(k)})$. Since $T(\pi)T(\pi^{-1}) = T(\pi\pi^{-1}) = I$, this homomorphism is an isomorphism.

LEMMA 9.4. *Let σ be a subset of $\{1, \dots, k\}$ with $s > 0$ elements, let S_σ be a covariant functor of s abelian groups such that $S_\sigma(B_1, \dots, B_s) = 0$ whenever one $B_i = 0$, and let*

$$g_\sigma: S_\sigma(A_{\sigma_1}, \dots, A_{\sigma_s}) \rightarrow T(A_1 + \dots + A_k)$$

be a homomorphism which is natural when both sides are regarded as covariant functors of the k groups $A_1, \dots, A_k$. Then

$$g_\sigma S_\sigma(A_{\sigma_1}, \dots, A_{\sigma_s}) \subset T(\psi_\sigma \tau \dots \tau \psi_{\sigma_s})T(A).$$

PROOF. Let τ be any non-void subset of σ , with t elements. Define endomorphisms $\alpha_i: A_i \rightarrow A_i$, for $i = 1, \dots, k$ by taking α_i to be the identity or zero according as $i \in \tau$ or $i \notin \tau$. Consider the diagram

$$\begin{array}{ccc} S_\sigma(A_{\sigma_1}, \dots, A_{\sigma_s}) & \xrightarrow{g_\sigma} & T(A_1 + \dots + A_k) \\ \downarrow S_\sigma(\alpha_{\sigma_1}, \dots, \alpha_{\sigma_s}) & & \downarrow T(\sum_{j \in \tau} \psi_j) \\ S_\sigma(A_{\sigma_1}, \dots, A_{\sigma_s}) & \xrightarrow{g_\sigma} & T(A_1 + \dots + A_k). \end{array}$$

Since $T(\sum_{j \in \tau} \psi_j) = T(\alpha_1 + \dots + \alpha_k)$, this diagram is commutative in virtue of the naturality of g_σ . If τ is a proper subset of σ , then $S_\sigma(\alpha_{\sigma_1}, \dots, \alpha_{\sigma_s}) = 0$ because one argument is zero, while if $\tau = \sigma$, then $S_\sigma(\alpha_{\sigma_1}, \dots, \alpha_{\sigma_s})$ is the identity because each argument is the identity. Now write down the commutativity relation for the diagram above for each τ and add with the signs $\wp(s - t)$; using (9.1), we get

$$g_\sigma = [\sum_\tau \wp(s - t)T(\sum_{j \in \tau} \psi_j)]g_\sigma = T(\psi_{\sigma_1} \tau \dots \tau \psi_{\sigma_s})g_\sigma.$$

This establishes the desired conclusion.

THEOREM 9.5. *For covariant functors T and U any natural homomorphism $h: T \rightarrow U$ induces a natural homomorphism*

$$h: T(A_1 | \dots | A_k) \rightarrow U(A_1 | \dots | A_k).$$

PROOF. By definition, $T(A_1 | \dots | A_k) \subset T(A)$, hence the injection

$$\mu: T(A_1 | \dots | A_k) \rightarrow T(A_1 + \dots + A_k)$$

is a natural homomorphism (onto a direct summand, by Theorem 9.1). Also

$$h(A_1 + \cdots + A_k): T(A_1 + \cdots + A_k) \rightarrow U(A_1 + \cdots + A_k)$$

is a natural homomorphism. The result follows by applying Lemma 9.4 to the composite $g = h(A)_\mu$, with $S(A_1, \cdots, A_k)$ replaced by $T(A_1 | \cdots | A_k)$ and T replaced by U .

The cross-effect functors may now be characterized by their appearance in a direct sum decomposition like that of Theorem 9.1.

THEOREM 9.6. *A natural isomorphism*

$$g: \sum_{\sigma} S_{\sigma}(A_{\sigma_1}, \cdots, A_{\sigma_s}) \cong T(A_1 + \cdots + A_k),$$

where the sum is taken over all non-void subsets $\sigma = \{\sigma_1, \cdots, \sigma_s\}$ of $\{1, \cdots, k\}$ and where each S_{σ} is a covariant functor of s arguments, zero when one argument is zero, carries $S_{\sigma}(A_{\sigma_1}, \cdots, A_{\sigma_s})$ isomorphically onto $T(\psi_{\sigma_1} \tau \cdots \tau \psi_{\sigma_s})T(A)$, and hence yields a natural isomorphism

$$T(q_{\sigma})g: S_{\sigma}(A_{\sigma_1}, \cdots, A_{\sigma_s}) \cong T(A_{\sigma_1} | \cdots | A_{\sigma_s}),$$

where q_{σ} is defined as in (9.8).

PROOF. The second conclusion follows from the first, in view of Theorem 9.1. Since gS_{σ} is contained in $T(\psi_{\sigma_1} \tau \cdots \tau \psi_{\sigma_s})T(A)$ by the Lemma, and since $T(A)$ is the direct sum of these groups, by Theorem 9.1 again, the first conclusion is also immediate.

In subsequent applications this theorem will allow us to identify the cross-effect functors of T with suitable functors S_{σ} , upon construction of an isomorphism g .

COROLLARY 9.7. *If $\rho = \{1, \cdots, k\}$, then, under the hypotheses of Theorem 9.6, g maps $S_{\rho}(A_1, \cdots, A_k)$ isomorphically onto $T(A_1 | \cdots | A_k)$.*

We now consider iterated cross-effects.

LEMMA 9.8. *If $T(A, B)$ is a covariant functor of two abelian groups, which vanishes when any one argument is zero, and if S and R are the cross-effect functors,*

$$S(A_1, A_2, B) = T(A_1 | A_2, B), \quad R(A, B_1, B_2) = T(A, B_1 | B_2),$$

then

$$S(A_1, A_2, B_1 | B_2) = R(A_1 | A_2, B_1, B_2).$$

The common value of these two functors will be denoted by

$$T(A_1 | A_2, B_1 | B_2).$$

PROOF. We use the endomorphisms $\psi_i: A_1 + A_2 \rightarrow A_1 + A_2$, $i = 1, 2$, as above, as well as the analogous $\phi_i: B_1 + B_2 \rightarrow B_1 + B_2$, $i = 1, 2$. Then in $T(A_1 + A_2, B_1 + B_2)$ we have the endomorphism,

$$T(\psi_1 \tau \psi_2, \phi_1 \tau \phi_2) = T(I, \phi_1 \tau \phi_2)T(\psi_1 \tau \psi_2, I),$$

where I is the appropriate identity map. Hence

$$\begin{aligned}\operatorname{Im} T(\psi_1 \tau \psi_2, \phi_1 \tau \phi_2) &= T(I, \phi_1 \tau \phi_2)[\operatorname{Im} T(\psi_1 \tau \psi_2, I)] \\ &= T(I, \phi_1 \tau \phi_2)[S(A_1, A_2, B_1 + B_2)].\end{aligned}$$

Now the restriction of $T(I, \phi_1 \tau \phi_2)$ to S is, by definition, $S(I_1, I_2, \phi_1 \tau \phi_2)$, where $I_j: A_j \rightarrow A_j$ is the identity. Therefore

$$\begin{aligned}\operatorname{Im} T(\psi_1 \tau \psi_2, \phi_1 \tau \phi_2) &= \operatorname{Im} S(I_1, I_2, \phi_1 \tau \phi_2) \\ &= S(A_1, A_2, B_1 | B_2).\end{aligned}$$

By symmetry this proves the desired equality and yields a symmetric description of $T(A_1 | A_2, B_1 | B_2)$ as the image of $T(\psi_1 \tau \psi_2, \phi_1 \tau \phi_2)$.

LEMMA 9.9. *If $T(A)$ is a covariant functor with cross effect $S(A_1, A_2) = T(A_1 | A_2)$, then*

$$S(A_1 | A_2, A_3) = S(A_1, A_2 | A_3) = T(A_1 | A_2 | A_3).$$

The proof is similar, but more elementary.

We may regard these two lemmas as statements of commutative and associative laws for the iterated formation of cross-effect functors. These laws may be combined in the familiar fashion. Thus, given a covariant functor

$$T(A_1, \dots, A_m)$$

of m abelian groups, which vanishes when any one argument is zero, we have the iterated cross-effect functor

$$T(A_{11} | \dots | A_{1k_1}, A_{21} | \dots | A_{2k_2}, \dots, A_{m1} | \dots | A_{m,k_m}),$$

which may be described directly as an appropriate subgroup of

$$T(A_{11} + \dots + A_{1k_1}, \dots, A_{m1} + \dots + A_{m,k_m}).$$

LEMMA 9.10. *If T is a covariant functor and A a fixed group such that the k -fold cross-effect $T(A | \dots | A) = 0$, then for any k homomorphisms $f_i: A \rightarrow B$, we have*

$$T(f_1 \tau \dots \tau f_k) = 0.$$

Similarly, for any k homomorphisms $g: C \rightarrow A$,

$$T(g_1 \tau \dots \tau g_k) = 0.$$

PROOF. Take $A^{(k)} = A + \dots + A$, with k terms, and let d be the diagonal map $d: A \rightarrow A^{(k)}$ with $d(a) = (a, a, \dots, a)$. The k homomorphisms f_i then determine a homomorphism $f: A^{(k)} \rightarrow B$ by

$$f(a_1, \dots, a_k) = f_1 a_1 + \dots + f_k a_k, \quad a_i \in A.$$

Then, using the maps ψ_i of (9.3) for the direct sum $A^{(k)}$, we have $f_i = f\psi_i d$, and hence, by (9.2) and (9.2'),

$$\begin{aligned} T(f_1 \tau \cdots \tau f_k) &= T(f\psi_1 d \tau \cdots \tau f\psi_k d) \\ &= T(f)T(\psi_1 \tau \cdots \tau \psi_k)T(d) = 0, \end{aligned}$$

since $T(\psi_1 \tau \cdots \tau \psi_k) = 0$ by hypothesis.

THEOREM 9.11. *For any positive integer k , the following two properties of the functor T are equivalent:*

$$(i) \quad T(A_1 | \cdots | A_k) = 0,$$

for any k abelian groups $A_1, \cdots, A_k$;

$$(ii) \quad T(f_1 \tau \cdots \tau f_k) = 0,$$

for any two abelian groups A and B and any k homomorphisms $f_i: A \rightarrow B$.

PROOF. It is trivial that (ii) implies (i), for $T(A_1 | \cdots | A_k)$ is defined as the image of $T(\psi_1 \tau \cdots \tau \psi_k)$. The converse follows from Lemma 9.10.

COROLLARY 9.12. *$T(A_1 | \cdots | A_k)$ is identically zero if and only if it is zero when $A_1 = \cdots = A_k$.*

PROOF. Given $T(A | \cdots | A) = 0$ for all A , Lemma 9.10 gives (ii) above, and Theorem 9.11 gives (i). The converse is trivial.

A functor T satisfying either of the equivalent conditions (i) or (ii) of this theorem is said to be a functor of degree less than k . This means that every functor T , with $T(0) = 0$, has a well defined degree, which is a non-negative integer or infinity. A functor of degree 0 is one which is identically zero. Similarly, a non-zero functor of degree 1 is characterized according to (ii) by the property that $T(f_1 + f_2) = T(f_1) + T(f_2)$ for every pair of mappings

$$f_1, f_2: A \rightarrow B.$$

According to (i) and Theorem 9.1 a non-zero functor of degree 1 is also characterized by the property that

$$T(\kappa_1) + T(\kappa_2): T(A_1) + T(A_2) \rightarrow T(A_1 + A_2)$$

is an isomorphism, where each $\kappa_i: A_i \rightarrow A_1 + A_2$ is the natural injection. Finally, by the corollary above, such a functor is also characterized by the property that $T(A | A) = 0$.

A covariant functor $T(A_1, \cdots, A_k)$ of k variables, zero when one argument is zero, clearly has a finite or infinite degree with respect to each variable separately.

10. Comparison of functors

The cross-effect functors may be used to determine effectively when a natural homomorphism of functors is an isomorphism. In stating the method we use the following terminology. A *monomorphism* $f: A \rightarrow B$ is a homomorphism with

kernel zero; an *epimorphism* $f:A \rightarrow B$ is a homomorphism with $f(A) = B$. Thus “epimorphism” means “homomorphism onto”, while “isomorphism” is reserved for its proper meaning, “isomorphism onto.”

THEOREM 10.1. *Let*

$$(10.1) \quad \theta(A):S(A) \rightarrow T(A)$$

be a natural homomorphism between the covariant functors S and T , each defined for abelian groups and assumed to commute with the formation of direct limits. Then $\theta(A)$ is a mono-, epi- or isomorphism for all abelian groups A if and only if the induced maps on the cross-effect functors

$$(10.2) \quad \theta(Z_1, \dots, Z_k):S(Z_1 \mid \dots \mid Z_k) \rightarrow T(Z_1 \mid \dots \mid Z_k)$$

are mono-, epi-, or isomorphisms, respectively, for all positive integers k and for all cyclic groups $Z_1, \dots, Z_k$.

The same conclusion holds if “cyclic group” is replaced by “cyclic group of infinite or prime power order.” A corresponding theorem holds for functors of two or more arguments.

PROOF. The “only if” is clear. Conversely, assume that all the cross-effect mappings (10.2) have one of the stated properties (mono, epi, or iso). The representation of $S(Z_1 + \dots + Z_k)$ as a direct sum of its cross-effects of orders $1, \dots, k$ will then yield the same property for $\theta(Z_1 + \dots + Z_k)$; that is, for $\theta(A)$ when A is any finitely generated group. Since an arbitrary abelian group may be represented as the direct limit of its finitely generated subgroups, the desired property of $\theta(A)$ in general then follows by standard arguments from the assumption that S and T commute with the formation of direct limits.

For functors of finite degree a refinement of this theorem is possible, depending on the following lemma:

LEMMA 10.2. *Let $T(A, B)$ be a covariant functor of two abelian groups, zero when either argument is zero. If r and s are relatively prime integers, while A and B are abelian groups with $rA = 0$, $sB = 0$, then $T(A \mid A, B) = 0$ and $T(A, B \mid B) = 0$ imply $T(A, B) = 0$.*

PROOF. Let I_A, I_B be the respective identity maps of A and B , let A', B' be any abelian groups, and let $f_1, f_2:A \rightarrow A'$ and $g_1, g_2:B \rightarrow B'$ be any homomorphisms. The hypotheses then imply, by Lemma 9.10, that

$$(10.3) \quad T(f_1 \mp f_2, I_B) = 0, \quad T(I_A, g_1 \mp g_2) = 0.$$

Now choose integers p and q with $pr + qs = 1$. Then $qsI_A = I_A$ and $qsI_B = 0$; hence, by the additivity rule (10.3)

$$\begin{aligned} T(I_A, I_B) &= T(qsI_A, I_B) = qsT(I_A, I_B) \\ &= T(I_A, qsI_B) = T(I_A, 0) = 0; \end{aligned}$$

it follows that $T(A, B) = 0$.

THEOREM 10.3. *Let $T(A, B)$ be a covariant functor of two abelian groups which*

is of finite degree in each of its arguments. If r and s are relatively prime integers, while A and B are abelian groups with $rA = 0$, $sB = 0$, then $T(A, B) = 0$.

The proof is by induction on the total degree; that is, the sum of the degrees of T in its two arguments. For total degree 2 or less, the conclusion follows from the lemma. For T of total degree greater than 2, introduce the functors:

$$R(A_1, A_2, B) = T(A_1 | A_2, B), \quad S(A, B_1, B_2) = T(A, B_1 | B_2).$$

Now R , regarded as a functor of A_2 and B , is of lower total degree than T ; hence, by the induction assumption, $R(A_1, A, B) = 0$ for any A_1 , so that in particular $T(A | A, B) = 0$. A similar argument with S yields $T(A, B | B) = 0$. The conclusion $T(A, B) = 0$ now follows from the lemma.

For any prime number p , let a p -cyclic group be a cyclic group Z of order infinity or some power of p . Then Theorem 10.1 may be refined as follows:

THEOREM 10.4. *Let $\theta(A)$ be a natural homomorphism, as in (10.1), between functors S and T of finite degree commuting with direct limits. Then $\theta(A)$ is a mono-, epi-, or isomorphism for all abelian groups A if and only if for every integer k and every prime p the induced maps $\theta(Z_1, \dots, Z_k)$ of (10.2) are mono-, epi-, or isomorphisms, respectively, for all choices of p -cyclic groups $Z_1, \dots, Z_k$.*

PROOF. As observed below Theorem 10.1, it suffices to consider $\theta(Z_1, \dots, Z_k)$ for the case when the Z_i are all of prime power order, or infinity. By Theorem 10.3, the case when two of the Z_i are cyclic of different prime power order is trivial; hence the conclusion.

11. The torsion product

The torsion product of two abelian groups is used in the treatment of the Künneth relations. The usual description of the torsion product (cf. [10] or [6], p. 161) is more conveniently replaced for our purposes by a direct definition of the torsion product in terms of generators and relations.

For abelian groups A and B we take $\text{Tor}(A, B)$ to be the abelian group whose generators are the symbols $\tau_h(a, b)$ for all positive integers h and all pairs of elements $a \in A, b \in B$ such that $ha = 0$ and $hb = 0$. These generators are subject to the relations

$$(11.1) \quad \tau_h(a, b_1 + b_2) = \tau_h(a, b_1) + \tau_h(a, b_2), \quad ha = 0 = hb_1 = hb_2,$$

$$(11.1') \quad \tau_h(a_1 + a_2, b) = \tau_h(a_1, b) + \tau_h(a_2, b), \quad ha_1 = ha_2 = 0 = hb,$$

$$(11.2) \quad \tau_{hk}(a, b) = \tau_h(ka, b), \quad hka = 0 = hb,$$

$$(11.2') \quad \tau_{hk}(a, b) = \tau_h(a, kb), \quad ha = 0 = hkb.$$

Note that (11.2) and (11.2') hold whenever the right hand sides are defined. From (11.1) and (11.1') it follows that $\tau_h(a, 0) = 0$, $\tau_h(0, b) = 0$ for $ha = 0 = hb$. Hence also

$$\text{Tor}(A, 0) = 0, \quad \text{Tor}(0, B) = 0.$$

Furthermore $\text{Tor } (A, B) = 0$ if one of A or B is a torsion-free group. For any abelian groups A and B there is a natural isomorphism

$$\text{Tor } (A, B) \cong \text{Tor } (B, A)$$

given by mapping $\tau_h(a, b)$ into $\tau_h(b, a)$.

The cross-effect functor of Tor is zero, as a consequence of

THEOREM 11.1. *For abelian groups A, B , and C the natural homomorphisms*

$$\text{Tor } (A + B, C) \xrightleftharpoons[g]{f} \text{Tor } (A, C) + \text{Tor } (B, C)$$

defined by

$$\begin{aligned} f\tau_h(a + b, c) &= \tau_h(a, c) + \tau_h(b, c) & ha = hb = 0 = hc, \\ g\tau_h(a, c) &= \tau_h(a, c), & g\tau_h(b, c) &= \tau_h(b, c) \end{aligned}$$

are isomorphisms and are inverses of each other.

PROOF. It is clear that f and g preserve the defining relations (11.1)–(11.2'), hence are homomorphisms, and both composites fg and gf are clearly identities.

For cyclic groups we will systematically adopt the notation

$$Z_h(c) = \text{cyclic group with generator } c \text{ and order } h;$$

in particular, $Z_\infty(c)$ will denote the infinite cyclic group generated by c , while $Z_1(c) = 0$.

The computation of Tor for cyclic groups is given by

THEOREM 11.2. *For cyclic groups of finite order*

$$\text{Tor } (Z_r(c), Z_s(d)) = Z_{(r,s)}(\tau_{[r,s]}(c, d)),$$

while if one or both of r or s is ∞ ,

$$\text{Tor } (Z_r(c), Z_s(d)) = 0.$$

PROOF. Observe that each generator of $\text{Tor } (Z_r(c), Z_s(d))$ has the form $\tau_h(ic, jd)$, where $hi \equiv 0 \pmod{r}$ and $hj \equiv 0 \pmod{s}$ and hence $hij \equiv 0$ modulo the least common multiple $[r, s]$. We define mappings

$$\text{Tor } (Z_r(c), Z_s(d)) \xrightleftharpoons[g]{f} Z_{(r,s)}(b)$$

by setting

$$\begin{aligned} f\tau_h(ic, jd) &= (hij/[r, s])b, \\ gb &= \tau_{[r,s]}(c, d), \end{aligned}$$

and verifying that f and g are homomorphisms with both composites the identity.

Finally, we note that the character of the defining relations for $\text{Tor } (A, B)$ readily shows that this functor commutes with the formation of direct limits.

We next show that the torsion product, so defined, agrees with the usual definition. If the groups A, B are represented as $A = F/R, B = F'/R'$, where F and F' are free groups with the respective subgroups R and R' , then the injection maps on tensor products yield a commutative diagram

$$\begin{array}{ccc} R \otimes F' & \xrightarrow{i} & F \otimes F' \\ \uparrow & \nearrow j & \uparrow i \\ R \otimes R' & \longrightarrow & F \otimes R'. \end{array}$$

Cartan-Eilenberg [10] define $\text{Tor}_1(A, B)$ (relative to the ring $J = \mathbb{Z}$ of integers) as the factor group

$$\text{Tor}_1(A, B) = [i(R \otimes F') \cap i'(F \otimes R')]/j(R \otimes R').$$

The result is independent, up to isomorphism, of the choice of representations $A = F/R, B = F'/R'$. In particular we choose F to be the free abelian group with generators $[a]$, for all $a \in A$, and R to be the subgroup of F generated by all $[a_1] - [a + a_2] + [a_2]$, for $a_1, a_2 \in A$; the correspondence $[a] \rightarrow a$ then induces an isomorphism $F/R \cong A$.

Using this "standard" representation for F and F' , we next define a homomorphism

$$(11.3) \quad \theta: \text{Tor}(A, B) \rightarrow \text{Tor}_1(A, B)$$

by setting

$$(11.4) \quad \theta\tau_h(a, b) = h([a] \otimes [b]) + j(R_1 \otimes R_2),$$

where $ha = 0$ in A and $hb = 0$ in B . To justify this definition, observe first that $ha = 0$ implies $h[a] \equiv [ha] \equiv 0 \pmod{R}$, hence

$$h([a] \otimes [b]) = (h[a]) \otimes [b] \equiv [ha] \otimes [b] \equiv 0 \pmod{i(R \otimes F')}.$$

Therefore $\theta\tau_h(a, b)$ lies in $\text{Tor}_1(A, B)$.

To show that θ is a homomorphism, we must show that the relations (11.1)–(11.2') are preserved. As for (11.1),

$$\begin{aligned} \theta\tau_h(a, b_1 + b_2) - \theta\tau_h(a, b_1) - \theta\tau_h(a, b_2) \\ = h\{[a] \otimes [b_1 + b_2] - [a] \otimes [b_1] - [a] \otimes [b_2]\} \\ = (h[a]) \otimes ([b_1 + b_2] - [b_1] - [b_2]) \equiv 0 \pmod{j(R \otimes R')}. \end{aligned}$$

The treatment of (11.1') is analogous. As for (11.2),

$$\begin{aligned} \theta\tau_{hk}(a, b) - \theta\tau_h(ka, b) &= hk[a] \otimes [b] - [ka] \otimes h[b] \\ &= (k[a] - [ka]) \otimes h[b]. \end{aligned}$$

Now $k[a] - [ka] \in R$ and $hb = 0$ implies $h[b] \in R'$, hence the result is in $j(R \otimes R')$. Similarly, (11.2') is preserved.

THEOREM 11.3. *The homomorphism θ of (11.3) establishes a natural isomorphism between Tor and Tor_1 .*

PROOF. It is known that Tor_1 , like Tor , commutes with the formation of direct limits and is zero when one argument is zero, hence this theorem can be established by application of Theorem 10.3 on the comparison of cross-effects. By Theorem 11.1, the cross-effects of Tor are all zero; the same is known to hold for Tor_1 . Therefore it suffices to prove that θ is an isomorphism in the case when A and B are cyclic groups. Since both Tor and Tor_1 vanish when one argument is an infinite cyclic group, it suffices to consider only groups A and B of prime power order for the same prime; thus we may assume that A and B are cyclic of orders r and s , respectively, where r divides s (and both are prime powers). The fact that θ is an isomorphism in this case follows from the observation that θ maps the generators of Tor onto the generators of Tor_1 as given by the following lemma.

LEMMA 11.4. *If A and B are cyclic groups with generators a_0 and b_0 of order r and s , where r divides s , then $\text{Tor}_1(A, B)$ is cyclic of order r , with generator the coset of $s[a_0] \otimes [b_0]$.*

PROOF. Take the standard free group F for A , with free generators $[ka_0]$, $k = 0, \dots, r-1$. Consider the set G of $r-1$ elements

$$G = \{[0], [2a_0] - 2[a_0], \dots, [(r-1)a_0] - (r-1)[a_0]\}.$$

Observe first that

$$F \text{ has free generators } [a_0] \text{ and } g, \quad g \in G;$$

$$R \text{ has free generators } r[a_0] \text{ and } g, \quad g \in G.$$

Indeed, the statement as to F is immediate. All the cited generators of R lie in R , while modulo these generators we have $[ka_0] = [(k+h)a_0] \equiv k[a_0]$ for any k , and hence

$$[ka_0] - [(k+l)a_0] + [la_0] \equiv 0$$

for any given element of R . We use the corresponding sets of generators for F' and R' .

Now we have at once the following sets of free generators:

$$\begin{array}{llll} i(R \otimes F'): & r[a_0] \otimes [b_0], & r[a_0] \otimes g', & g \otimes [b_0], \quad g \otimes g'; \\ i'(F \otimes R'): & s[a_0] \otimes [b_0], & [a_0] \otimes g', & sg \otimes [b_0], \quad g \otimes g'; \\ i(R \otimes F') \cap i'(F \otimes R'): & s[a_0] \otimes [b_0], & r[a_0] \otimes g', & sg \otimes [b_0], \quad g \otimes g'; \\ j(R \otimes R'): & rs[a_0] \otimes [b_0], & r[a_0] \otimes g', & sg \otimes [b_0], \quad g \otimes g', \end{array}$$

where g and g' range over G and G' . Comparison of the last two lines gives the desired conclusion.

The argument involved in this lemma is essentially just a special case of the fact that the definition of Tor_1 is independent of the choice of the representation F/R .

12. Cross caps and Künneth relations

Let K_1 and K_2 be two free chain complexes, and let $H(K_i)$ denote the (graded) homology group of K_i , $i = 1, 2$. The analysis of the homology of $K_1 \otimes K_2$ starts with the homomorphism

$$(12.1) \quad \alpha: H(K_1) \otimes H(K_2) \rightarrow H(K_1 \otimes K_2),$$

with $\alpha(\mathbf{z}_1 \otimes \mathbf{z}_2)$ defined as the homology class of $z_1 \otimes z_2$, where z_1, z_2 are cycles in the classes $\mathbf{z}_1, \mathbf{z}_2$.

If the complexes K_1 and K_2 are free, then the following three facts can be established and are known as the *Künneth relations*:

- (i) The map α is a monomorphism. We regard this map as an identification and treat $H(K_1) \otimes H(K_2)$ as a subgroup of $H(K_1 \otimes K_2)$.
- (ii) $H(K_1) \otimes H(K_2)$ is a direct summand of $H(K_1 \otimes K_2)$.
- (iii) The factor group $H(K_1 \otimes K_2)/H(K_1) \otimes H(K_2)$ is isomorphic with $\text{Tor}(H(K_1), H(K_2))$. This isomorphism lowers the degree by one.

We shall first give an explicit form of the isomorphism involved in (iii) using the description of Tor from the preceding section.

Consider elements $\mathbf{z}_1 \in H_{p_1}(K_1)$, $\mathbf{z}_2 \in H_{p_2}(K_2)$ such that $hz_1 = 0 = hz_2$ for some integer h . Let z_1, z_2 be cycles in the classes $\mathbf{z}_1, \mathbf{z}_2$, respectively, and let

$$b_1 \in K_{1,p_1+1}, \quad b_2 \in K_{2,p_2+1}, \quad \partial b_1 = hz_1, \quad \partial b_2 = hz_2$$

be appropriate "caps". We consider the cross-cap

$$(12.2) \quad \chi_h(z_1, z_2) = z_1 \otimes b_2 + \wp(p_1 + 1)b_1 \otimes z_2 = (1/h)\partial(b_1 \otimes b_2).$$

This clearly is a cycle of dimension $q = p_1 + p_2 + 1$ in $K_1 \otimes K_2$. If b_2 is replaced by another cap b'_2 , then $b_2 - b'_2$ is a $(p_2 + 1)$ -cycle of K_2 and

$$\chi_h(z_1, z_2) - \chi'_h(z_1, z_2) = z_1 \otimes (b_2 - b'_2).$$

Similarly, if b_1 is replaced by another cap, we find that $\chi_h(z_1, z_2)$ is modified by $\wp(p_1 + 1)(b_1 - b'_1) \otimes z_2$. Consequently the homology class of $\chi_h(z_1, z_2)$ determines a unique element of the group

$$(12.3) \quad H_q(K_1 \otimes K_2)/[H_{p_1}(K_1) \otimes H_{p_2+1}(K_2) + H_{p_1+1}(K_1) \otimes H_{p_2}(K_2)],$$

where $q = p_1 + p_2 + 1$. Henceforth we shall regard $\chi_h(z_1, z_2)$ as an element of the group (12.3).

It is clear that $\chi_h(z_1, z_2)$ is additive in z_1 and z_2 . Indeed, if z'_1 is another cycle with $\partial b'_1 = hz'_1$, then taking $b_1 + b'_1$ as a cap for $z_1 + z'_1$ we find $\chi_h(z_1 + z'_1, z_2) = \chi_h(z_1, z_2) + \chi_h(z'_1, z_2)$. Further, if $z_1 = \partial c$ is a bounding cycle, then choosing the cap $b_1 = hc$ we find $\chi_h(z_1, z_2) = (1/h)\partial(hc \otimes b_2) = \partial(c \otimes b_2)$, so that

$\chi_h(z_1, z_2)$ yields the zero element of (12.3). Therefore, we may regard $\chi_h(z_1, z_2)$ as a function of the homology classes z_1, z_2 and write $\chi_h(z_1, z_2)$.

Given integers h, k and homology classes z_1, z_2 with $hkz_1 = 0, hz_2 = 0$, we choose cycles z_1, z_2 and caps b_1, b_2 with $\partial b_1 = hkz_1, \partial b_2 = hz_2$. Then

$$\chi_{hk}(z_1, z_2) = (1/hk)\partial(b_1 \otimes kb_2) = (1/h)\partial(b_1 \otimes b_2) = \chi_h(kz_1, z_2).$$

Thus

$$(12.4) \quad \chi_{hk}(z_1, z_2) = \chi_h(kz_1, z_2) \quad \text{if} \quad hkz_1 = 0 = hz_2.$$

Similarly we find

$$(12.4') \quad \chi_{hk}(z_1, z_2) = \chi_h(z_1, kz_2) \quad \text{if} \quad hz_1 = 0 = hkz_2.$$

Relations (12.4) and (12.4') are replicas of the defining relations (11.2) and (11.2') of the torsion product. Thus, summarizing we obtain

THEOREM 12.1. *The torsion product $\text{Tor}(H_{p_1}(K_1), H_{p_2}(K_2))$ is mapped homomorphically into the group (12.3) using the map χ which to each generator $\tau_h(z_1, z_2)$ assigns the cross-cap $\chi_h(z_1, z_2)$.*

We can give a rapid proof of the Künneth relations (i)–(iii) assuming that K_1 and K_2 are free. Since a subgroup of a free group is free, it follows that the boundaries $B(K_1)$ form a free group. Thus from the exact sequence $0 \rightarrow Z(K_1) \rightarrow K_1 \rightarrow B(K_1) \rightarrow 0$ we deduce the existence of a projection $K_1 \rightarrow Z(K_1)$. Composing this with the natural map $Z(K_1) \rightarrow H(K_1)$ we find a map $\lambda_1: K_1 \rightarrow H(K_1)$ which induces the identity map $\lambda_1^*: H(K_1) \rightarrow H(K_1)$. A similar map $\lambda_2: K_2 \rightarrow H(K_2)$ also is obtained. It then follows from the naturality of the map α that the composition

$$H(K_1) \otimes H(K_2) \xrightarrow{\alpha} H(K_1 \otimes K_2) \xrightarrow{(\lambda_1 \otimes \lambda_2)^*} H(K_1) \otimes H(K_2)$$

is the identity. This yields (i) and (ii).

To establish (iii) we first replace it by a more precise statement

(iii') If K_1 and K_2 are free complexes, then the cross cap map χ yields a natural isomorphism

$$\chi: \text{Tor}(H(K_1), H(K_2)) \cong H(K_1 \otimes K_2) / (H(K_1) \otimes H(K_2)).$$

Since the functors involved commute with direct limits we may assume that K_1 and K_2 are finitely generated. A finitely generated free complex is known to be isomorphic to a direct sum of elementary complexes. Here an elementary complex E is either (a) a complex in which the chain group on one dimension is infinite cyclic and the others are all zero or, (b) a complex in which two successive chain groups are infinite cyclic and connected by a non-zero boundary homomorphism, while the remaining chain groups are all zero. Since χ is natural and the functors involved in (iii') are additive (i.e., of degree 1), it suffices to prove (iii') for elementary complexes. This verification is left to the reader.

We now consider a skew-commutative graded ∂ -ring G . The multiplication $\circ$ in the ring G yields a map

$$\pi: G \otimes G \rightarrow G, \quad \pi(a \otimes b) = a \circ b,$$

which is a homomorphism of graded ∂ -rings. The composite map

$$H(G) \otimes H(G) \xrightarrow{\alpha} H(G \otimes G) \xrightarrow{\pi^*} H(G)$$

converts $H(G)$ into a skew-commutative graded ∂ -ring. Composing the map χ with π^* yields a map

$$(12.5) \quad \chi_G: \text{Tor}(H_{p_1}(G), H_{p_2}(G)) \rightarrow H_q(G)/F, \quad q = p_1 + p_2 + 1,$$

where

$$(12.6) \quad F = H_{p_1}(G) \circ H_{p_2+1}(G) + H_{p_1+1}(G) \circ H_{p_2}(G).$$

In most applications of the map χ_G , the subgroup F of $H_q(G)$ will be zero. The image of $\tau_h(z_1, z_2)$ under χ_G will be written as $\chi_{G,h}(z_1, z_2)$, with G omitted whenever possible without ambiguity. If z_1, z_2 are cycles in G and $\partial b_1 = h z_1$, $\partial b_2 = h z_2$, then $\chi_{G,h}(z_1, z_2)$ is the class determined by the cycle $(1/h)\partial(b_1 \circ b_2)$. From the skew-commutativity of G we deduce readily

$$(12.7) \quad \chi_h(z_1, z_2) = \wp((p_1 + 1)(p_2 + 1))\chi_h(z_2, z_1).$$

We also have the rules

$$(12.8) \quad \begin{aligned} \chi_h(z, z) &= 0 & p_1 = p_2 &\equiv 0 \pmod{2} \\ \chi_h(z, z) &\equiv 0 \pmod{2} & p_1 = p_2 &\equiv 1 \pmod{2}. \end{aligned}$$

Indeed, if $\partial b = h z$ with z in G , then

$$\begin{aligned} (1/h)\partial(b \circ b) &= z \circ b + \wp(p_1 + 1)b \circ z \\ &= [1 + \wp(p_1 + 1 + p_1(p_1 + 1))]z \circ b, \end{aligned}$$

which implies (12.8).

In subsequent considerations it is important to know when suspensions of cycles are zero in the bar construction (I.7.9). To consider this, let G be any augmented skew-commutative graded ∂ -ring. The augmentation yields a chain transformation

$$\nu: G \otimes G \rightarrow G, \quad \nu(a \otimes b) = \alpha(a)b + \alpha(b)a.$$

THEOREM 12.2. *There is a chain homotopy*

$$D: S\nu \simeq S\pi: G \otimes G \rightarrow B(G),$$

where $S: G \rightarrow B(G)$ is the suspension for the bar construction $B(G)$.

PROOF. For homogeneous elements a, b of G , with $a \in G_p$ we set $D(a \otimes b) = \wp(p)[a \mid b]$. Straightforward applications of the boundary formulas in $B(G)$ and

$G \otimes G$ yield

$$(\partial D - D\partial)(a \otimes b) = (S\nu - S\pi)(a \otimes b),$$

as desired; the minus sign appears here in $\partial D - D\partial$ because $S\nu$ and $S\pi$ are chain transformations raising the dimensions by 1.

We use several corollaries. Since $\nu(a \otimes b)$ is zero when both a and b have positive dimension, we get

COROLLARY 12.3. *If z_1 and z_2 are cycles of positive dimension in G , then $S(z_1 \circ z_2) \sim 0$ in $B(G)$.*

Although cross caps $\chi_h(z_1, z_2)$ in G are defined only modulo certain products, this corollary shows that their suspensions are defined uniquely as homology classes. Furthermore these cross caps are obtained from cycles in $G \otimes G$ by application of π . Hence we have

COROLLARY 12.4. *If z_1 and z_2 are cycles of positive dimension with $hz_i \sim 0$ in G , then $S\chi_h(z_1, z_2) \sim 0$ in $B(G)$.*

13. Examples of functors of degree 2

Following J. H. C. Whitehead [18], we introduce for each abelian group Π the abelian group $\Gamma_4(\Pi)$ (for explanation of the subscript 4, see §19), which is generated by symbols $\gamma(x)$, one for each $x \in \Pi$, subject to the relations

$$(13.1) \quad \gamma(-x) = \gamma(x),$$

$$(13.2) \quad \gamma(x \tau y \tau z) = 0.$$

This last relation asserts that the expression $\gamma(x \tau y) = \gamma(x + y) - \gamma(x) - \gamma(y)$ is bilinear or equivalently that

$$(13.2') \quad \gamma(x + y + z) - \gamma(x + y) - \gamma(x + z) - \gamma(y + z) \\ + \gamma(x) + \gamma(y) + \gamma(z) = 0.$$

Setting $x = y = z = 0$ in (13.2') yields $\gamma(0) = 0$.

Among the consequences of the relations (13.1) and (13.2) we cite

$$(13.3) \quad \gamma(mx) = m^2\gamma(x)$$

$$(13.4) \quad \text{if } hx = 0 \quad \text{then} \quad h(h, 2)\gamma(x) = 0,$$

where $(h, 2)$ denotes the greatest common divisor of the integers h and 2. We prove (13.3) by induction, expanding $\gamma(mx + x - x)$ according to (13.2'). To verify (13.4) we note that $h^2\gamma(x) = \gamma(hx) = \gamma(0) = 0$ and that $(h + 1)^2\gamma(x) = \gamma((h + 1)x) = \gamma(x)$. This implies $h^2\gamma(x) = 2h\gamma(x) = 0$.

For cyclic groups the functor Γ_4 has the following values

$$(13.5) \quad \Gamma_4(Z_\infty(c)) = Z_\infty(\gamma(c)),$$

$$(13.6) \quad \Gamma_4(Z_h(c)) = Z_{h(h,2)}(\gamma(c)), \quad h < \infty.$$

To verify (13.6) we set up homomorphisms

$$\Gamma_4(Z_h(c)) \underset{g}{\overset{f}{\rightleftarrows}} Z_{h(h,2)}(d)$$

by setting $gd = \gamma(c)$ and $f(\gamma(ic)) = i^2d$. It readily follows that f and g are well defined and that both composites gf and fg are the identity. The proof of (13.5) is similar.

Since $\gamma(0) = 0$, it follows that $\Gamma_4(0) = 0$, so that Γ_4 is a covariant functor to which the cross-effect theory of §9 may be applied. In particular, Corollary 9.7 will give

THEOREM 13.1. *For an element $(x_1, x_2) \in \Pi_1 + \Pi_2$ the definition*

$$g_\Gamma(x_1 \otimes x_2) = \gamma((x_1, x_2)) - \gamma((x_1, 0)) - \gamma((0, x_2))$$

provides an isomorphism

$$(13.7) \quad g_\Gamma : \Pi_1 \otimes \Pi_2 \cong \Gamma_4(\Pi_1 | \Pi_2).$$

PROOF. Consider the maps

$$\Gamma_4(\Pi_1 + \Pi_2) \underset{g}{\overset{f}{\rightleftarrows}} \Gamma_4(\Pi_1) + \Gamma_4(\Pi_2) + \Pi_1 \otimes \Pi_2,$$

given for $(x_1, x_2) \in \Pi_1 + \Pi_2$ by the formulas

$$\begin{aligned} f\gamma((x_1, x_2)) &= \gamma(x_1) + \gamma(x_2) + x_1 \otimes x_2, \\ g\gamma(x_1) &= \gamma((x_1, 0)), \quad g\gamma(x_2) = \gamma((0, x_2)), \\ g(x_1 \otimes x_2) &= \gamma((x_1, 0) \tau (0, x_2)), \end{aligned}$$

the last of which agrees with g_Γ . We verify that both f and g are homomorphisms, and that both composites are the identity maps.

To conclude the discussion of Γ_4 we note the exact sequence

$$(13.8) \quad 0 \rightarrow \Pi \otimes \Pi \xrightarrow{t} \Gamma_4(\Pi) \xrightarrow{s} \Pi/2\Pi \rightarrow 0$$

with

$$t(x \otimes y) = \gamma(x \tau y), \quad s\gamma(x) = x + 2\Pi.$$

The fact that (13.8) is exact for a cyclic group Π may be seen directly. The general case is most conveniently proved using cross-effects.

Another peculiar functor Ω will be of use; it is related to the torsion product in the same way that Γ_4 is related to the tensor product. We define the group $\Omega(\Pi)$ to be the abelian group generated by the symbols $\omega_h(x)$, for positive integers

h and elements $x \in \Pi$ with $hx = 0$, subject to the relations

$$(13.9) \quad \omega_{hk}(x) = k\omega_h(x), \quad hx = 0,$$

$$(13.10) \quad k\omega_{hk}(x) = \omega_h(kx), \quad h kx = 0,$$

$$(13.11) \quad \omega_h(kx \tau y) = \omega_{hk}(x \tau y), \quad h kx = hy = 0,$$

$$(13.12) \quad \omega_h(x \tau y \tau z) = 0, \quad hx = hy = hz = 0.$$

The last relation implies $\omega_h(0) = 0$. In the presence of this condition the first three relations are equivalent to the single relation

$$(13.13) \quad \omega_{hk}(x + y) + (k - 1)\omega_{hk}(x) = \omega_h(kx + y) + (k - 1)\omega_h(y).$$

The relations (13.11) and (13.12) imply that the correspondence $\tau_h(x, y) \rightarrow \omega_h(x \tau y)$ yields a natural homomorphism

$$\text{Tor}(\Pi, \Pi) \rightarrow \Omega(\Pi).$$

Among the elementary consequences of these relations we cite

$$(13.14) \quad \omega_h(x \tau x) = 2\omega_h(x), \quad hx = 0,$$

$$(13.15) \quad \omega_h(mx \tau y) = |m| \omega_h(x \tau y), \quad hx = hy = 0,$$

$$(13.16) \quad \omega_h(mx) = m^2 \omega_h(x), \quad hx = 0.$$

The last two are proved for positive m by induction, and thence for negative m by first deriving $\omega_h(-x) = \omega_h(x)$ from (13.12). It follows that the correspondence $\gamma(x) \rightarrow \omega_h(x)$ for $hx = 0$ provides a natural homomorphism

$$\Gamma_4({}_h\Pi) \rightarrow \Omega(\Pi),$$

where ${}_h\Pi$ denotes the subgroup of all x with $hx = 0$ in Π .

Clearly $\Omega(\Pi) = 0$ for any torsion-free group Π and hence, in particular, for an infinite cyclic group. For finite cyclic groups we have

$$(13.17) \quad \Omega(Z_m(c)) = Z_m(\omega_m(c)).$$

Indeed, we may set up homomorphisms

$$\Omega(Z_m(c)) \begin{matrix} \xrightarrow{f} \\ \xleftarrow{g} \end{matrix} Z_m(d)$$

by setting $gd = \omega_m(c)$ and

$$f\omega_h(ic) = (hi^2/m) d, \quad hi \equiv 0 \pmod{m}.$$

It readily follows that f and g are well defined, and that both composites fg and gf are identities.

The cross-effects are determined by

THEOREM 13.2. *For an element (x_1, x_2) in $\Pi_1 + \Pi_2$ with $h(x_1, x_2) = 0$ the definition*

$$g_\Omega \tau_h(x_1, x_2) = \omega_h((x_1, x_2)) - \omega_h((x_1, 0)) - \omega_h((0, x_2))$$

provides an isomorphism

$$(13.18) \quad g_\Omega : \text{Tor}(\Pi_1, \Pi_2) \cong \Omega(\Pi_1 | \Pi_2).$$

PROOF. Consider the maps

$$\Omega(\Pi_1 + \Pi_2) \xrightleftharpoons[g]{f} \Omega(\Pi_1) + \Omega(\Pi_2) + \text{Tor}(\Pi_1, \Pi_2)$$

defined by the formulas

$$\begin{aligned} f\omega_h((x_1, x_2)) &= \omega_h(x_1) + \omega_h(x_2) + \tau_h(x_1, x_2), \\ g\omega_h(x_1) &= \omega_h((x_1, 0)), \quad g\omega_h(x_2) = \omega_h((0, x_2)), \\ g\tau_h(x_1, x_2) &= \omega_h((x_1, 0) \tau (0, x_2)), \end{aligned}$$

the last of which agrees with g_Ω . We verify that both f and g are homomorphisms, and that both composites fg and gf are identities. By Corollary 9.7 this gives the conclusion.

CHAPTER III. COMPUTABILITY OF $H(\Pi, n)$

14. The contraction for infinite cyclic groups

For the infinite cyclic group Z_∞ it is well known that the homology groups of $A(Z_\infty, 1)$ are infinite cyclic in dimension 1 and zero in all higher dimensions; indeed, the computations leading to this result provide a contraction of $A(Z_\infty, 1)$ to a simpler graded ∂ -ring $M(\infty, 1)$. Explicitly, take $M(\infty, 1)$ to be the exterior algebra $E(u)$ generated by the identity 1 and one element u of dimension 1, with product $u^2 = 1$, boundary $\partial u = 0$, and augmentation $\alpha(1) = 1$, $\alpha(u) = 0$. Take Z_∞ to be the additive group of integers, and define homomorphisms

$$A(Z_\infty, 1) \xrightleftharpoons[g]{f} M(\infty, 1)$$

by the formulas

$$(14.1) \quad f[i] = iu, \quad fc_q = 0, \quad q > 1, i \in Z_\infty, c_q \in A_q;$$

$$(14.2) \quad gu = [1].$$

THEOREM 14.1. *For the infinite cyclic group Z_∞ , the maps f and g defined above are graded ∂ -ring homomorphisms, and induce the projection and injection of a contraction*

$$(14.3) \quad (f, g, \Phi) : A_N(Z_\infty, 1) \rightarrow M(\infty, 1).$$

PROOF. The requisite properties of f and g are immediate. The known calculation of the homology groups of $A(Z_\infty, 1)$, as made in [4, §7], provides a homotopy Φ with $\partial\Phi + \Phi\partial = gf - I$, given for all integers $x, x_2, \dots, x_q$ by the recursive formula

$$(14.4) \quad \Phi[x + 1, x_2, \dots, x_q] = [1, x, x_2, \dots, x_q] + \Phi[x, x_2, \dots, x_q].$$

(Here and subsequently we use the usual notation $[x_1, \dots, x_q]$ rather than the bar notation $[x_1 | \dots | x_q]$ to describe cells of $A(\Pi, 1)$.) The annihilation properties $f\Phi = 0$ and $\Phi g = 0$ for this homotopy are readily established.

For higher levels n we introduce the graded ∂ -rings

$$M(\infty, n) = B_N^{n-1}M(\infty, 1).$$

By Theorem I.12.1_N we have for each n a contraction

$$(f_n, g_n, \Phi_n): A_N(Z_\infty, n) \rightarrow M(\infty, n)$$

with $f_n = B_N^{n-1}(f)$, $g_n = B_N^{n-1}(g)$. The last bar used in the construction of $M(\infty, n)$ will be denoted by $|_n$, just as in the case of $A_N(Z_\infty, n)$.

The explicit description of $M(\infty, 2) = B_N M(\infty, 1) = B_N E(u)$ is as follows. Its cells may be denoted by the integers p , as

$$(14.5) \quad (p) = [u |_2 \cdots |_2 u], \quad p \text{ factors } u,$$

with

$$(14.6) \quad \dim(p) = 2p, \quad \partial(p) = 0.$$

The product $(p) \star_2 (q)$ will be the sum of all shuffles of p terms u through q terms u ; since $d_B u = 2$, all terms of the shuffle are positive, and the number of terms is the number of combinations of $p + q$ things p at a time; hence

$$(14.7) \quad (p) \star_2 (q) = \binom{p+q}{p} (p+q).$$

In the graded ∂ -ring $M(\infty, 3)$ the cells are

$$(14.8) \quad \sigma = [(p_1) |_3 \cdots |_3 (p_k)], \quad d_B \sigma = \sum_{i=1}^k (2p_i + 1),$$

for all k and all sets of k positive integers $p_1, \dots, p_k$. The boundary is

$$(14.9) \quad \partial\sigma = \sum_{i=1}^{k-1} \wp(i) \binom{p_i + p_{i+1}}{p_i} [(p_1) |_3 \cdots |_3 (p_i + p_{i+1}) |_3 \cdots |_3 (p_k)].$$

If $\tau = [(q_1) |_3 \cdots |_3 (q_m)]$ is another cell, and if the (k, m) shuffles are represented as permutations π of the set $(r_1, \dots, r_{k+m}) = (p_1, \dots, p_k, q_1, \dots, q_m)$, the product in $M(\infty, 3)$ is given as in (I, 7.10) by the formula

$$(14.10) \quad \sigma \star_3 \tau = \sum_{\pi} \wp(\varepsilon(\pi)) [(r_{\pi(1)}) |_3 \cdots |_3 (r_{\pi(k+m)})],$$

where the sum is taken over all (k, m) shuffles and $\varepsilon(\pi)$ is the (standard) sign of the permutation π .

15. The contraction for finite cyclic groups

For a finite cyclic group Z_h of order h the homology groups of $A(Z_h, 1)$ are known [11, §11]; they are cyclic of order h in odd dimensions and zero in all positive even dimensions. The computations leading to this result may also be used to find the product in the homology ring of $A(Z_h, 1)$, and then to provide a contraction of $A_N(Z_h, 1)$ to a suitably simplified graded ∂ -ring $M(h, 1)$. By iteration, we then have contractions of $A_n(Z_h, n)$ to $M(h, n) = B_N^{n-1}M(h, 1)$.

As an additive group, $M(h, 1)$ is defined to be the free abelian group with generators $1, v_k$, and w_k , with dimensions and boundary given by

$$(15.1) \quad \begin{aligned} dv_k &= 2k, & \partial v_k &= hw_{k-1}, & k &= 1, 2, \dots; \\ dw_k &= 2k + 1, & \partial w_k &= 0, & k &= 0, 1, \dots \end{aligned}$$

The product in the ring $M(h, 1)$ is determined, using binomial coefficients, by the formulas

$$(15.2) \quad \begin{aligned} v_k v_m &= \binom{k+m}{k} v_{k+m}, & w_k w_m &= 0, \\ v_k w_m &= \binom{k+m}{k} w_{k+m} = w_m v_k. \end{aligned}$$

As a ring, $M(h, 1)$ is generated by w_0 and the v_k , with $w_k = w_0 v_k$. The subring generated by w_0 is the exterior algebra $E(w_0)$ with one generator w_0 of dimension 1. The subring $V(h, 1)$ generated by the v_k is a known ring; its tensor product with the rationals is the polynomial ring in an indeterminate t of dimension 2, with $v_k = t^k/k!$. As a graded ring $M(h, 1)$ is the tensor product $E(w_0) \otimes V(h, 1)$. It is readily verified that the boundary operator introduced in (15.1) and (15.2) makes $M(h, 1)$ a graded ∂ -ring.

Take Z_h to be the additive group of integers, modulo h . We introduce the cocycle $s^2 \in Z^2(Z_h, 1; J)$ defined, for $0 \leq i, j < h$, by

$$(15.3) \quad \begin{aligned} s^2(i, j) &= 0, & i + j &< h, \\ s^2(i, j) &= 1, & i + j &\geq h. \end{aligned}$$

Incidentally, it is known that the cohomology group $H^2(Z_h, 1; J)$ is cyclic of order h with s^2 determining a generator. We also introduce a one-dimensional cochain $s^1 \in C^1(Z_h, 1; J)$ by

$$(15.4) \quad s^1(i) = i, \quad 0 \leq i < h.$$

It is readily verified that

$$(15.5) \quad \delta s^1 = h s^2.$$

A projection $f: A(Z_h, 1) \rightarrow M(h, 1)$ may be defined by

$$(15.6) \quad \begin{aligned} f[x_1, y_1, \dots, x_m, y_m] &= \{\prod_{i=1}^m s^2(x_i, y_i)\} v_m, \\ f[x_1, y_1, \dots, x_m, y_m, z] &= \{s^1(z) \prod_{i=1}^m s^2(x_i, y_i)\} w_m, \end{aligned}$$

for arguments $x_i, y_i, z \in Z_h$. An injection $g: M(h, 1) \rightarrow A(Z_h, 1)$ may be defined, using the generator 1 of Z_h , by $gw_0 = [1]$ and

$$(15.7) \quad \begin{aligned} gw_k &= \sum_{x_1, \dots, x_k} [1, x_1, \dots, 1, x_k], & x_i \in Z_h \\ gw_k &= \sum_{x_1, \dots, x_k} [1, x_1, \dots, 1, x_k, 1] & x_i \in Z_h, \end{aligned}$$

where each x_i is summed over the group Z_h .

THEOREM 15.1. *If Z_h is a cyclic group of order h , then f and g , as defined above, are respectively the projection and the injection of a contraction*

$$(15.8) \quad (f, g, \Phi): A_N(Z_h, 1) \rightarrow M(h, 1).$$

PROOF. We first establish properties of the injection g , which by (15.7) may also be defined by the recursive formulas

$$(15.9) \quad gw_0 = [1], \quad gv_{k+1} = \sum_x [gw_k, x], \quad gw_k = [gv_k, 1].$$

As an auxiliary step, we first show by induction on the dimension that

$$(15.10) \quad (gw_k)*[1] = 0, \quad (gv_k)*[1] = [gv_k, 1],$$

where $*$ = $*_1$ is the product of the ring $A(Z_h, 1)$. The proofs for w_0 and v_1 are immediate, while more generally

$$(gw_k)*[1] = [gv_k, 1]*[1] = [gv_k, 1, 1] - [(gv_k)*1, 1] = 0,$$

by the induction assumption on v_k , and

$$\begin{aligned} (gv_{k+1})*[1] &= \sum_x [gw_k, x]*[1] \\ &= \sum_x [gw_k, x, 1] - \sum_x [(gw_k)*1, x] = [gv_{k+1}, 1]. \end{aligned}$$

Next we show by induction that $\partial g = g\partial$. For argument w_0 , this is immediate. For argument v_1 of dimension 2,

$$\begin{aligned} \partial gv_1 &= \partial \sum_x [1, x] = \sum_x [x] - \sum_x [1 + x] + \sum_x [1] \\ &= \sum_x [1] = hgw_0 = g\partial v_1. \end{aligned}$$

For argument w_k , (15.10) and induction give

$$\partial gw_k = \partial((gv_k)*[1]) = \partial gv_k*[1] = hgw_{k-1}*[1] = 0.$$

For argument v_{k+1} , write $gv_{k+1} = \sum_x [gv_k, 1, x]$ and use the formula

$$(15.11) \quad \partial[c, x, y] = [\partial(c, x), y] + [c, y] - [c, x + y] + [c, x],$$

valid for any even-dimensional chain c of $A(Z_h, 1)$. Then

$$\partial gv_{k+1} = \sum_x \{[\partial gw_k, x] + [gv_k, x] - [gv_k, 1 + x] + [gv_k, 1]\}.$$

The first term is zero, by induction; the second and third summations cancel, and the final term yields $hgw_k = g\partial v_{k+1}$, as desired.

Finally we show that $g(ab) = ga*gh$ by induction on $da + db$. If b , or a , is

w_0 , this result is that already found in (15.10). For $a = v_k$, $b = w_m$, or vice versa, (15.10) and induction give

$$gv_k * gw_m = gv_k * gv_m * [1] = \binom{k+m}{k} gv_{k+m} * [1] = \binom{k+m}{k} gw_{k+m}.$$

For $a = v_{k+1}$, $b = v_{m+1}$, the $*$ -product is

$$\begin{aligned} gv_{k+1} * gv_{m+1} &= (\sum_x [gw_k, x] * (\sum_y [gw_m, y]) \\ &= \sum_y [gv_{k+1} * gw_m, y] + \wp(2m) \sum_x [gw_k * gv_{m+1}, x]. \end{aligned}$$

Application of the induction assumption to each term yields the result

$$\left\{ \binom{k+m+1}{k+1} + \binom{k+m+1}{k} \right\} gv_{k+m+2} = \binom{k+m+2}{k+1} gv_{k+m+2},$$

as desired. For $a = w_k$, $b = w_m$, we have by (15.10)

$$gw_k * gw_m = gv_k * [1] * gv_m * [1] = 0,$$

since $[1] * [1] = [1, 1] - [1, 1] = 0$.

Secondly, we establish properties of the projection f . If we assign to each cell σ of $A(Z_h, 1)$ an integer $s(\sigma)$ according to the recursive formulas

$$s[x] = s^1(x), \quad s[x, y] = s^2(x, y), \quad s[x, y, \sigma] = s^2(x, y)s(\sigma),$$

the definition of f may be rewritten as

$$\begin{aligned} f\sigma &= s(\sigma)v_m, & d\sigma &= 2m, & m &= 1, 2, \dots, \\ f\sigma &= s(\sigma)w_m, & d\sigma &= 2m+1, & m &= 0, 1, \dots. \end{aligned}$$

The required fact that $\partial f = f\partial$ now reduces to the assertions

$$s(\partial\sigma) = 0, \quad d\sigma \equiv 1 \pmod{2}; \quad s(\partial\sigma) = hs(\sigma), \quad d\sigma \equiv 0 \pmod{2}.$$

For σ of dimensions 2 and 3, these follow from (15.5). For σ of higher dimension, write $\sigma = [x, y, z, \tau]$; then

$$(15.12) \quad \partial[x, y, z, \tau] = [\partial(x, y, z), \tau] + [x, y, \partial(z, \tau)]$$

and, using $\delta s^2 = 0$ again,

$$s(\partial\sigma) = s(\partial[x, y, z])s(\tau) + s[x, y]s(\partial[z, \tau]) = s[x, y]s(\partial[z, \tau]),$$

which yields the result by induction.

To show that $f(\sigma * \tau) = (f\sigma)(f\tau)$ for cells $\sigma = [x_1, \dots, x_p]$, $\tau = [y_1, \dots, y_q]$, consider any term

$$T = \pm [t_1, \dots, t_{p+q}]$$

in the shuffle product $\sigma * \tau$. Call T *mixed* if there is an index r such that t_{2r-1} is an x and t_{2r} is a y , or vice versa. Choose the first such index r for each mixed T , and let T' be the term obtained from T by interchanging t_{2r-1} and t_{2r} . Since

$s^2(x, y)$ is symmetric, $f(T) = f(T')$. Since T and T' have opposite signs, the results cancel, and $f(\sigma \star \tau) = \sum \pm f(T)$ with summation taken only over the unmixed terms. In these unmixed terms, the sign due to the shuffle is always plus. If p and q are both odd, there are no unmixed terms, and $f(\sigma \star \tau) = 0$, in agreement with the fact that $w_k w_m = 0$. If p and q are both even, with $p = 2k$, $q = 2m$, the unmixed terms T are had by taking all shuffles of the k pairs $(x_1, x_2), \dots, (x_{2k-1}, x_{2k})$ through the m pairs $(y_1, y_2), \dots, (y_{2m-1}, y_{2m})$. For each such shuffle

$$\begin{aligned} f(T) &= s(x_1, x_2) \cdots s(x_{2k-1}, x_{2k}) s(y_1, y_2) \cdots s(y_{2m-1}, y_{2m}) v_{k+m} \\ &= s(\sigma) s(\tau) v_{k+m}, \end{aligned}$$

and the number of such shuffles is $\binom{k+m}{k}$; hence

$$f(\sigma \star \tau) = \binom{k+m}{k} s(\sigma) s(\tau) v_{k+m} = f(\sigma) f(\tau),$$

as desired. For $p = 2k$, $q = 2m + 1$, the unmixed terms T are as above with the last argument y_{2m+1} always at the end; the remaining case $p = 2k + 1$, $q = 2m$ is treated similarly.

Third, consider the composites of f and g . Clearly

$$fgw_0 = f[1] = w_0 \quad \text{and} \quad fgv_1 = f(\sum_x [1, x]) = s^2(\sum_x [1, x]) v_1 = v_1;$$

similar arguments show easily that fg is the identity. To compute gf introduce the two-dimensional chain

$$(15.13) \quad a = \sum_x [1, x], \quad x \in Z_h.$$

Then $s^2 a = 1$, and g is given by $gw_0 = [1]$ and

$$gv_1 = a, \quad gw_k = [a, gw_{k-1}], \quad gv_{k+1} = [a, gv_k],$$

for $k = 1, 2, \dots$. An easy induction shows that

$$(15.14) \quad \begin{aligned} gf[x] &= s(x)[1], & gf[x, y] &= s(x, y)a, \\ gf[x, y, \sigma] &= s(x, y)[a, gf\sigma] = [gf(x, y), gf\sigma]. \end{aligned}$$

To define the homotopy Φ of the desired contraction, introduce for each i , $0 \leq i < h$, the 2-chain

$$C(i) = \sum_{t=0}^{i-1} [1, t].$$

We then define Φ by recursion as $\Phi 1 = 0$ and

$$(15.15) \quad \Phi[x] = C(x), \quad \Phi[x, y] = [C(x), y],$$

$$(15.16) \quad \Phi[x, y, \sigma] = [C(x), y, \sigma] + s(x, y)[a, \Phi\sigma].$$

We wish to prove that, modulo norms,

$$\partial\Phi + \Phi\partial = gf - I.$$

For argument a 1-cell $[x]$ the proof is immediate by the definition of C . For argument a 2-cell $[x, y]$, the proof is also direct, using the normalization and with division into two cases according as $s^2(x, y)$ is 0 or 1. For the general case we first show by direct computation that for any chain c

$$(15.17) \quad \partial[a, c] = [a, \partial c].$$

Now take an argument $\tau = [x, y, \sigma] = [x, y, z, \rho]$ of dimension 3 or greater. The boundary may be written in either of the forms

$$\partial\tau = [\partial(x, y), \sigma] - [x, \partial(y, \sigma)] = [\partial(x, y, z), \rho] + [x, y, \partial\sigma],$$

while by (15.16) $\Phi\tau$ may be written in either of the forms

$$\begin{aligned} \Phi\tau &= [\Phi(x), y, \sigma] + s(x, y)[a, \Phi\sigma] \\ &= [\Phi(x, y), \sigma] + [gf(x, y), \Phi\sigma], \end{aligned}$$

which are equal term by term. Hence, using (15.17), we have

$$\partial\Phi\tau = [\partial\Phi(x, y), \sigma] + [\Phi(x), \partial(y, \sigma)] + [gf(x, y), \partial\Phi\sigma],$$

while on the other hand, using both forms of $\partial\tau$,

$$\Phi\partial\tau = [\Phi\partial(x, y), \sigma] - [\Phi(x), \partial(y, \sigma)] + [gf\partial(x, y, z), \Phi\rho] + [gf(x, y), \Phi\partial\sigma].$$

The terms starting with $\Phi(x)$ cancel. The term $gf\partial(x, y, z) = g\partial f(x, y, z) = 0$, because $\partial w_1 = 0$, and $f(x, y, z)$ is a multiple of w_1 . The induction assumption applied to the cells $[x, y]$ and σ then yields the desired conclusion.

Finally, the requisite annihilation properties of Φ hold; $f\Phi = 0$ follows by induction, and $\Phi g = 0$, mod norms, follows because $C(1)$ is a norm. The proof of Theorem 15.1 is now complete.

16. Computability and naturality

Let Π be a finitely generated abelian group. We may represent Π as a direct sum of cyclic groups Z_{h_i} of orders h_i , $i = 1, \dots, t$ (with $h_i = \infty$ admitted as a possibility). The direct product Theorem 6.1a asserts that

$$H_q(\Pi, n) \cong H_q(A_N(Z_{h_1}, n) \otimes \dots \otimes A_N(Z_{h_t}, n)).$$

Using the contractions of the preceding two sections and applying Lemma 3.1, we therefore obtain

$$H_q(\Pi, n) \cong H_q(M(h_1, n) \otimes \dots \otimes M(h_t, n)).$$

Since each of the complexes $M(h_i, n)$ is free and finitely generated in each dimension, the same holds for the complex $M(h_1, n) \otimes \dots \otimes M(h_t, n)$. We thus see that the groups $H_q(\Pi, n)$ are *computable* for a finitely generated group Π .

Let $\phi: \Pi \rightarrow \Pi'$ be a homomorphism of finitely generated groups. We shall consider the induced homomorphism $\phi_*: H_q(\Pi, n) \rightarrow H_q(\Pi', n)$. Since the representations of Π and Π' as direct sums of cyclic groups may be so chosen that ϕ carries each summand of Π into a summand of Π' , the problem of explicitly

computing ϕ_* leads to the following problem for cyclic groups: Given cyclic groups $Z_h(c)$ and $Z_{h'}(c')$ with generators c and c' of orders h and h' , respectively, and given a homomorphism

$$\phi: Z_h(c) \rightarrow Z_{h'}(c'),$$

defined by $\phi(c) = rc'$ with $r \neq 0$, find a map $\phi_{M,n}$ such that the diagram

$$(16.1) \quad \begin{array}{ccc} A(Z_h(c), n) & \xrightarrow{f} & M(h, n) \\ \downarrow A(\phi, n) & & \downarrow \phi_{M,n} \\ A(Z_{h'}(c'), n) & \xrightarrow{f'} & M(h', n) \end{array}$$

is commutative up to a homotopy, where the maps f and f' are those of the contractions described in §§14, 15.

We first define the maps

$$\phi_M = \phi_{M,1}: M(h, 1) \rightarrow M(h', 1)$$

as follows: If $h = \infty$, then r is arbitrary, and we set

$$(16.2) \quad \begin{array}{ll} \phi_M u = ru', & h' = \infty, \\ \phi_M u = rw'_0, & h' < \infty. \end{array}$$

If $h < \infty$, then also $h' < \infty$ and r must satisfy $hr \equiv 0 \pmod{h'}$. We choose r and then t so that

$$(16.3) \quad hr = th', \quad 0 \leq r < h,$$

and set

$$(16.4) \quad \phi_M v_k = t^k v'_k, \quad \phi_M w_k = rt^k w'_k.$$

In all cases it is readily verified that ϕ_M is a graded ∂ -ring homomorphism. With the final definition

$$\phi_{M,n} = B_N^{n-1}(\phi_{M,1}),$$

we assert that the diagram (16.1) is commutative up to a homotopy.

Since (16.1) for general n is obtained from the diagram for $n = 1$ by application of B_N^{n-1} , it suffices to consider the case $n = 1$. We shall consider only the most complicated case, when $h < \infty$ and hence $h' < \infty$.

As a preliminary we prove

$$(16.5) \quad \sum_{i=0}^{h-1} s'(r, ri) = t,$$

where s' is the basic 2-cocycle for the group $Z_{h'}$. Indeed, if we write $ir = u_i h' + v_i$, with $0 \leq v_i < h'$, then

$$s'(r, ri) = u_{i+1} - u_i, \quad i = 0, \dots, h-1.$$

In the sum (16.5) we may omit the term with $i = 0$. The sum then becomes $u_h = u_1$. Since r is chosen less than h , $u_1 = 0$; by (16.3) $u_h = t$; this gives the result.

We now consider the diagram

$$\begin{array}{ccc} A(Z_h(c), 1) & \xrightarrow{g} & M(h, 1) \\ \downarrow A(\phi, 1) & & \downarrow \phi_M \\ A(Z_{h'}(c'), 1) & \xrightarrow{f'} & M(h', 1) \end{array}$$

where g is the injection of the contraction of A to M , and calculate the composite $f'A(\phi, 1)g$. For argument w_0 we have

$$f'A(\phi, 1)gw_0 = f'A(\phi, 1)[1] = f'[r] = rw'_0 = \phi_M w_0,$$

since $0 \leq r < h$. For argument v_1 we have

$$\begin{aligned} f'A(\phi, 1)gv_1 &= f'A(\phi, 1) \sum_{i=0}^{h-1} [1, i] = f' \sum_{i=0}^{h-1} [r, ri] \\ &= \sum_{i=0}^{h-1} s'(r, ri)v'_1 = tv'_1 = \phi_M v_1. \end{aligned}$$

A similar computation applies to other arguments w_k, v_k and yields the commutativity of the diagram above. Then

$$f'A(\phi, 1) \simeq f'A(\phi, 1)gf = \phi_M f,$$

which is the desired homotopy for the diagram (16.1) for $n = 1$. Inspection shows that the homotopy here and in the case $g'\phi_M \simeq A(\phi, 1)g$ is not trivial.

In general, we have the diagram

$$(16.6) \quad \begin{array}{ccc} A(Z_h(c), n) & \xrightleftharpoons[g]{f} & M(h, n) \\ \downarrow A(\phi, n) & & \downarrow \phi_{M,n} \\ A(Z_{h'}(c'), n) & \xrightleftharpoons[g']{f'} & M(h', n), \end{array}$$

commutative up to homotopies for all n .

In case $h = h'$ the maps $\phi_{M,n}$ may be calculated conveniently in terms of a weight function present in each $M(h, n)$.

By a *weight function* w in a graded ∂ -ring G we understand a (new) grading of G , which thus assigns to certain w -homogeneous elements a of G an integral weight $w(a)$, and which has the following properties

- (i) If $a \in G_0$, then $w(a) = 0$,
- (ii) $w(\partial a) = w(a)$, if $w(a)$ is defined,
- (iii) $w(a \circ b) = w(a) + w(b)$, if $w(a), w(b)$ defined.

Such a weight function mechanically carries over into the homology ring $H(G)$. Furthermore, it carries over to the bar construction $B(G)$ by setting

$$w_B(1) = 0$$

$$w_B[a_1 | \cdots | a_m] = w(a_1) + \cdots + w(a_m), \quad w(a_i) \text{ defined.}$$

Property (i) for w_B is immediate, and property (iii) follows from the shuffle definition of the product in $B(G)$. As for property (ii) we have $w_B(\partial_s u) = w_B(u)$ in virtue of (ii) for w . In the simplicial boundary $\partial_s u$, the initial face $F_0 u$ is present only when a_1 has degree zero and hence weight zero; in this case $w_B(F_0 u) = w_B(u)$. The same argument applies to the last face, while for the intermediate faces F_i , $w_B(F_i u) = w_B(u)$ follows from (iii).

For the complex $M(\infty, 1)$ we define a weight by setting $w(u) = 1$ and $w(1) = 0$. We then have weights in all the complexes $M(\infty, n)$. Specifically, in $M(\infty, 2)$, $w(p) = p$ (notation of (14.5) above), and in $M(\infty, 3)$

$$w[(p_1) |_3 \cdots |_3 (p_k)] = p_1 + \cdots + p_k.$$

In the complexes $M(h, 1)$, h finite, weights may be defined by setting

$$w(v_k) = k, \quad w(w_k) = k + 1.$$

Conditions (i)–(iii) are then satisfied, so that a weight function is defined in each of the complexes $M(h, n)$.

The transformation $\phi_M: M(h, 1) \rightarrow M(h, 1)$ induced by $\phi: Z_h(c) \rightarrow Z_h(c)$ is given by $\phi_M a = r^{w(a)} a$ if $w(a)$ is defined. By an easy induction one proves that

$$(16.7) \quad \phi_{M,n} u = r^{w(u)} u, \quad w(u) \text{ defined.}$$

17. Typical cycles and their properties

In the complex $A(\Pi, n)$ we introduce for each $x \in \Pi$ and each integer $t \geq 1$ the chain

$$(17.1) \quad \gamma_{nt}(x, n) = [x |_n \cdots |_n x] \quad (t \text{ terms } x)$$

of dimension nt .

THEOREM 17.1. *For n even and $x \in \Pi$, $\gamma_{nt}(x, n)$ is a cycle of $A(\Pi, n)$ which satisfies the equality*

$$(17.2) \quad \gamma_{ns}(x, n) * \gamma_{nt}(x, n) = \binom{s+t}{s} \gamma_{n(s+t)}(x, n)$$

and the following homologies

$$(17.3) \quad \gamma_{nt}(rx, n) \sim r^t \gamma_{nt}(x, n), \quad r \text{ an integer,}$$

$$(17.4) \quad \gamma_{nt}(x_1 \tau \cdots \tau x_k, n) \sim \sum \gamma_{np_1}(x_1, n) * \cdots * \gamma_{np_k}(x_k, n),$$

where the sum is taken over all sets of positive integers p_i with

$$p_1 + p_2 + \cdots + p_k = t.$$

For n odd, $n > 1$, $\gamma_{nt}(x, n)$ is a cycle modulo 2, and the same equality and homologies are valid, modulo 2.

PROOF. The equation (17.2) is immediate. If $n > 1$, then

$$\begin{aligned} [x] *_{n-1} [x] &= [x |_{n-1} x] + \wp((n-1)(n-1))[x |_{n-1} x] \\ &= 0, & n \equiv 0 \pmod{2}, \\ &= 2[x |_{n-1} x], & n \equiv 1 \pmod{2}. \end{aligned}$$

Upon calculating the boundary $\partial_B \gamma_{nt}(x, n)$ we find at once that the residual boundary $\partial_r \gamma_{nt}(x, n) = 0$, and that each term of the simplicial boundary $\partial_s \gamma_{nt}(x, n)$ involves a product $[x] *_{n-1} [x]$ computed above. Therefore $\gamma_{nt}(x, n)$ is a cycle or a cycle modulo 2 according as n is even or odd. In the rest of this proof we assume n even, the results for n odd are found from these by replacing homologies by homologies modulo 2 throughout.

To prove (17.3) first take $\Pi = Z_\infty(c)$ and define $\phi: Z_\infty(c) \rightarrow Z_\infty(c)$ by setting $\phi(c) = rc$. Contraction to $M(\infty, n)$ carries $\gamma_{nt}(c, n)$ into $[u |_n \cdots |_n u]$, an element of weight t , and hence carries $A(\phi, n) \gamma_{nt}(c, n) = \gamma_{nt}(rc, n)$ into $\phi_{M,n}[u |_n \cdots |_n u] = r^t[u |_n \cdots |_n u]$. Since contraction preserves homologies, we have $\gamma_{nt}(rc, n) \sim r^t \gamma_{nt}(c, n)$. After applying the map $Z_\infty(c) \rightarrow \Pi$ carrying c into an arbitrary $x \in \Pi$, we obtain the desired relation (17.3).

For $k = 1$, (17.4) is an identity. We next prove it for $k = 2$, writing x and y in place of x_1 and x_2 . Take two isomorphic copies Π_1 and Π_2 of the group Π , and let x_1 be the copy of x in Π_1 , y_2 the copy of y in Π_2 . The direct product Theorem 6.1 then yields a chain equivalence

$$A(\Pi_1 + \Pi_2, n) \xrightleftharpoons[g_A]{f_A} A(\Pi_1, n) \otimes A(\Pi_2, n).$$

Hence we have the homology

$$\begin{aligned} \gamma_{nt}(x_1 + y_2, n) &\sim g_A f_A \gamma_{nt}(x_1 + y_2, n) \\ &= g_A f_A [x_1 + y_2 |_n \cdots |_n x_1 + y_2]. \end{aligned}$$

Upon applying the definition (6.4) of f_A of this becomes

$$\begin{aligned} \gamma_{nt}(x_1 + y_2, n) &\sim g_A \sum_{p=0}^t [x_1 |_n \cdots |_n x_1] \otimes [y_2 |_n \cdots |_n y_2], \\ &= g_A \sum_{p=0}^t \gamma_{np}(x_1, n) \otimes \gamma_{n(t-p)}(y_2, n), \\ &= \sum_{p=0}^t \gamma_{np}(x_1, n) *_{n\gamma_{n(t-p)}}(y_2, n). \end{aligned}$$

Upon transposing the terms with $p = 0$ and $p = t$ to the left, this is

$$\gamma_{nt}(x_1 + y_2, n) \sim \sum_{p=1}^{t-1} \gamma_{np}(x_1, n) *_{n\gamma_{n(t-p)}}(y_2, n).$$

Now map $\Pi_1 + \Pi_2$ into Π by the homomorphism identifying each element of Π_1 or of Π_2 with its original in Π . This homomorphism applied to the above homology yields (17.4) for $k = 2$.

For $k > 2$ we proceed by induction. By the deviation formulas

$$\begin{aligned} \gamma_{ni}(x_1 \tau \cdots \tau x_k \tau x_{k+1}, n) &= \gamma_{ni}(x_1 \tau \cdots \tau x_{k-1} \tau (x_k + x_{k+1}), n) \\ &\quad - \gamma_{ni}(x_1 \tau \cdots \tau x_{k-1} \tau x_k, n) - \gamma_{ni}(x_1 \tau \cdots \tau x_{k-1} \tau x_{k+1}, n). \end{aligned}$$

Apply the formula (17.4) for k arguments to each of the three terms on the right to get the homologous cycle

$$\sum \gamma_{np_1}(x_1, n) *_{n_1} \cdots *_{n_k} \gamma_{np_{k-1}}(x_{k-1}, n) *_{n_k} E,$$

where

$$E = \gamma_{np_k}(x_k + x_{k+1}, n) - \gamma_{np_k}(x_k, n) - \gamma_{np_k}(x_{k+1}, n).$$

The formula (17.4) for the already established case $k = 2$ applies to this expression E ; upon substitution of the result we obtain (17.4) for $k + 1$, as desired.

The method of argument used in the proof of the case $k = 2$ above will be called the "method of splitting variables." Giving in $A(\Pi, n)$ a cycle $z(x_1, \dots, x_t)$ which depends on t arguments $x_i \in \Pi$, take t copies $\Pi_1, \dots, \Pi_t$ of Π , locating x_i in the i^{th} copy, and apply the direct product theorem to the complex $A(\Pi_1 + \cdots + \Pi_t, n)$ to obtain the homology

$$z(x_1, \dots, x_t) \sim g_A f_A z(x_1, \dots, x_t).$$

After calculation of the right hand side by the formulas for g_A and f_A , and application of the homomorphism $\Pi_1 + \cdots + \Pi_t \rightarrow \Pi$ identifying each copy Π_i of Π with the original, there results a homology for $z(x_1, \dots, x_t)$ in $A(\Pi, n)$. In the sequel this method will be applied without detailed computation.

COROLLARY 17.2. For $n > 1$ and $k > t$,

$$(17.5) \quad \gamma_{ni}(x_1 \tau \cdots \tau x_k) \sim 0,$$

while for $k = t$,

$$(17.6) \quad \gamma_{ni}(x_1 \tau \cdots \tau x_t) \sim [x_1] *_{n_1} \cdots *_{n_t} [x_t].$$

COROLLARY 17.3. For $n > 1$, the suspended cycles $S\gamma_{ni}(x, n)$ satisfy

$$(17.7) \quad S\gamma_{ni}(x + y, n) \sim S\gamma_{ni}(x, n) + S\gamma_{ni}(y, n),$$

$$(17.8) \quad \begin{aligned} pS\gamma_{ni}(x, n) &\sim 0, & t = p^e, p \text{ prime,} \\ S\gamma_{ni}(x, n) &\sim 0, & t > 1, \text{ not a prime power.} \end{aligned}$$

The first corollary is immediate. The homology (17.7) follows directly from (17.4) and Corollary 12.3. The same corollary and (17.2) show that $\binom{t}{i} S\gamma_{ni}(x, n) \sim 0$, for $i = 1, \dots, t - 1$. But the greatest common divisor of the binomial coefficients $\binom{t}{i}$ for $i = 1, \dots, t - 1$ is known [13, Lemma 6] to be p when $t = p^e$ is a prime power, and zero otherwise; hence (17.8).

From (17.7) and (17.3) we can also deduce

$$(r^t - r)S\gamma_{nt}(x, n) \sim 0$$

for all integers r . However, this is a consequence of (17.8).

For n even and $s \in \Pi$ with $2s = 0$ we introduce the cycle

$$\begin{aligned}\zeta(s, n) &= (1/2)\partial[s, s \mid_n s, s] \\ &= [s, s \mid_n s] - [s \mid_n s, s] - [s, s \mid_{n-1} s, s],\end{aligned}$$

of dimension $2n + 2$, where we denote $[s \mid s]$ by $[s, s]$. If $n = 2$, the last term is then simply $[s, s, s, s]$.

THEOREM 17.4. *For n even and elements $s, t \in \Pi$ with $2s = 2t = 0$, we have the homology*

$$\begin{aligned}(17.9) \quad \zeta(s \tau t, n) &\sim [s, s]_*[t] - [s]_*[t, t] \\ &= (1/2)\partial([s, s]_*[t, t]).\end{aligned}$$

PROOF. Apply the process of splitting variables to the arguments s and t in $\zeta(s + t, n)$. It yields

$$\begin{aligned}g_A f_A \zeta(s + t, n) &= (1/2)\partial\{[t, t \mid_n t, t] + [s, s]_*[t, t] + [s, s \mid_n s, s]\} \\ &= \zeta(s, n) + \zeta(t, n) + (1/2)\partial([s, s]_*[t, t]),\end{aligned}$$

as desired.

Note that $\partial[s, s] = -2[s] + [0] \equiv -2[s]$, modulo norms, so that $-[s, s]$ is a 2-cap for the cycle $[s]$, and the right side of (17.9) has the form of a cross-cap for the cycles $[s]$ and $[t]$.

18. The ring $\Gamma(\Pi)$

Motivated by Theorem 17.1 we introduce for each abelian group Π the commutative graded ring $\Gamma(\Pi)$ which is generated by the elements $\gamma_t(x)$ for each $x \in \Pi$ and each non-negative integer t , of degree $2t$, subject to the relations

$$(18.1) \quad \gamma_0(x) = 1,$$

$$(18.2) \quad \gamma_t(rx) = r^t \gamma_t(x),$$

$$(18.3) \quad \gamma_t(x + y) = \sum_{i+j=t} \gamma_i(x) \gamma_j(y),$$

$$(18.4) \quad \gamma_s(x) \gamma_t(x) = \binom{s+t}{s} \gamma_{s+t}(x).$$

From these one readily proves by induction that $\gamma_t(0) = 0$ if $t > 0$. The relation (18.3) can be written

$$\gamma_t(x \tau y) = \sum_{i=1}^{t-1} \gamma_i(x) \gamma_{t-i}(y),$$

from which we have, by induction

$$(18.5) \quad \gamma_t(x_1 \tau \cdots \tau x_k) = \sum \gamma_{p_1}(x_1) \cdots \gamma_{p_k}(x_k),$$

where the sum is taken over all sets of positive integers $p_1, \dots, p_k$ such that $p_1 + \dots + p_k = t$. The relations (18.1)–(18.4) are parallel to those just established in Theorem 17.1. Hence for n even the mapping

$$\gamma_t(x) \rightarrow \text{homology class of } \gamma_{nt}(x, n)$$

yields a ring homomorphism

$$(18.6) \quad \gamma: \Gamma(\Pi) \rightarrow H(\Pi, n) \quad (n \text{ even}),$$

in which all degrees are multiplied by $n/2$.

We next determine the ring $\Gamma(\Pi)$ for cyclic groups Π .

THEOREM 18.1. *There is an isomorphism*

$$(18.7) \quad f: M(\infty, 2) \cong \Gamma(Z_\infty(c))$$

given in the notation (14.5) by

$$f((t)) = \gamma_t(c), \quad f^{-1}\gamma_t(rc) = r^t(t).$$

The proof is immediate. For the next theorem, we denote by (t, h^∞) the constant value of the greatest common divisor (t, h^e) for large e .

THEOREM 18.2. *The ring $\Gamma(Z_h(c))$ for h finite is generated by the elements $a_t = \gamma_t(c)$ for non-negative t , subject to the relations*

$$(18.8) \quad a_0 = 1, \quad h(t, h^\infty)a_t = 0, \quad t > 0,$$

$$(18.9) \quad a_s a_t = \binom{s+t}{t} a_{s+t}.$$

Furthermore, the additive group of elements of dimension $2t$ in $\Gamma(Z_h(c))$ is the cyclic group of order $h(t, h^\infty)$ generated by a_t .

The proof requires the following number-theoretic lemma.

LEMMA 18.3. *For positive integers t and h let*

$$F_{t,h} = \text{g.c.d. } \{(h+k)^t - k^t\}, \quad k = 1, 2, \dots.$$

Then

$$(18.10) \quad h(t, h^\infty) = \text{g.c.d. } \left\{ \binom{t}{j} F_{j,h} \right\}, \quad j = 1, \dots, t.$$

In particular, $h(t, h^\infty)$ is a divisor of $F_{t,h}$.

PROOF. Let $R_{t,h}$ denote the right hand side of (18.10). Introduce the polynomial

$$f_k(x) = [x + (h+k)]^t - (x+k)^t$$

with integral coefficients, the coefficient of x^{t-j} being $\binom{t}{j} [(h+k)^j - k^j]$. The content of $f_k(x)$ is the greatest common divisor of its coefficients; hence

$$R_{t,h} = \text{g.c.d. } \{\text{content } f_k(x)\}, \quad k = 1, 2, \dots.$$

But the content of a polynomial is not changed by the translation $x \rightarrow x - k$ which carries $f_k(x)$ into $f_0(x)$. Therefore

$$R_{t,h} = \text{content } f_0(x) = h \left[\text{g.c.d. } \left\{ \binom{t}{j} h^{j-1} \right\} \right], \quad j = 1, \dots, t.$$

We are therefore reduced to proving that

$$(t, h^\infty) = \text{g.c.d.} \left\{ h^{t-1}, \binom{t}{t-1} h^{t-2}, \dots, t \right\}.$$

Since the greatest common divisor on the right clearly divides (t, h^∞) , it suffices to prove for each prime p with $p \mid h$ and $p^e \mid t$ that p^e divides each term

$$\binom{t}{j} h^{j-1} = (t/j) \binom{t-1}{j-1} h^{j-1}$$

on the right. If $(j, p) = 1$, the result is immediate. If j is divisible by p^f and no higher power of p , then $j - 1 \geq f$, the result is again immediate, and the lemma is proved.

To prove the theorem, let A denote the abstract ring generated by the elements a_i subject to the relations (18.8) and (18.9), and define mappings

$$\Gamma(Z_h(c)) \xrightleftharpoons[g]{f} A$$

by setting

$$f\gamma_i(kc) = k^i a_i, \quad ga_i = \gamma_i(c).$$

A change of k modulo h changes k^i to $(h + k)^i$; the difference is a multiple of $F_{i,h}$ and hence of the order $h(t, h^\infty)$ of a_i ; therefore f is well defined. It is then easy to check that f preserves the defining relations (18.1)–(18.4) for Γ . To show g well defined, note that

$$0 = \gamma_i((h + k)c) - \gamma_i(kc) = [(h + k)^i - k^i]\gamma_i(c).$$

Hence $F_{i,h}\gamma_i(c) = 0$, and therefore, by (18.4) $\binom{t}{j} F_{i,j}\gamma_i(c) = 0$, for $j = 1, \dots, t$.

Therefore, by the lemma, $h(t, h^\infty)\gamma_i(c) = 0$, which proves that g is well defined. It is immediate that g preserves the relations (18.9).

To show that $h(t, h^\infty)$ is the exact additive order of a_i it suffices to show that the relation (18.9) introduces no lower order for a_{s+t} , and hence to show that

$$(s + t, h^\infty) \mid \binom{s + t}{s} (s, h^\infty).$$

Write $s + t = (s + t, h^\infty)r'$ and $s = (s, h^\infty)s'$, where r' and s' are prime to h . Then

$$\begin{aligned} \binom{s + t}{s} (s, h^\infty) &= (s + t/s) \binom{s + t - 1}{s - 1} (s, h^\infty) \\ &= (s + t, h^\infty)(r'/s') \binom{s + t - 1}{s - 1}. \end{aligned}$$

Since s' is prime to h , it is also prime to $(s + t, h^\infty)$. Therefore $(r'/s') \binom{s + t - 1}{s - 1}$ is an integer, which yields the result.

There is a tensor product theorem for the ring Γ .

THEOREM 18.4. *For two abelian groups Π_1 and Π_2 the mappings*

$$(18.11) \quad \Gamma(\Pi_1 + \Pi_2) \underset{g}{\overset{f}{\rightleftarrows}} \Gamma(\Pi_1) \otimes \Gamma(\Pi_2)$$

defined by

$$\begin{aligned} f\gamma_t((x_1, x_2)) &= \sum_{i+j=t} \gamma_i(x_1) \otimes \gamma_j(x_2), \\ g[\gamma_s(x_1) \otimes \gamma_t(x_2)] &= \gamma_s((x_1, 0))\gamma_t((0, x_2)) \end{aligned}$$

are isomorphisms of graded rings and are inverses of each other.

PROOF. It is readily verified that f preserves the defining relations for Γ , using in connection with (18.4) the identity

$$\sum_{i+j=r} \binom{m}{i} \binom{n}{j} = \binom{r+s}{r}, \quad m+n=r+s$$

for binomial coefficients (with $0 \leq i \leq m$, $0 \leq j \leq n$), proved by expanding $(x+y)^{r+s} = (x+y)^m(x+y)^n$. It is trivial that g preserves the defining relations and that both composites are identities.

Now let $\Gamma_{2t}(\Pi)$ be the abelian group consisting of all elements of degree $2t$ in the graded ring $\Gamma(\Pi)$. We have at once

COROLLARY 18.5. *For two abelian groups Π_1 and Π_2 , the mappings f and g defined above yield isomorphisms*

$$(18.12) \quad f: \Gamma_{2t}(\Pi_1 | \Pi_2) \cong \sum_{r+s=t} \Gamma_{2r}(\Pi_1) \otimes \Gamma_{2s}(\Pi_2).$$

In low dimensions the groups $\Gamma_{2t}(\Pi)$ are readily determined. Clearly $J = \Gamma_0(\Pi)$ where J is the ring of integers.

THEOREM 18.6. *If $\Gamma'_4(\Pi)$ denotes the group introduced in §13 and denoted there by $\Gamma_4(\Pi)$, then there are isomorphisms*

$$(18.13) \quad f: \Pi \cong \Gamma_2(\Pi), \quad f: \Gamma'_4(\Pi) \cong \Gamma_4(\Pi)$$

defined by

$$f(x) = \gamma_1(x), \quad f\gamma(x) = \gamma_2(x).$$

PROOF. The group $J + \Pi + \Gamma'_4(\Pi)$ will be converted into a commutative ring, defining the multiplication by the requirement that $1 \in J$ is the identity and

$$xy = \gamma(x \tau y), \quad \gamma(x)\gamma(y) = 0, \quad x\gamma(y) = 0.$$

This is compared with the ring $\Gamma(\Pi)_4$ obtained from the ring $\Gamma(\Pi)$ by dividing by the ideal generated by all elements of degree 6 or greater. The map f above defines a homomorphism

$$f: J + \Pi + \Gamma'_4(\Pi) \rightarrow \Gamma(\Pi)_4.$$

A homomorphism g in the opposite direction is obtained by setting

$$g1 = 1, \quad g\gamma_1(x) = x, \quad g\gamma_2(x) = \gamma(x).$$

Both f and g are ring homomorphisms, and the composites are identities.

19. The ring $\Lambda(\Pi)$; homology with rational coefficients

THEOREM 19.1. For $x_1, \dots, x_t$ in Π the cycle

$$(19.1) \quad [x_1] * _n \dots * _n [x_t] \in Z_{nt}(\Pi, n)$$

determines a homology class in $H_{nt}(\Pi, n)$ which is a multilinear function of its arguments $x_1, \dots, x_t$, symmetric if n is even and alternating if n is odd.

PROOF. In $A(\Pi, 1)$ and hence in $A(\Pi, n)$ we have the homology

$$0 \sim [x] + [y] - [x + y] = \partial[x \mid_1 y].$$

This implies the multilinearity of (19.1). Since $[x]$ has dimension n in $A(\Pi, n)$,

$$[x] * _n [y] = \wp(n^2)[y] * _n [x],$$

$$[x] * _n [x] = [x \mid_n x] + \wp(n^2)[x \mid_n x].$$

The first gives the symmetry of (19.1) for n even; the second gives the alternating property— $[x] * _n [x] = 0$ —for n odd.

The case n odd suggests the introduction of the graded “exterior” ring $\Lambda(\Pi)$ which is generated by the identity and the symbols (x) of degree 1 for $x \in \Pi$, subject to the relations

$$(19.2) \quad (x + y) = (x) + (y),$$

$$(19.3) \quad (x)(x) = 0.$$

These relations imply $(x)(y) + (y)(x) = 0$. In the sequel it is convenient to write x instead of (x) and $x \wedge y$ instead of $(x)(y)$. The elements of degree t in $\Lambda(\Pi)$ constitute a group denoted by $\Lambda_t(\Pi)$. Clearly

$$(19.4) \quad \Lambda_t(\Pi) \cong (\Pi \otimes \dots \otimes \Pi) / D, \quad (t \text{ factors}),$$

where D is the subgroup of the tensor product generated by all elements $x_1 \otimes \dots \otimes x_t$ with $x_i = x_j$ for some $i \neq j$.

THEOREM 19.2. For two abelian groups Π_1 and Π_2 , the mappings

$$(19.5) \quad \Lambda(\Pi_1 + \Pi_2) \underset{g}{\overset{f}{\longleftrightarrow}} \Lambda(\Pi_1) \otimes \Lambda(\Pi_2)$$

defined by

$$\begin{aligned} f(x_1, x_2) &= x_1 \otimes 1 + 1 \otimes x_2, \\ g(x_1 \otimes x_2) &= (x_1, 0) \wedge (0, x_2) \end{aligned}$$

are isomorphisms of graded rings and are inverses of each other.

The proof is immediate.

THEOREM 19.3. For any abelian group Π there is a ring monomorphism

$$(19.6) \quad \lambda: \Lambda(\Pi) \rightarrow H(\Pi, n), \quad n \text{ odd},$$

defined by setting

$$\lambda(x) = \text{homology class of } [x],$$

in which the degrees are multiplied by n .

PROOF. That λ is a ring homomorphism is clear from Theorem 19.1. Since both functors in (19.6) commute with the formation of direct limits, it suffices to establish (19.6) for finitely generated groups. For a finitely generated group Π we shall establish more; to wit, that λ is a monomorphism of $\Lambda(\Pi)$ into a direct summand of $H(\Pi, n)$. If $\Pi = Z_h(c)$ is cyclic, then $\Lambda(Z_h(c))$ is just the direct sum $J + Z_h(c)$ with 1 the identity and the product otherwise zero. The map λ carries J into the homology group $J = H_0(\Pi, n)$ and $Z_h(c)$ isomorphically onto the homology group $H_n(\Pi, n) \cong H_n(M(h, n)) = Z_h(w_0)$, so that the image of λ is immediately a direct summand of $H(\Pi, n)$.

Suppose now that $\lambda_i: \Lambda(\Pi_i) \rightarrow H(\Pi_i, n)$ is a monomorphism into a direct summand for two abelian groups Π_1 and Π_2 . In the commutative diagram

$$\begin{array}{ccc} \Lambda(\Pi_1) \otimes \Lambda(\Pi_2) & \xrightarrow{\lambda_1 \otimes \lambda_2} & H(\Pi_1, n) \otimes H(\Pi_2, n) \\ \downarrow g & & \downarrow \pi_* \\ \Lambda(\Pi_1 + \Pi_2) & \xrightarrow{\lambda} & H(\Pi_1 + \Pi_2, n) \end{array}$$

where g is the isomorphism of Theorem 19.2 above, it then follows that $\lambda_1 \otimes \lambda_2$ is a monomorphism into a direct summand. Also π_* in the diagram is the mapping induced by the multiplication in $H(\Pi_1 + \Pi_2, n)$, and is a monomorphism onto a direct summand in virtue of parts (i) and (iii) of the Künneth relations of §12. Therefore λ is also a monomorphism onto a direct summand.

REMARK. The analogue of the ring $\Lambda(\Pi)$ for even n can be defined quite similarly, using commutativity in place of skew-commutativity. It turns out to be nothing else than the subring of $\Gamma(\Pi)$ generated by elements of degree 2 in $\Gamma(\Pi)$. The monomorphism asserted in Theorem 19.3 may be established by the same methods in this case.

Abelian groups may be replaced by modules throughout our study of FD -complexes, the bar construction, and the functors Γ and Λ . Specifically, if C is any commutative ring (with unit) and M any C -module, we may define $\Gamma(M)$ as an algebra over M exactly as in §18 above, except that the scalar r in (18.2) is now an element of C and not an integer. Similarly, if G is a graded ∂ -algebra over C , we may carry out the bar construction $B(G) = B_C(G)$ "over C "—using scalars in C , tensor products over C , etc. The normalization and contraction theorems of I apply to the bar construction over C .

We now consider the homology ring $H(\Pi, n; R)$ with coefficients in the field R of rational numbers. First observe that for any graded ∂ -ring G there are natural isomorphisms

$$B(G \otimes R) \cong B(G) \otimes R,$$

$$H(G \otimes R) \cong H(G) \otimes R,$$

where $G \otimes R$ is regarded as a graded ∂ -algebra over R with an augmentation $\alpha: G \otimes R \rightarrow R$. Consequently we have, for any abelian group Π ,

$$H(\Pi, n; R) \cong H(\Pi, n) \otimes R.$$

Similarly, $\Pi \otimes R$ is an R -module, and for the functions Γ and Λ applied to R -modules there are natural isomorphisms

$$\Gamma(\Pi \otimes R) \cong \Gamma(\Pi) \otimes R, \quad \Lambda(\Pi \otimes R) \cong \Lambda(\Pi) \otimes R.$$

In the sequel all the isomorphisms above will be regarded as identifications.

The tensor product $\Pi \otimes R$ may be regarded as a vector space V over R . Then $\Lambda(\Pi \otimes R)$ is clearly just the ordinary exterior algebra of V over R . As to $\Gamma(\Pi \otimes R)$, the change of basis $\gamma_t(x) = (1/t!)x^t$ replaces the product formula (18.4) by the formula $x^s x^t = x^{s+t}$, while formula (18.3) becomes the usual binomial expansion of $(x + y)^t$. Thus $\Gamma(\Pi \otimes R)$ becomes the ordinary polynomial algebra of V over R in which every element of V is given degree 2. This means that if the set of elements x_α is a basis of the vector space V , then $\Gamma(\Pi \otimes R)$ is the usual polynomial ring $R[x_\alpha]$, with doubled degrees.

THEOREM 19.4. *For any abelian group Π , the homomorphisms (18.6) and (19.6) yield isomorphisms*

$$(19.7) \quad \gamma: \Gamma(\Pi \otimes R) \cong H(\Pi, n; R), \quad n \text{ even},$$

$$(19.8) \quad \lambda: \Lambda(\Pi \otimes R) \cong H(\Pi, n; R), \quad n \text{ odd}$$

In (19.7) the degrees are multiplied by $n/2$; in (19.8) by n .

PROOF. Since all functors involved commute with the formation of direct limits, it suffices to consider the case of a finitely generated group Π . Since all functors involved satisfy the tensor product theorem, it then suffices to consider the case when Π is cyclic.

If $\Pi = Z_h(c)$ is a finite cyclic group, and if R is regarded as a graded ∂ -ring with every element of degree zero, then clearly

$$\Gamma(Z_h(c) \otimes R) = R, \quad \Lambda(Z_h(c) \otimes R) = R.$$

Now inspection of the boundary formulas for $M(h, 1)$ shows that $M(h, 1) \otimes R \cong B(R)$. Since $B_N(R) = R$, we obtain a contraction of $M(h, 1) \otimes R$ to R . Therefore, by iteration, there result contractions of $M(h, n) \otimes R$ to R . This proves that

$$H(Z_h(c), n; R) \cong R,$$

and that the mappings γ and λ of (19.7) and (19.8) are isomorphisms in this case.

If $\Pi = Z_\infty(c)$ is an infinite cyclic group, then we have isomorphisms

$$M(\infty, 1) \cong \Lambda(Z_\infty(c)),$$

$$B_N(\Lambda(Z_\infty(c) \otimes R)) \cong \Gamma(Z_\infty(c) \otimes R),$$

the latter under the mapping

$$[c \mid \cdots \mid c] \rightarrow \gamma_t(c) = (1/t!)c^t, \quad t \text{ factors } c.$$

We also have a contraction

$$(f, g, \Phi): B_N(\Gamma(\Pi \otimes R)) \rightarrow \Lambda(\Pi \otimes R)$$

defined in terms of the exponential notation introduced above for $\Gamma(\Pi \otimes R)$ by the formulas

$$f[c] = (c) \in \Lambda(\Pi \otimes R), \quad fu = 0,$$

for u any cell of dimension 2 or greater in $B_N(\Gamma(\Pi \otimes R))$,

$$g(c) = [c]$$

$$\Phi[c] = 0, \quad \Phi[c^t] = [c \mid c^{t-1}], \quad t \geq 2,$$

$$\Phi[c^t \mid u] = [\Phi c^t \mid u], \quad t \geq 1,$$

with u any cell of $B_N(\Gamma(\Pi \otimes R))$. Note here that g multiplies the degrees by 3, while f divides degrees by 3. Combining these contractions with the known contraction $B \rightarrow B_N$ we establish contractions

$$M(\infty, n) \otimes R \rightarrow \Gamma(Z_\infty(c) \otimes R), \quad n \text{ even},$$

$$M(\infty, n) \otimes R \rightarrow \Lambda(Z_\infty(c) \otimes R), \quad n \text{ odd},$$

compatible with the mappings γ and λ of the theorem. This completes the proof.

COROLLARY 19.5. *For any abelian group Π , the natural mapping of Π into $\Pi \otimes R$ induces isomorphisms*

$$H(\Pi, n; R) \cong H(\Pi \otimes R, n; R).$$

For analogous considerations of the cohomology ring in the case $\Pi = J$, see Serre [9, p. 501].

CHAPTER IV. INVARIANT DESCRIPTIONS FOR HOMOLOGY AND COHOMOLOGY

20. Cross-effects for $H(\Pi, n)$

Using the direct product theorem, we can obtain an expression for the cross-effect functors of $H_q(\Pi, n) = H_q(A(\Pi, n))$. We denote by $A_+(\Pi, n)$ the ideal spanned by all elements of positive dimension in $A(\Pi, n)$. Then $A(\Pi, n)$ as a chain complex is a direct sum

$$(20.1) \quad A(\Pi, n) = J + A_+(\Pi, n),$$

where J denotes the additive group of integers.

THEOREM 20.1. *For positive dimensions q there is an isomorphism*

$$(20.2) \quad g_A: H_q(A_+(\Pi_1, n) \otimes \cdots \otimes A_+(\Pi_t, n)) \cong H_q(\Pi_1 \mid \cdots \mid \Pi_t, n),$$

given by the injection g_A of the direct sum theorem as

$$(20.3) \quad g_A(a_1 \otimes \cdots \otimes a_t) = a_1 *_{n-1} \cdots *_{n-1} a_t, \quad a_i \in A_+(\Pi_i, n).$$

PROOF. For brevity, let $L_i = A_+(\Pi_i, n)$. Then, by (20.1), we have the direct sum decomposition

$$\begin{aligned} A(\Pi_1, n) \otimes \cdots \otimes A(\Pi_t, n) \\ = J + \sum_i L_i + \sum_{i < j} L_i \otimes L_j + \cdots + (L_1 \otimes \cdots \otimes L_t). \end{aligned}$$

By the direct product theorem the complex on the left is a contraction of $A(\Pi_1 + \cdots + \Pi_t, n)$. Therefore, for $q > 0$, we obtain the isomorphism

$$H_q(\Pi_1 + \cdots + \Pi_t, n) \cong \sum_i H_q(L_i) + \sum_{i < j} H_q(L_i \otimes L_j) + \cdots + H_q(L_1 \otimes \cdots \otimes L_t).$$

Application of Corollary 9.7 shows that the injection g_A of $H_q(L_1 \otimes \cdots \otimes L_t)$ into $H_q(\Pi_1 + \cdots + \Pi_t, n)$ is indeed an isomorphism onto the cross-effect group $H_q(\Pi_1 | \cdots | \Pi_t, n)$.

THEOREM 20.2. *Suspensions of cross-effects are zero; that is, for all $q > 0$, the composite map*

$$H_q(\Pi_1 | \Pi_2, n) \rightarrow H_q(\Pi_1 + \Pi_2, n) \xrightarrow{S} H_{q+1}(\Pi_1 + \Pi_2, n+1)$$

is zero, and similarly for higher cross-effects.

PROOF. Apply Theorem 12.2 to $G \otimes G$, where $G = A(\Pi_1 + \Pi_2, n)$. On the subcomplex $A_+(\Pi_1, n) \otimes A_+(\Pi_2, n)$ of $G \otimes G$ the map ν defined by the augmentation as in Theorem 12.2 is zero, while the map π defined by the product is precisely g_A . Hence $Sg_A \simeq 0$, which gives the conclusion.

For convenience we summarize the trivial and well known facts as to the homology groups $H(\Pi, n)$ in the beginning dimensions.

THEOREM 20.3. *For any abelian group Π , we have*

$$(20.4) \quad \begin{aligned} H_0(\Pi, n) &= J, & H_q(\Pi, n) &= 0, & 0 < q < n, \\ \theta_n: \Pi &\cong H_n(\Pi, n), \end{aligned}$$

where θ_n maps $x \in \Pi$ into the homology class of $[x]$.

In the sequel we identify Π and $H_n(\Pi, n)$ by θ_n . We next establish

THEOREM 20.4. *For abelian groups $\Pi_1, \cdots, \Pi_t$*

$$(20.5) \quad H_q(\Pi_1 | \cdots | \Pi_t, n) = 0, \quad 0 < q < tn,$$

$$(20.6) \quad \Pi_1 \otimes \cdots \otimes \Pi_t \cong H_{nt}(\Pi_1 | \cdots | \Pi_t, n),$$

where the second isomorphism maps $x_1 \otimes \cdots \otimes x_t$ into the homology class of $[x_1] * \cdots * [x_t]$.

PROOF. For free complexes K_1 and K_2 with $H_{p_i}(K_i) = 0$ for $p_i < m_i$, $i = 1, 2$, the Künneth relations yield an isomorphism

$$(20.7) \quad \alpha: H_{m_1}(K_1) \otimes H_{m_2}(K_2) \cong H_{m_1+m_2}(K_1 \otimes K_2).$$

Now $H_q(A_+(\Pi_i, n)) = 0$ for $q < n$ and $\theta_0: \Pi_i \cong H_n(A_+(\Pi_i, n))$. An iterated application of (20.7) to (20.2) yields the desired conclusion.

THEOREM 20.5. *For any abelian group Π*

$$(20.8) \quad H_{n+1}(\Pi, n) = 0 \quad n > 1.$$

PROOF. By the suspension theorem it suffices to consider $n = 2$. Inspection of the complexes $M(h, 2)$ proves (20.8) for cyclic groups Π . Since all cross-

effects are zero by (20.5), the conclusion follows for finitely generated groups and hence, by direct limits, for all groups.

THEOREM 20.6. *For groups Π_1 and Π_2 we have the isomorphism*

$$(20.9) \quad \chi: \text{Tor}(\Pi_1, \Pi_2) \cong H_{2n+1}(\Pi_1 | \Pi_2, n), \quad n > 1,$$

under the cross-cap mapping χ , and the isomorphism

$$(20.10) \quad \Pi_1 \otimes H_{n+2}(\Pi_2, n) + H_{n+2}(\Pi_1, n) \otimes \Pi_2 \cong H_{2n+2}(\Pi_1 | \Pi_2, n), \quad n > 1,$$

given by the product $\star_n$. Hence, if $n > 2$, $H_{2n+2}(\Pi_1 | \Pi_2 | \Pi_3, n) = 0$.

PROOF. Apply the Künneth relations to the tensor product $A_+(\Pi_1, n) \otimes A_+(\Pi_2, n)$ of (20.2). In dimension $2n + 1$ the part given by the Künneth map α is zero in virtue of (20.8). This proves (20.9). In dimension $2n + 2$, the part given by the torsion product is zero, again in virtue of (20.8). Consequently, the map α is an isomorphism. Using (20.8) once more, this reduces to the cited isomorphism (20.10).

In the next four sections we calculate invariant formulas for the groups $H_{n+k}(\Pi, n)$ for low values of k . Our method consists in first computing these groups explicitly for cyclic groups Π , using the complexes $M(h, n)$. Only the final results of these computations will be cited as they are needed. The passage from cyclic groups to arbitrary groups uses the basic comparison Theorem 10.1, with cross-effect groups determined by the theorems just above. This indicates the two types of limitations on our method. On the one hand, the explicit computations for $M(h, n)$ rapidly become cumbersome. On the other hand, the theorems above exhaust all the cases in which the cross-effects are computable without the complications inherent in the use of the full Künneth formula. In §§21–25 we therefore limit ourselves to the dimensions $q \leq 2n + 2$ and $q \leq 3n$ covered by the cross-effect theorems alone. This accounts for the omission of the group $H_7(\Pi, 2)$. Explicit computations have convinced us that this group is much more complicated than the groups which we do consider.

We have omitted here the groups $H_q(\Pi, 1)$, for which complete results will be given in a separate publication.

21. The embedding of $\Gamma(\Pi)$

THEOREM 21.1. *The ring homomorphism*

$$(21.1) \quad \theta_\Gamma: \Gamma(\Pi) \rightarrow H(\Pi, n), \quad n \text{ even},$$

defined as in §18 by setting

$$(21.2) \quad \theta_\Gamma(\gamma_t(x)) = \gamma_{nt}(x, n) = [x |_n \cdots |_n x], \quad t \text{ factors } n,$$

is a monomorphism on the subgroup of elements of $\Gamma(\Pi)$ of degree ≤ 6 . If $n = 2$, it is an isomorphism between the subgroup of elements of $\Gamma(\Pi)$ of degree ≤ 6 and the subgroup of elements of $H(\Pi, 2)$ of even degree ≤ 6 .

NOTE. In (21.2) and subsequently in similar cases it is convenient to let the cycle $\gamma_{nt}(x, n)$ denote also the homology class of this cycle.

PROOF. Since both functors $\Gamma(\Pi)$ and $H(\Pi, n)$ commute with the formation of direct limits, it suffices to establish the result for a finitely generated group Π . For such groups Π we shall prove more; to wit, that θ_Γ in the cited dimensions is a monomorphism of $\Gamma(\Pi)$ onto a direct summand of $H(\Pi, n)$, which is onto the whole of $H(\Pi, n)$, if $n = 2$.

Suppose first that this result holds for the maps $\theta_{i,\Gamma}$, $i = 1, 2$, belonging to two finitely generated groups Π_1 and Π_2 . Consider the commutative diagram

$$\begin{array}{ccc} \Gamma(\Pi_1) \otimes \Gamma(\Pi_2) & \xrightarrow{\theta_{1,\Gamma} \otimes \theta_{2,\Gamma}} & H(\Pi_1, n) \otimes H(\Pi_2, n) \\ \downarrow g & & \downarrow \pi_* \\ \Gamma(\Pi_1 \otimes \Pi_2) & \xrightarrow{\theta_\Gamma} & H(\Pi_1 + \Pi_2, n) \end{array}$$

in which g is the isomorphism from the tensor product Theorem 18.4 for $\Gamma(\Pi)$, while π_* is the homomorphism induced by the multiplication in $H(\Pi_1 + \Pi_2, n)$. By parts (i) and (ii) of the Künneth relations for the complex $A(\Pi_1, n) \otimes A(\Pi_2, n)$, π_* is a monomorphism onto a direct summand in all dimensions. By the assumption $\theta_{1,\Gamma}$ and $\theta_{2,\Gamma}$ and hence their tensor product is a monomorphism onto a direct summand for dimensions up to six inclusive. The same therefore follows for θ_Γ , since g is an isomorphism.

It remains to establish this property for a p -cyclic group. We first take $n = 2$. Calculation on the complexes $M(h, 2)$ shows that

$$\begin{aligned} H_2(Z_h(c), 2) &= Z_h([c]), \\ H_4(Z_h(c), 2) &= Z_h(\gamma_4(x, 2)), & h = \infty \text{ or } (h, 2) = 1, \\ &= Z_{2h}(\gamma_4(x, 2)), & h = 2^e, \\ H_6(Z_h(c), 2) &= Z_h(\gamma_6(x, 2)), & h = \infty \text{ or } (h, 3) = 1, \\ &= Z_{3h}(\gamma_6(x, 2)), & h = 3^e. \end{aligned}$$

On the other hand, Theorem 18.2 shows that

$$\begin{aligned} \Gamma_2(Z_h(c)) &= Z_h(\gamma_1(c)), \\ \Gamma_4(Z_h(c)) &= Z_{h(2,h)}(\gamma_2(c)), \\ \Gamma_6(Z_h(c)) &= Z_{h(3,h)}(\gamma_3(c)). \end{aligned}$$

The results check, and this concludes the proof for $n = 2$.

Suppose now that $n > 2$, and consider the composite map θ' ,

$$\Gamma(Z_h(c)) \xrightarrow{\theta_\Gamma} A(Z_h(c), n) \xrightarrow{B_n^{n-1}(f)} M(h, n),$$

obtained using the projection f of the contraction of $A(Z_h(c), 1)$ to $M(h, 1)$. We now aim to construct a homomorphism

$$\psi: M_{n,t}(h, n) \rightarrow \Gamma_t(Z_h(c)), \quad t = 0, 1, 2, 3,$$

such that

$$\psi\partial = 0, \quad \psi\theta' = \text{identity}.$$

Then if ψ is defined as zero on other dimensions, it is a chain transformation of $M(h, n)$ into the complex $\Gamma(Z_h(c))$, with boundary operator 0, and the composite

$$\Gamma_t(Z(c)) \xrightarrow{\theta_\Gamma} H_{nt}(Z_h(c), n) \xrightarrow{\psi_*} \Gamma_t(Z_h(c))$$

is the identity for $t = 0, 1, 2$, and 3. Hence θ_Γ is a monomorphism onto a direct summand of $H_{nt}(Z_h(c), n)$ for the indicated values of t .

For finite h , we use the notation $w = w_0$, $v = v_1$, and $vw = [v \mid_{n-1} w]$ for the indicated cells of $M(h, n-1)$ and define ψ by setting

$$\begin{aligned} \psi_1 &= 1, & \psi[w] &= \gamma_1(c), \\ \psi[w \mid_n w] &= \gamma_2(c), & \psi[v \mid_n w] &= -h\gamma_2(c), \\ \psi[w \mid_n w \mid_n w] &= \gamma_3(c), & \psi[vw \mid_n w] &= -h\gamma_3(c), \\ \psi[w \mid_n vw] &= -2h\gamma_3(c), \end{aligned}$$

with ψ zero on all cells of $M(h, n)$ not listed. It is immediate that $\psi\theta'$ is the identity. The proof that $\psi\partial\sigma_q = 0$ for cells σ_q of dimensions $q = n+1, 2n+1$, and $3n+1$ in $M(h, n)$ is given by direct computation, noting that the only such cells σ_q for which ψ does not vanish on each term of $\partial\sigma_q$ are the cells

$$\begin{aligned} q = 1: & [v], \\ q = 2: & [w \mid_n v], [v \mid_n w], [vw], \\ q = 3: & [v \mid_n w \mid_n w], [w \mid_n v \mid_n w], [w \mid_n w \mid_n v], [vw \mid_n w], \\ & [w \mid_n vw], [vw \mid_n v], [v \mid_n vw]. \end{aligned}$$

In each of these cases $\psi\partial\sigma_q = 0$, in view of the properties

$$h\gamma_1(c) = 0, \quad h(2, h)\gamma_2(c) = 0, \quad h(3, h)\gamma_3(c) = 0,$$

established in Theorem 18.2.

If $h = \infty$, we apply the same construction, replacing w by the cell u of $M(h, 1)$ and v by zero. This completes the proof of Theorem 21.1.

Theorem 21.1 includes a description of the group $H_6(\Pi, 2)$ which we may make explicit by describing the six-dimensional component $\Gamma_6(\Pi)$ as follows:

THEOREM 21.2. *If $F(\Pi)$ is the free abelian group generated by symbols $\sigma(x)$ for $x \in \Pi$, then there is an epimorphism*

$$(21.3) \quad \rho: \Pi \otimes \Gamma_4(\Pi) + F(\Pi) \rightarrow \Gamma_6(\Pi)$$

defined by setting

$$\begin{aligned} \rho(x \otimes \gamma_2(y)) &= \gamma_1(x)\gamma_2(y), \\ \rho(\sigma(x)) &= \gamma_3(x). \end{aligned}$$

The kernel of ρ is the subgroup generated by the relations

$$(21.4) \quad \sigma(x \tau y) = x \otimes \gamma_2(y) + y \otimes \gamma_2(x),$$

$$(21.5) \quad 3\sigma(x) = x \otimes \gamma_2(x).$$

The proof is like the determination of $\Gamma_4(\Pi)$ in Theorem 18.6. We denote the quotient of $\Pi \otimes \Gamma_4(\Pi) + F(\Pi)$ by the suggested kernel by $\Gamma'_6(\Pi)$. Then convert

$$R = J + \Pi + \Gamma_4(\Pi) + \Gamma'_6(\Pi)$$

into a commutative ring by defining the multiplication by

$$xy = \gamma_2(x \tau y), \quad x\gamma_2(y) = x \otimes \gamma_2(y),$$

with all other products (except by 1) zero. This ring will be compared with the ring $\Gamma(\Pi)_6$ obtained from the ring $\Gamma(\Pi)$ by dividing by the ideal generated by all elements of degree 8 or greater. The map ρ above then clearly defines a homomorphism $\rho: R \rightarrow \Gamma(\Pi)_6$; a homomorphism g in the opposite direction is defined by setting

$$g1 = 1, \quad g\gamma_1(x) = x, \quad g\gamma_2(x) = \gamma_2(x), \quad g\gamma_3(x) = \sigma(x).$$

The composites are easily seen to be the identity, using the relations (21.4) and (21.5)

22. The ring $H(\Pi, 2)$ in low dimensions

By the last section we have already determined $H_q(\Pi, 2)$, for $q \leq 6$, except for the group $H_5(\Pi, 2)$. We now determine this group. The description uses the cycle

$$\zeta(s, 2) = (1/2)\partial[s, s \parallel s, s], \quad s \in {}_2\Pi$$

which was discussed in §17. In addition we shall use the cross-cap

$$\chi_h(x, y) = \chi_h([x], [y]), \quad hx = hy = 0,$$

which is well defined in $H_5(\Pi, 2)$ because $H_3(\Pi, 2) = 0$. Direct computation for p -cyclic groups yields

$$\begin{aligned} H_5(Z_h(c), 2) &= 0, & h &= \infty \text{ or } (h, 2) = 1, \\ &= Z_2(\zeta(2^{e-1}c, 2)), & h &= 2^e. \end{aligned}$$

THEOREM 22.1. *Let ${}_2\Pi$ denote the subgroup of all elements $s \in \Pi$ with $2s = 0$. There is a natural epimorphism*

$$(22.1) \quad \theta: \text{Tor}(\Pi, \Pi) + \Gamma_4({}_2\Pi) \rightarrow H_5(\Pi, 2)$$

defined by setting

$$\begin{aligned} \theta_s \tau_h(x, y) &= \chi_h(x, y), & hx &= hy = 0, \\ \theta_s \gamma_2(s) &= \zeta(s, 2) & s &\in {}_2\Pi. \end{aligned}$$

The kernel of θ_5 is the subgroup generated by the relations

$$(22.2) \quad \tau_h(x, x) = 0, \quad hx = 0,$$

$$(22.3) \quad \gamma_2(s \tau t) = \tau_2(s, t), \quad s, t \in {}_2\Pi.$$

PROOF. Since the identities on Tor match those on the cross cap, the homomorphism θ_5 is well defined on $\text{Tor}(\Pi, \Pi)$. Furthermore, since the cycle $[x]$ has even dimension, equation (12.8) gives the homology

$$(22.4) \quad \chi_h(x, x) = 0, \quad hx = 0.$$

By Theorem 17.4 on $\zeta(s, 2)$ we have the homology

$$(22.5) \quad \zeta(s \tau t, 2) \sim \chi_2(s, t), \quad s, t \in {}_2\Pi.$$

Since χ_2 is additive in its arguments, this implies that $\zeta(r \tau s \tau t, 2) \sim 0$ for $r, s, t \in {}_2\Pi$. Since $s = -s$, $\zeta(-s, 2) = \zeta(s, 2)$. Therefore θ_5 preserves the identities used to define $\Gamma_4({}_2\Pi)$ and hence is a homomorphism. By (22.4) and (22.5) the kernel also contains the relations (22.2) and (22.3).

For the purposes of this section let $R(\Pi)$ denote the quotient group of $\text{Tor}(\Pi, \Pi) + \Gamma({}_2\Pi)$ by the relations (22.2) and (22.3). We wish to prove that the induced homomorphism

$$\bar{\theta}_5: R(\Pi) \rightarrow H_5(\Pi, 2)$$

is an isomorphism.

For a cyclic group $\Pi = Z_h(c)$ we know that $\text{Tor}(\Pi, \Pi)$ is generated by $\chi_h(c, c)$, which is zero in $R(\Pi)$ by (22.2), and that $\Gamma_4({}_2\Pi)$ is zero if h is odd and is generated by $\gamma_2((h/2)c)$ if h is even. Hence, for a p -cyclic group,

$$R(Z_h(c)) = 0, \quad h = \infty \text{ or } h \text{ odd},$$

$$R(Z_h(c)) = Z_2(\gamma_2(2^{e-1}c)), \quad h = 2^e.$$

These results check with those above for $H_5(Z_h, 2)$, and hence $\bar{\theta}_5$ is an isomorphism for Π p -cyclic.

The cross-effect functor of R is determined by

LEMMA 22.2. The map g_R defined by

$$g_R: \tau_h(x_1, y_2) = \tau_h((x_1, 0), (0, y_2)), \quad x_1 \in \Pi_1, y \in \Pi_2$$

yields a natural isomorphism

$$(22.6) \quad g_R: \text{Tor}(\Pi_1, \Pi_2) \cong R(\Pi_1 \mid \Pi_2).$$

PROOF. Consider the maps

$$R(\Pi_1 + \Pi_2) \xrightleftharpoons[g]{f} R(\Pi_1) + R(\Pi_2) + \text{Tor}(\Pi_1, \Pi_2)$$

defined for $hx_i = hy_i = 0 = 2s_i$ in Π_i , $i = 1, 2$, by

$$\begin{aligned} f\tau_h((x_1, x_2), (y_1, y_2)) &= \tau_h(x_1, y_1) + \tau_h(x_2, y_2) + [\tau_h(x_1, y_2) - \tau_h(y_1, x_2)], \\ f\gamma_2((s_1, s_2)) &= \gamma_2(s_1) + \gamma_2(s_2) + \tau_2(s_1, s_2), \\ g\tau_h(x_1, y_1) &= \tau_h((x_1, 0), (y_1, 0)), & g\gamma_2(s_1) &= \gamma_2((s_1, 0)), \\ g\tau_h(x_2, y_2) &= \tau_h((0, x_2), (0, y_2)), & g\gamma_2(s_2) &= \gamma_2((s_2, 0)), \\ g\tau_h(x_1, y_2) &= \tau_h((x_1, 0), (0, y_2)). \end{aligned}$$

We verify that f and g are homomorphisms, and then that both composites fg and gf are identities. This gives the conclusion, by Corollary 9.7.

By Theorem 20.6, the cross-effect group $H_5(\Pi_1 | \Pi_2, 2)$ is also $\text{Tor}(\Pi_1, \Pi_2)$. Thus to compare cross-effects it suffices to check the obvious commutativity in the diagram

$$\begin{array}{ccc} \text{Tor}(\Pi_1, \Pi_2) & \xrightarrow{g_R} & R(\Pi_1 | \Pi_2) \\ \downarrow \text{Tor}(\theta_2, \theta_2) & & \downarrow \bar{\theta}_5 \\ \text{Tor}(H_2(\Pi_1, 2), H_2(\Pi_2, 2)) & \xrightarrow{\chi} & H_5(\Pi_1 | \Pi_2, 2), \end{array}$$

where θ_2 is the identification of Π_i with $H_2(\Pi_i, 2)$, χ is the cross-effect map, and $\bar{\theta}_5$ is the map induced by θ_5 on the cross-effects. Since both functors R and $H^5(\Pi, 2)$ are of degree 2, the Theorem now follows from the basic comparison Theorem 10.4.

23. The ring $H(\Pi, 3)$ in low dimensions

In this section we determine the homology ring $H(\Pi, 3)$ in dimensions up to 7, inclusive. The following cycles are involved:

$$\begin{aligned} \gamma_2(x, 2) &= [x], \\ \gamma_4(x, 2) &= [x \parallel x], \\ \gamma_6(x, 2) &= [x \parallel x \parallel x], \\ \zeta(s, 2) &= (1/2)\partial[s, s \parallel s, s], & s \in {}_2\Pi, \\ \sigma(u) &= [b(u), b(u), b(u)] + 3\{[u \parallel b(u), b(u)] - b(u)*_2[u \parallel u] + 3[u \parallel u \parallel b(u)]\}, \\ & & u \in {}_3\Pi, \end{aligned}$$

where $b(u)$ is the 3-cap for $[u]$ defined by

$$b(u) = [u, u] + [u, u^2].$$

All these cycles are suspended from the complex $A(\Pi, 2)$; the notation will systematically omit the symbol for suspension. A new cycle appears, as follows. Take $x \in \Pi$ with $hx = 0$; there is then a chain $b \in A_2(\Pi, 1)$ with $\partial b = h[x]$. We set

$$(23.1) \quad \mu_h(x) = [x]*_3b = (1/h)\partial[b \mid_3 b].$$

The homology class of $\mu_h(x)$ is independent of the choice, since a different choice b' yields a cycle $b - b'$ in $Z_4(\Pi, 3)$. Since $H_4(\Pi, 3) = 0$, $[x] *_3 b \sim [x] *_3 b'$. In addition we shall use the cross-cap of dimension 7

$$\chi_h(x, y) = \chi_h([x], [y]), \quad hx = hy = 0.$$

This cross-cap is formed relative to the multiplication $*_3$ in the ring $A(\Pi, 3)$; it is well defined because $H_4(\Pi, 3) = 0$. It has no relation with the 5-dimensional cross-cap in $A(\Pi, 2)$ used in the previous section.

Direct computation for p -cyclic groups yields the following results

$$\begin{aligned} H_3(Z_h(c), 3) &= Z_h([c]), \\ H_4(Z_h(c), 3) &= 0, \\ H_5(Z_h(c), 3) &= Z_2(\gamma_4(c, 2)), & h = \infty \text{ or } h = 2^e, \\ &= 0, & h \neq \infty, (h, 2) = 1, \\ H_6(Z_h(c), 3) &= Z_2(\zeta(2^{e-1}c, 2)), & h = 2^e, \\ &= 0, & h = \infty \text{ or } (h, 2) = 1, \\ H_7(Z_h(c), 3) &= Z_3(\gamma_6(c, 2)), & h = \infty, \\ &= Z_h(\mu_h(c)), & h < \infty, h \not\equiv 0 \pmod{3}, \\ &= Z_3(\gamma_6(c, 2)) + Z_h(\mu_h(c)), & h = 3^e, \\ H_8(Z_h(c), 3) &= Z_3(\sigma(3^{e-1}c)), & h = 3^e, \\ &= Z_3([c] *_3 [c \parallel c]), & h = \infty \text{ or } h = 2^e, \\ &= 0, & h < \infty, (h, 6) = 1. \end{aligned}$$

By Theorems 20.3 and 20.5 we already have

$$(23.2) \quad \Pi = H_3(\Pi, 3), \quad H_4(\Pi, 3) = 0.$$

THEOREM 23.1. *The map $x \rightarrow \gamma_4(x, 2)$ induces a natural isomorphism*

$$(23.3) \quad \theta_5: \Pi/2\Pi \cong H_5(\Pi, 3) \quad (\text{stable}).$$

Here and in the sequel the label "(stable)" is applied whenever the homology group remains invariant under suspension.

THEOREM 23.2. *There is a natural isomorphism*

$$(23.4) \quad \theta_6: {}_2\Pi + \Lambda_2(\Pi) \cong H_6(\Pi, 3)$$

defined by setting

$$\begin{aligned} \theta_6(s) &= \zeta(s, 2), & s \in {}_2\Pi, \\ \theta_6(x \wedge y) &= [x] *_3 [y], & x, y \in \Pi. \end{aligned}$$

Here ${}_2\Pi$ denotes the subgroup of all elements of order 2 in Π , while $\Lambda_2(\Pi)$ is the two-dimensional component of the ring $\Lambda(\Pi)$ and is the quotient of the group $\Pi \otimes \Pi$ by the relation $x \otimes x = 0$.

THEOREM 23.3. *There is a natural isomorphism*

$$(23.5) \quad \theta_7: \Omega(\Pi) + \Pi/3\Pi \cong H_7(\Pi, 3),$$

where $\Omega(\Pi)$ is the group introduced in §13 and θ_7 is defined by

$$\theta_7\omega_h(x) = \mu_h(x), \quad hx = 0,$$

$$\theta_7(x + 3\Pi) = \gamma_6(x, 2).$$

THEOREM 23.4. *There is a natural isomorphism*

$$\theta_8: \Pi \otimes (\Pi/2\Pi) + {}_3\Pi \rightarrow H_8(\Pi, 3)$$

defined by setting

$$\theta_8(x \otimes (y + 2\Pi)) = [x]_*[y \parallel y],$$

$$\theta_8(u) = \sigma(u), \quad u \in {}_3\Pi.$$

Here ${}_3\Pi$ denotes the subgroup of all elements of order 3 in Π .

The only non-trivial products up to dimension 8 are those given by $[x]_*[y] = \theta_6(x \wedge y)$ and $[x]_*[y \parallel y] = \theta_8(x \otimes (y + 2\Pi))$.

PROOF OF THEOREM 23.1. Consider the diagram

$$\begin{array}{ccc} H_4(\Pi, 2) & \xrightarrow{S} & H_5(\Pi, 3) \\ \uparrow \theta_7 & & \uparrow \theta_6 \\ \Gamma_4(\Pi) & \xrightarrow{s} & \Pi/2\Pi, \end{array}$$

where s is the homomorphism with $s\gamma(x) = x + 2\Pi$ described in §13. To determine the suspension S , observe that the only relevant added cell is the cell $[x]_*[y]$ of dimension 6 with boundary

$$-[x]_*[y] \sim -\gamma_4(x \tau y, 2).$$

These homology classes span the kernel of S and are mapped by θ_7^{-1} onto elements spanning the kernel of s . This proves the diagram commutative and shows that θ_6 is an isomorphism.

The suspension map $H_5(\Pi, 2) \rightarrow H_6(\Pi, 3)$ is as follows. The subgroup of $H_5(\Pi, 2)$ given by $\text{Tor}(\Pi, \Pi)$ suspends into zero, while the subgroup $\Gamma_4({}_2\Pi)$ suspends into ${}_2\Pi$ by the natural epimorphism, setting $\gamma(x \tau y) = 0$.

The suspension map $H_6(\Pi, 2) \rightarrow H_7(\Pi, 3)$ is as follows: The subgroup of $H_6(\Pi, 2)$ given by $\Pi \otimes \Gamma_4(\Pi)$ suspends into zero, while the subgroup given by $F(\Pi)$ in Theorem 21.2 suspends into $\Pi/3\Pi$ by the natural map $\sigma(x) \rightarrow x + 3\Pi$.

PROOF OF THEOREM 23.2. First observe that θ_6 is a homomorphism. Since $[x \tau y] \sim 0$ in $A(\Pi, 1)$, the homology class of the cycle $[x]_*[y]$ is clearly bilinear in its arguments x and y and vanishes when $x = y$; hence θ_6 defines a homomorphism on $\Lambda_2(\Pi)$. In view of the homology (22.5) and the fact that the cross-cap appearing there suspends into a boundary in $A(\Pi, 3)$, by Corollary 12.4, we have

$$\zeta(s + t, 2) \sim \zeta(s, 2) + \zeta(t, 2), \quad s, t \in {}_2\Pi.$$

Hence θ_6 defines a homomorphism on the group ${}_2\Pi$.

For a cyclic group $\Pi = Z_h(c)$ the group $\Lambda_2(\Pi)$ is zero. The subgroup ${}_2\Pi$ is zero, if $h = \infty$ or h is odd, while if $h = 2^e$, it is cyclic of order 2 with generator $2^{e-1}c$. Comparison with the values of $H_6(Z_h(c), 3)$ tabulated above shows that θ_6 is an isomorphism for Π p -cyclic.

Since the functor ${}_2\Pi$ is additive, the cross-effects for the functor $S(\Pi) = {}_2\Pi + \Lambda_2(\Pi)$ coincide with the cross-effect $\Lambda_2(\Pi_1 | \Pi_2) \cong \Pi_1 \otimes \Pi_2$ for $\Lambda_2(\Pi)$ implied by Theorem 19.2. For the functor $H_6(\Pi, 3)$ the cross-effect is given by Theorem 20.4 and is also $\Pi_1 \otimes \Pi_2$. Hence to compare cross-effects it suffices to verify the obvious commutativity in the diagram

$$\begin{array}{ccc} \Pi_1 \otimes \Pi_2 & \xrightarrow{g} & \Lambda_2(\Pi_1 | \Pi_2) \\ \downarrow \theta_3 \otimes \theta_3 & & \downarrow \theta'_6 \\ H_3(\Pi_1, 3) \otimes H_3(\Pi_2, 3) & \rightarrow & H_6(\Pi_1 | \Pi_2, 3) \end{array}$$

in which g is the map $g(x_1 \otimes x_2) = x_1 \wedge x_2$ given in the computation of the cross-effect for $\Lambda_2(\Pi)$, while θ'_6 is the cross-effect map induced by θ_6 . Since $\Lambda_2(\Pi)$ and $H_6(\Pi, 3)$ are both functors of degree 2, this completes the proof of the theorem.

PROOF OF THEOREM 23.3. To show that θ_7 is a homomorphism on the summand $\Pi/3\Pi$, we use the homologies

$$\begin{aligned} \gamma_6(x \tau y, 2) &\sim [x] *_2 \gamma_4(y, 2) + \gamma_4(x, 2) *_2 [y], \\ 3\gamma_6(x, 2) &\sim [x] *_2 \gamma_4(x, 2) \end{aligned}$$

in $A(\Pi, 2)$ given by Theorem 17.1. Since products suspend into zero, these give in $A(\Pi, 3)$ the homologies

$$\gamma_6(x \tau y, 2) \sim 0, \quad 3\gamma_6(x, 2) \sim 0,$$

which show that θ_7 is a homomorphism on $\Pi/3\Pi$.

To show that θ_7 is a homomorphism on $\Omega(\Pi)$ we first derive some properties of the cycles $\mu_h(x)$. We assert that

$$(23.6) \quad \mu_{hk}(x) \sim k\mu_h(x), \quad hx = 0.$$

Indeed, if b is an h -cap for $[x]$, then kb is an hk -cap for $[x]$, and the result follows from the definition (23.1). We note that an explicit h -cap for $[x]$ is the chain

$$b_h(x) = \sum_{i=0}^{h-1} [x | ix], \quad hx = 0.$$

We next assert that

$$(23.7) \quad k\mu_{hk}(x) \sim \mu_h(kx) \quad h kx = 0.$$

In view of the homomorphism $Z_{hk}(c) \rightarrow \Pi$ given by mapping c into x , it suffices to establish (23.7) for the generator c of $Z_{hk}(c)$. If we use the homomorphism

$$\phi: Z_h(a) \rightarrow Z_{hk}(c), \quad \phi(a) = kc,$$

then, as in (16.6), the diagram

$$\begin{array}{ccc} A(Z_h(a), 3) & \xleftarrow{g} & M(h, 3) \\ \downarrow A(\phi, 3) & & \downarrow \phi_{M,3} \\ A(Z_{hk}(c), 3) & \xleftarrow{g'} & M(hk, 3) \end{array}$$

is commutative up to a homotopy. Here the basic cells w_0 and v_1 on dimensions 1 and 2 in $M(h, 1)$ are mapped by g into $[a]$ and its cap $b_h(a)$, respectively. Therefore

$$A(\phi, 3) g(w_0 *_3 v_1) = \mu_h(kc).$$

On the other hand, the definition of ϕ_M given in §16 has $\phi_M w_0 = kw'_0$ and $\phi_M v_1 = v'_1$, where w'_0 and v'_1 are the cells in $M(hk, 1)$ in dimensions 1 and 2, with

$$g' \phi_{M,3} (w_0 *_3 v_1) = k\mu_{hk}(c).$$

Since the diagram is commutative up to a homotopy, this proves the desired homology

$$k\mu_{hk}(c) \sim \mu_h(kc).$$

Finally, we have the homology

$$(23.8) \quad \mu_h(x \tau y) \sim \chi_h(x, y) \quad hx = hy = 0.$$

where χ_h is the cross-cap introduced above. This is proved by splitting variables in the explicit formula

$$\begin{aligned} \mu_h(x + y) &= [x + y] *_3 b_h(x + y) \\ &= \sum_{i=0}^{h-1} [x + y] *_3 [x + y; ix + iy]. \end{aligned}$$

The process yields

$$\begin{aligned} \mu_h(x + y) &\sim g_A f_A \mu_h(x + y) \\ &\sim \mu_h(x) + \mu_h(y) + [x] *_3 b_h(y) + b_h(y) *_3 [x]. \end{aligned}$$

Since $b_h(x)$ and $b_h(y)$ are h -caps for $[x]$ and $[y]$, the cross-cap is given explicitly, as

$$\chi_h(x, y) = [x] *_3 b_h(y) + b_h(y) *_3 [x],$$

and the result (23.8) is established.

The homologies (23.6), (23.7) and (23.8) give a complete list of the requisite properties of the cycle $\mu_h(x)$. Indeed the last one implies, by properties of the cross-cap, that

$$\begin{aligned} \mu_h(x \tau y \tau z) &\sim 0, & hx = hy = hz &= 0, \\ \mu_h(kx \tau y) &\sim \mu_{hk}(x \tau y), & h k x &= h y = 0. \end{aligned}$$

Comparison with the defining relations of the generator $\omega_h(x)$ of $\Omega(\Pi)$ shows that θ_7 is a homomorphism on $\Omega(\Pi)$.

If Π is a p -cyclic group $Z_h(c)$, then $\Pi/3\Pi$ is cyclic of order 3, generator $c + 3\Pi$, if $h = \infty$ or $h = 3^e$, and is otherwise zero. On the other hand, we know by (13.17) that

$$\Omega(Z_h(c)) = Z_h(\omega_h(c)), h \text{ finite}; \quad \Omega(Z_\infty(c)) = 0.$$

Since these results compare exactly under θ_7 with those tabulated above for $H_7(Z_h(c), 3)$, we conclude that θ_7 is an isomorphism in this case.

The functor $\Pi/3\Pi$ is additive (cross-effects zero), and both functors $\Omega(\Pi)$ and $H_7(\Pi, 3)$ are of degree 2. Hence it suffices to consider cross-effects involving two groups Π_1 and Π_2 . The cross-effect for $\Omega(\Pi)$ is given as $\text{Tor}(\Pi_1, \Pi_2)$ under the map $g_\Omega \tau_h(x_1, x_2) = \omega_h((x_1, 0) \tau (0, x_2))$ of Theorem 13.2. For the functor $H_7(\Pi, 3)$ the cross-effect given by Theorem 20.6 is also $\text{Tor}(\Pi_1, \Pi_2)$. Hence to conclude the proof it suffices to verify, using (23.8), the commutativity in the diagram

$$\begin{array}{ccc} \text{Tor}(\Pi_1, \Pi_2) & \xrightarrow{g_\Omega} & \Omega(\Pi_1 | \Pi_2) \\ \downarrow \text{Tor}(\theta_3, \theta_3) & & \downarrow \theta'_7 \\ \text{Tor}(H_3(\Pi_1, 3), H_3(\Pi_2, 3)) & \rightarrow & H_7(\Pi_1 | \Pi_2, 3), \end{array}$$

where θ_3 is the identification $\Pi_i = H_3(\Pi_i, 3)$.

PROOF OF THEOREM 23.4. We already know that the map $y \rightarrow [y || y]$ induces a homomorphism on $\Pi/2\Pi$. On the other hand, the cycle $\sigma(u)$ is suspended from $A(\Pi, 2)$. The process of splitting variables then trivially yields the homology $\sigma(u + v) \sim \sigma(u) + \sigma(v)$ for $u, v \in {}_3\Pi$. This completes the proof that the mapping θ_8 of the Theorem is a homomorphism.

For cyclic groups the tables above make it clear that θ_8 is an isomorphism. The proof is completed by showing that θ_8 is an isomorphism on cross-effects by consideration of the commutative diagram

$$\begin{array}{ccc} \Pi_1 \otimes (\Pi_2/2\Pi_2) + \Pi_2 \otimes (\Pi_1/2\Pi_1) & \xrightarrow{g} & T(\Pi_1 | \Pi_2) \\ \downarrow & & \downarrow \\ H_3(\Pi_1, 3) \otimes H_5(\Pi_2, 3) + H_3(\Pi_2, 3) \otimes H_5(\Pi_1, 3) & \rightarrow & H_8(\Pi_1 | \Pi_2, 3) \end{array}$$

where $T(\Pi) = \Pi \otimes (\Pi/2\Pi)$, and g is the obvious map.

24. The ring $H(\Pi, 4)$ in low dimensions

In this section we determine the homology ring $H(\Pi, 4)$ in dimensions up to 8 inclusive. The suspended cycles $\gamma_2(x, 2)$, $\gamma_4(x, 2)$, $\gamma_6(x, 2)$, $\zeta(s, 2)$, and $\sigma(u)$ from $H(\Pi, 3)$ are still present, but the cycles $\mu_h(x)$ and $\chi_h(x, y)$ used there suspend into zero. In addition we have the cycles

$$\begin{aligned} (24.1) \quad \gamma_8(x, 4) &= [x |_4 x], \\ \zeta(s, 4) &= (1/2)\partial[s, s |_4 s, s], \quad 2s = 0, \end{aligned}$$

and the cross-cap $\chi_h(x, y)$ of dimension 9 formed from the cycles $[x]$ and $[y]$ for $hx = hy = 0$. Direct computation for p -cyclic groups yields

$$\begin{aligned}
 H_7(Z_h(c), 4) &= Z_2(\zeta(2^{e-1}c, 2)), & h &= 2^e, \\
 &= 0, & h &= \infty \text{ or } (h, 2) = 1. \\
 H_8(Z_h(c), 4) &= Z_\infty(\gamma_8(c, 4)) + Z_3(\gamma_6(c, 2)), & h &= \infty, \\
 &= Z_h(\gamma_8(c, 4)), & (h, 6) &= 1, \\
 &= Z_{2^{e+1}}(\gamma_8(c, 4)), & h &= 2^e. \\
 &= Z_{3^e}(\gamma_8(c, 4)) + Z_3(\gamma_6(c, 2)), & h &= 3^e. \\
 H_9(Z_h(c), 4) &= Z_2(\zeta(2^{e-1}c, 4)), & h &= 2^e, \\
 &= Z_3(\sigma(3^{e-1}c)), & h &= 3^e. \\
 &= 0, & h &= \infty \text{ or } (h, 6) = 1
 \end{aligned}$$

By the suspension theorem and the stable result (Theorem 23.1) for $H(\Pi, 3)$ we have isomorphisms

$$\begin{aligned}
 \Pi &\cong H_4(\Pi, 4), \\
 0 &= H_5(\Pi, 4), \\
 \theta_6: \Pi/2\Pi &\cong H_6(\Pi, 4).
 \end{aligned}$$

THEOREM 24.1. *There is a natural isomorphism*

$$(24.2) \quad \theta_7: {}_2\Pi \cong H_7(\Pi, 4) \quad (\text{stable}),$$

defined by setting

$$\theta_7(s) = \zeta(s, 2), \quad s \in {}_2\Pi$$

The stable result (24.2) formulated in this theorem has been obtained by geometrical means by G. W. Whitehead [17, Theorem 4].

THEOREM 24.2. *There is a natural isomorphism*

$$(24.3) \quad \theta_8: \Gamma_4(\Pi) + \Pi/3\Pi \cong H_8(\Pi, 4)$$

defined by setting

$$\theta_8\gamma(x) = \gamma_8(x, 4), \quad \theta_8(x + 3\Pi) = \gamma_6(x, 2).$$

THEOREM 24.3. *There is a natural isomorphism*

$$\theta_9: R(\Pi) + {}_3\Pi \cong H_9(\Pi, 4),$$

where $R(\Pi)$ is the functor used in Theorem 22.1 to describe $H_5(\Pi, 2)$. This map is defined by setting

$$\begin{aligned}
 \theta_9\tau_h(x, y) &= \chi_h(x, y), & hx &= hy = 0, \\
 \theta_9\gamma_2(s) &= \zeta(s, 4), & 2s &= 0, \\
 \theta_9u &= \sigma(u), & 3u &= 0.
 \end{aligned}$$

The only non-trivial product up to dimension 9 is given by Theorem 17.1 as

$$(24.4) \quad [x] \star_4 [y] \sim \gamma_8(x \tau y, 4).$$

To determine the suspension $H_6(\Pi, 3) \rightarrow H_7(\Pi, 4)$ observe that the only relevant added cell is the cell $[x \mid_4 y]$ of dimension 8, with boundary $[x] \star_3 [y]$. By Theorem 23.2, these cycles correspond exactly to the generators $x \wedge y$ of $\Lambda_2(\Pi)$. Therefore the combined suspension map

$${}_2\Pi + \Lambda_2(\Pi) \xrightarrow{\theta_6} H_6(\Pi, 3) \xrightarrow{S} H_7(\Pi, 4) \xrightarrow{\theta_7^{-1}} {}_2\Pi$$

may be described as the identity on ${}_2\Pi$ and the zero map on $\Lambda_2(\Pi)$. This remark suffices to prove Theorem 24.1.

Similarly, the combined suspension map

$$\Omega(\Pi) + \Pi/3\Pi \rightarrow \Gamma_4(\Pi) + \Pi/3\Pi$$

is the zero map on $\Omega(\Pi)$ and the identity on $\Pi/3\Pi$.

PROOF OF THEOREM 24.2. We already know from the proof of Theorem 23.3 that θ_8 is a homomorphism on $\Pi/3\Pi$, while by Theorem 17.1 it is a homomorphism on $\Gamma_4(\Pi)$. For cyclic groups the previous determination of $\Gamma_4(Z_h(c))$ and the tables of $H_8(Z_h(c), 4)$ show that θ_8 is an isomorphism. The cross-effects may be treated exactly as for $H_6(\Pi, 3)$.

In accordance with Theorem 21.1, note that the mapping θ_8 on $\Gamma_4(\Pi)$ is, indeed, a monomorphism onto a direct summand of $H_8(\Pi, 4)$.

As for Theorem 24.3 the proof is identical with that of Theorem 22.1 on $H_5(\Pi, 2)$ except for the presence of the direct summand ${}_3\Pi$ caused by the cycle $\sigma(u)$. This summand has no influence on cross-effects.

25. The rings $H(\Pi, 5)$ and $H(\Pi, 6)$

The first ring $H(\Pi, 5)$ will be determined up to dimension 10, inclusive, while $H(\Pi, 6)$ will be determined up to dimension 11. Since the previous results up to $H_7(\Pi, 4)$ are stable, we need only determine $H_9(\Pi, 5)$, $H_{10}(\Pi, 5)$, and $H_{11}(\Pi, 6)$.

THEOREM 25.1. *There is a natural isomorphism*

$$\theta_9: \Pi/2\Pi + \Pi/3\Pi \cong H_9(\Pi, 5) \quad (\text{stable}),$$

defined by setting

$$\theta_9(x + 2\Pi) = \gamma_8(x, 4), \quad \theta_9(x + 3\Pi) = \gamma_6(x, 2).$$

THEOREM 25.2. *There is a natural isomorphism*

$$\theta_{10}: \Lambda_2(\Pi) + {}_2\Pi + {}_3\Pi \cong H_{10}(\Pi, 5)$$

defined by setting

$$\begin{aligned} \theta_{10}(x \wedge y) &= [x] \star_5 [y], \\ \theta_{10}(s) &= \zeta(s, 4), & 2s &= 0, \\ \theta_{10}(u) &= \sigma(u), & 3u &= 0. \end{aligned}$$

Here $\Lambda_2(\Pi)$ is $\Pi \otimes \Pi$ modulo the diagonal, as in Theorem 23.2.

THEOREM 25.3. *There is a natural isomorphism*

$$\theta_{11}: {}_2\Pi + {}_3\Pi \cong H_{11}(\Pi, 6) \quad (\text{stable}),$$

defined for $s \in {}_2\Pi$ and $u \in {}_3\Pi$ by setting

$$\theta_{11}(s) = \zeta(s, 4), \quad \theta_{11}(u) = \sigma(u).$$

PROOF of THEOREM 25.1. Consider the diagram

$$\begin{array}{ccc} H_8(\Pi, 4) & \xrightarrow{S} & H_9(\Pi, 5) \\ \uparrow \theta_8 & & \uparrow \theta_9 \\ \Gamma_4(\Pi) + \Pi/3\Pi & \xrightarrow{s+i} & \Pi/2\Pi + \Pi/3\Pi, \end{array}$$

where $s: \Gamma_4(\Pi) \rightarrow \Pi/2\Pi$ is the map $\gamma(x) \rightarrow x + 2\Pi$ of §13. The only relevant added cell of $A(\Pi, 5)$ is the cell $[x \mid {}_s y]$ of dimension 10 and boundary

$$-[x]_* [y] \sim -\gamma_8(x \tau y, 4).$$

These homology classes of these cycles span the kernel of S and are mapped by θ_8^{-1} onto the kernel of s . This proves the diagram commutative and shows that θ_9 is an isomorphism.

PROOF of THEOREM 25.2. To show that θ_{10} is a homomorphism, we need only examine the summand ${}_2\Pi$. By Theorem 17.4 the deviation $\zeta({}_s \tau t, 4)$ is a sum of products in $A(\Pi, 4)$, hence is homologous to zero in $A(\Pi, 5)$; this proves θ_{10} a homomorphism.

For cyclic groups direct computation yields

$$\begin{aligned} H_{10}(Z_h(c), 5) &= Z_2(\zeta(2^{e-1}c, 4)), & h &= 2^e, \\ &= Z_3(\sigma(3^{e-1}c)), & h &= 3^e, \\ &= 0, & h &= \infty \text{ or } (h, 6) = 1. \end{aligned}$$

This shows at once that θ_{10} is an isomorphism for cyclic groups. The cross-effect diagram is the same as that of Theorem 23.2 for $H_6(\Pi, 3)$.

Theorem 25.3 is immediately obtained from Theorem 25.2 by suspension.

26. Cohomology in $H(\Pi, 2)$

This and the succeeding section are devoted to the determination in low dimensions of the cohomology groups $H(\Pi, n; G)$ with coefficients in an arbitrary abelian group G . It is well known that for each cocycle w in $Z^1(\Pi, 1; G)$ the map $x \rightarrow w[x]$ is a homomorphism $\Pi \rightarrow G$ and that this correspondence yields an isomorphism

$$(26.1) \quad H^1(\Pi, 1; G) \cong \text{Hom}(\Pi, G) \quad (\text{stable});$$

since the result is stable, it applies also to $H^2(\Pi, 2; G)$. In the next dimension each cocycle $w \in Z^2(\Pi, 1; G)$ or $Z^3(\Pi, 2; G)$ is defined by its values $f(x, y) =$

$w[x, y]$. The function f is a factor set; the construction of the group extension of G by Π with the factor set f yields the well known isomorphisms

$$(26.2) \quad H^2(\Pi, 1; G) \cong \text{Ext}(\Pi, G),$$

$$(26.3) \quad H^3(\Pi, 2; G) \cong \text{Extabel}(\Pi, G), \quad (\text{stable}),$$

with the groups of central respectively abelian group extensions.

Since $H_3(\Pi, 2) = 0$ and $H_4(\Pi, 2) \cong \Gamma_4(\Pi)$, the universal coefficient theorem yields at once

THEOREM 26.1. *For abelian groups Π and G there is an isomorphism*

$$(26.4) \quad H^4(\Pi, 2; G) \cong \text{Hom}(\Gamma_4(\Pi); G)$$

obtained by assigning to each cocycle $Z^4(\Pi, 2; G)$ the quadratic function q on Π to G with $q(x) = w[x \parallel x]$.

This result can be used to establish quickly a group-theoretic theorem due to Hughes [12]:

THEOREM 26.2. *For abelian groups Π and G a bilinear function d on $\Pi \times \Pi$ to G has $d(x, x) = 0$ for all $x \in \Pi$ if and only if there is a cocycle $f \in Z^2(\Pi, 1; G)$ with*

$$(26.5) \quad d(x, y) = f(x, y) - f(y, x), \quad x, y \in \Pi.$$

PROOF. In the complex $A_2 = A(\Pi, 2)$ we consider the subcomplex $SA_1 = SA(\Pi, 1)$ obtained by suspension. The relative cohomology sequence

$$H^3(A_2, SA_1; G) \rightarrow H^3(A_2; G) \rightarrow H^3(SA_1; G) \xrightarrow{\delta} H^4(A_2, SA_1; G) \xrightarrow{j^*} H^4(A_2; G)$$

is exact. Now SA_1 and A_2 have the same cells in dimensions 2 and 3, hence $H^3(A_2, SA_1; G) = 0$. In dimension 4 inspection of the boundary formulas in $A(\Pi, 2)$ shows that the assignment to each cocycle $Z^4(A_2, SA_1; G)$ of the bilinear function $d(x, y) = w[x \parallel y]$ provides an isomorphism

$$H^4(A_2, SA_1; G) \cong \text{Hom}(\Pi \otimes \Pi, G).$$

Using (26.2) and (26.3) the exact sequence above now becomes

$$0 \rightarrow \text{Extabel}(\Pi, G) \rightarrow \text{Ext}(\Pi, G)$$

$$\xrightarrow{\delta} \text{Hom}(\Pi \otimes \Pi, G) \xrightarrow{j^*} \text{Hom}(\Gamma_4(\Pi), G).$$

The proposition that image $\delta = \text{kernel } j^*$ gives the theorem.

Symmetric homology and cohomology groups related to $A(\Pi, 1)$ may be introduced as the groups of the quotient complex $A(\Pi, 1)/A^*(\Pi, 1)$, where $A^*(\Pi, 1)$ denotes the subcomplex spanned by all products $a \ast_1 b$ of chains a and b of positive dimension.

THEOREM 26.3.

$$H_2(A(\Pi, 1), A^*(\Pi, 1)) = 0, \quad H_3(A(\Pi, 1), A^*(\Pi, 1)) = 0, \quad \text{and}$$

$$H^3(A(\Pi, 1), A^*(\Pi, 1); G) = 0.$$

For example, the third conclusion asserts that a three dimensional cocycle $w \in Z^3(\Pi, 1)$ with $w(x*(y, z)) = 0$ for all x, y , and z is a coboundary $w = \delta v$, where $v(x*y) = 0$. This conclusion may also be had by a suitable argument on group extensions.

For the proof consider the diagram

$$\begin{array}{ccccccc}
 & & & & H_4(A_2, SA_1) & & \\
 & & & & \downarrow \partial & & \\
 H_2(SA_1^*) & \leftarrow & H_3(SA_1, SA_1^*) & \xleftarrow{j} & H_3(SA_1) & \xleftarrow{i} & H_3(SA_1^*) \\
 & & & & \downarrow & & \\
 & & & & H_3(A_2) & &
 \end{array}$$

Since $H_2(SA_1^*) = H_1(A_1^*) = 0$, j is an epimorphism. Since $H_3(A_2) = 0$, ∂ is an epimorphism. However, the 4-cells of A_2/SA_1 are the cells $[x \parallel y]$, and $\partial[x \parallel y] = [x]*[y]$; hence image ∂ = image i . Therefore i also is an epimorphism. Since both i and j are epimorphisms, $H_3(SA_1, SA_1^*) = H_2(A_1, A_1^*) = 0$.

Next take $w \in Z^3(A_1, A_1^*; G)$. We may then define a cochain $u \in Z^4(A_2; G)$ by setting

$$u[x, y, z] = w[x, y, z], \quad u[x \parallel y] = 0.$$

The symmetry of w and the boundary formulas in $A(\Pi, 2)$ shows that u is a cocycle in A_2 . But the quadratic function q assigned to this cocycle under the isomorphism (26.4) is zero; therefore, u is a coboundary and there exists a cochain $h \in C^3(A_2, G)$ with $\delta h = u$; that is, a function $h(x, y) \in G$ with $\delta h = w$ in A_1 and $h(x*y) = 0$. This proves that $H^3(A_1, A_1^*; G) = 0$ for all G , and this result implies that $H_3(A_1, A_1^*) = 0$.

A partial extension of this result to higher dimensions is given by

THEOREM 26.4. *If Π is free or if $q > 1$ is even then*

$$(26.6) \quad H_q(A^*(\Pi, 1)) \rightarrow H_q(A(\Pi, 1))$$

is an epimorphism, or, equivalently,

$$H_q(A(\Pi, 1)) \rightarrow H_q(A(\Pi, 1), A^*(\Pi, 1))$$

is zero.

If Π is cyclic the theorem is true since the complexes $M(h, 1)$ show that the group $H_q(A(\Pi, 1)) = 0$ precisely in the cases cited.

Suppose now that the theorem is valid for two groups Π_1 and Π_2 . Applying the map

$$gf: A(\Pi_1 + \Pi_2, 1) \rightarrow A(\Pi_1 + \Pi_2, 1)$$

from the direct sum theorem, we have for $x_i \in \Pi_1$, $y_i \in \Pi_2$,

$$gf[x_1 + y_1, \dots, x_m + y_m]$$

$$= [x_1, \dots, x_m] + [y_1, \dots, y_m] + \sum_{i=1}^{m-1} [x_1, \dots, x_i] * [y_{i+1}, \dots, y_m].$$

The sum here lies in the complex $A^*(\Pi_1 + \Pi_2, 1)$. Since gf is homotopic to the identity, we thus have for any cycle $z \in Z_q(A(\Pi, 1))$

$$z \sim gfz \sim A(\nu^{(1)}, 1)z + A(\nu^{(2)}, 1)z \quad (\text{mod } A^*),$$

for $\nu^{(i)}$ as in §6. By the assumption of the theorem for Π_1 and Π_2 , each of the cycles on the right is homologous to a symmetric cycle. This proves the result for $\Pi_1 + \Pi_2$, and hence for any finitely generated group. The general case is then obtained by direct limits.

Without the restrictions made in its hypotheses this theorem is not valid, in general, even for cyclic groups. Indeed, our contraction of the complex $A(Z_h(c), 1)$ to $M(h, 1)$ may be used to show that if $q = 2p^e + 1$, where p is a prime dividing h , then (26.6) is not an epimorphism.

For use in the next section we establish the following result for Extabel.

THEOREM 26.5. *If p is a prime and Π an abelian group with $p\Pi = 0$, there is a natural isomorphism*

$$(26.7) \quad \rho: \text{Extabel}(\Pi, G) \cong \text{Hom}(\Pi, G/pG).$$

If the group of abelian group extensions is represented as $H^3(\Pi, 2; G)$, then the isomorphism ρ is defined by assigning to each cocycle $f \in Z^3(\Pi, 2; G)$ the homomorphism h defined for $r \in \Pi$ as

$$(26.8) \quad h(r) = f\left(\sum_{i=0}^{p-1} [r, ir]\right) + pG.$$

PROOF. If we use the cap $b_p(r) = \sum [r, ir]$ for r in $A(\Pi, 2)$, with $\partial b_p(r) = p[r]$, the definition may be written

$$(26.9) \quad h(r) = fb_p(r) + pG.$$

By splitting variables in $b_p(r + s)$ we obtain in $A(\Pi, 1)$ the homology

$$(26.10) \quad b_p(r + s) - b_p(r) - b_p(s) + p[r, s] \sim (p(p-1)/2)[r*s].$$

The right side is homologous to 0 in $A(\Pi, 2)$. Hence we get $h(r + s) = h(r) + h(s)$, so that h is indeed a homomorphism. From (26.8) we also compute that h is zero if f is a coboundary. The assignment $\rho f = h$ thus provides a homomorphism for the groups in (26.7).

We now give an alternative (non-natural) description of ρ . Since every element of Π has prime order p , the group Π can be written as a direct sum of cyclic groups of order p with generators r_α . Given any abelian group extension $\phi: E \rightarrow \Pi$ of G by Π we choose to each r_α a representative $u_\alpha \in E$ with $\phi u_\alpha = r_\alpha$, and then to any element $r = \sum m_\alpha r_\alpha$ with integers m_α , $0 \leq m_\alpha < p$ the representative $u(r) = \sum m_\alpha u_\alpha$ in E with $\phi u(r) = r$. The factor set f for the extension E relative to this choice of basis is then determined by the elements $pu_\alpha = g_\alpha \in G$ as

$$(26.11) \quad f\left(\sum m_\alpha r_\alpha, \sum n_\alpha r_\alpha\right) = \sum_\beta g_\beta,$$

where the sum on the right is taken over all indices β with $m_\beta + n_\beta \geq p$. Since by (26.10) we have a homology

$$b_p(\sum m_\alpha r_\alpha) \sim \sum m_\alpha b_p(r_\alpha) \pmod{pA(\Pi, 2)};$$

the homomorphism $h = \rho f$ is given by the formula

$$h(\sum m_\alpha r_\alpha) = \sum m_\alpha g_\alpha + pG.$$

It follows that ρ is a monomorphism. For, if $h = 0$, then each $g_\alpha \in pG$, and the obvious new choice of representatives in E makes f cohomologous to 0. It also follows that ρ is an epimorphism. For, given $h: \Pi \rightarrow G/pG$ we have elements g_α with $hr_\alpha = g_\alpha + pG$. Defining f by (26.11) above, we obtain an f with $\rho f = h$.

In the case $p = 2$, formula (26.8) for h takes the simple form

$$(26.12) \quad h(r) = f(r, r) + 2G.$$

COROLLARY 26.6. *For each bilinear function $b: \Pi \otimes \Pi \rightarrow G$ such that $b(x \otimes x) \in 2G$, there exists a function a on Π to $G/2G$*

$$a(x \tau y) = b(x, y) + 2G.$$

PROOF. Consider the bilinear function $b': \Pi/2\Pi \otimes \Pi/2\Pi \rightarrow G/2G$ induced by b , and regard b' as a cochain $b' \in C^2(\Pi/2\Pi, 2; G/2G)$. Bilinearity of b' implies that $b'\partial[x, y, z] = 0$. The added condition $b'(x, x) = 0$ implies that

$$\begin{aligned} b'\partial[x \parallel y] &= b'([x, y] - [y, x]) = b'(x, y) + b'(y, x) \\ &= b'(x + y, x + y) = 0. \end{aligned}$$

Hence $\delta b' = 0$. Apply the isomorphism ρ of the theorem to the cohomology class of b' . Since $b'(x, x) = 0$, it follows that $\rho b' = 0$, so that $b' = \delta a'$ for some $a' \in C^2(\Pi/2\Pi, 2; G/2G)$. This yields the desired function a .

27. Higher cohomology groups

In this section we determine the cohomology groups $H^5(\Pi, 3)$, $H^6(\Pi, 3)$, $H^7(\Pi, 4)$, $H^8(\Pi, 4)$, and $H^9(\Pi, 5)$, with any coefficient group G . Just as in Theorem 26.1, we have an isomorphism

$$(27.1) \quad H^5(\Pi, 3; G) \cong \text{Hom}(\Pi/2\Pi; G) \quad (\text{stable}),$$

obtained by assigning to each cocycle $w \in Z^5(\Pi, 3; G)$ the homomorphism h with $h(x + 2\Pi) = w([x \parallel x])$.

The group $H^6(\Pi, 3; G)$ will be determined in terms of a group $L(\Pi, G)$ defined as the group of all pairs of functions (a, b) , where $b: \Pi \otimes \Pi \rightarrow G$ and a is a function on Π to $G/2G$, both subject to the conditions

$$(27.2) \quad a(x \tau y) = b(x, y) + 2G, \quad x, y \in \Pi,$$

$$(27.3) \quad b(x, x) = 0.$$

Each homomorphism $a: \Pi \rightarrow G/2G$ yields such a pair $(a, 0)$; each such pair (a, b) determines a homomorphism $b: \Lambda_2(\Pi) \rightarrow G$; these correspondences yield the sequence

$$(27.4) \quad 0 \rightarrow \text{Hom}(\Pi, G/2G) \xrightarrow{\lambda} L(\Pi, G) \xrightarrow{\kappa} \text{Hom}(\Lambda_2(\Pi), G) \rightarrow 0.$$

Corollary 26.6 shows that κ is an epimorphism; it then readily follows that the sequence is exact. It will follow from Theorem 27.1, and it can be proved directly, that the image of λ is a direct summand of $L(\Pi, G)$. We do not use this fact, and we cannot exhibit a natural decomposition of $L(\Pi, G)$.

THEOREM 27.1. *There is a natural isomorphism*

$$(27.5) \quad \theta^6: H^6(\Pi, 3; G) \cong L(\Pi, G) + \text{Hom}({}_2\Pi, G)$$

obtained by assigning to each cocycle $w \in Z^6(\Pi, 3, G)$ the element $(a, b) \in L(\Pi, G)$ and the homomorphism $c: {}_2\Pi \rightarrow G$, all determined by

$$(27.6) \quad a(x) = w[x \mid_3 x] + 2G,$$

$$(27.7) \quad b(x, y) = w([x] *_3 [y]),$$

$$(27.8) \quad c(s) = w(\zeta(s, 2)), \quad s \in {}_2\Pi.$$

PROOF. We first show that θ^6 is a homomorphism. The homology $\zeta(r \tau s, 2) \sim 0$, for $r, s \in {}_2\Pi$ in $A(\Pi, 3)$ shows that $c(s)$, as defined in (27.8), is a homomorphism. Since the homology class of $[x] *_3 [y]$ is additive in its arguments, the function b defined by (27.7) is bilinear. Finally, the homology modulo 2

$$\gamma_6(x \tau y, 3) \sim [x] *_3 [y], \quad \gamma_6(x, 3) = [x \mid_3 x]$$

of (17.4) shows that the functions a and b defined by (27.6) and (27.7) satisfy the defining equation (27.2) for $L(\Pi, G)$.

We now consider the following diagram, where E is short for the functor Extabel ,

$$\begin{array}{ccc} \begin{array}{c} 0 \\ \downarrow \\ E(H_3(\Pi, 3), G) \cong E(\Pi/2\Pi, G) \\ \downarrow \beta' \\ H^6(\Pi, 3; G) \\ \downarrow \alpha' \\ \text{Hom}(H_6(\Pi, 3), G) \\ \downarrow \\ 0 \end{array} & \begin{array}{c} \xrightarrow{\rho} \\ \\ \xrightarrow{\theta^6} \\ \\ \xrightarrow{\text{Hom}(\theta_6, G)} \end{array} & \begin{array}{c} 0 \\ \downarrow \\ \text{Hom}(\Pi, G/2G) \\ \downarrow \lambda \\ L(\Pi, G) + \text{Hom}({}_2\Pi, G) \\ \downarrow \kappa + i \\ \text{Hom}(\Lambda_2(\Pi), G) + \text{Hom}({}_2\Pi, G) \\ \downarrow \\ 0 \end{array} \end{array}$$

Here the left hand column is the exact sequence for the universal coefficient theorem for the complex $A(\Pi, 3)$. The map θ_6 appearing in the lower horizontal map is the homology isomorphism established in (23.4), while ρ is the isomorphism given by Theorem 26.5. The right hand column is the exact sequence trivially derived from (27.4) above by adjoining the direct summand $\text{Hom}({}_2\Pi, G)$ to the second and third terms. Since the top and bottom horizontal maps are known to be isomorphisms, the fact that the map θ^6 under consideration is an isomorphism will be established once we show commutativity of the diagram. The lower square is trivially commutative.

To prove the upper square commutative use the isomorphism $H_5(\Pi, 3) \cong \Pi/2\Pi$ to obtain the exact sequence

$$(27.9) \quad 0 \rightarrow B_5(\Pi, 3) \rightarrow Z_5(\Pi, 3) \rightarrow \Pi/2\Pi \rightarrow 0.$$

We shall compute a factor set f_0 of this extension. Choose representatives $v(r) \in \Pi$ for $r \in \Pi/2\Pi$, with $v(0) = 0$. Then $[v(r) \parallel v(r)]$ is a system of representatives in $Z_5(\Pi, 3)$. Then for $r, s \in \Pi/2\Pi$

$$f_0(r, s) = [v(r) \parallel v(r)] + [v(s) \parallel v(s)] - [v(r+s) \parallel v(r+s)].$$

Taking $r = s$ and noting that $v(r+s) = 0$, since $r+s = 0$ in $\Pi/2\Pi$, we obtain

$$(27.10) \quad f_0(r, r) = 2[v(r) \parallel v(r)] = -\partial[v(r) \mid_3 v(r)].$$

Next consider an element $e \in \text{Ext}(\Pi/2\Pi, G)$. It is well known that a map

$$\phi: B_5(\Pi, 3) \rightarrow G$$

exists, such that ϕf_0 is a cocycle in the class e . The homomorphism $h: \Pi/2\Pi \rightarrow G/2G$ associated with e according to the map then is

$$\begin{aligned} h(r) &= \phi f_0(r, r) + 2G = -\phi \partial[v(r) \mid_3 v(r)] + 2G \\ &= \phi \partial[v(r) \mid_3 v(r)] + 2G. \end{aligned}$$

Applying λ to h we find the element $(a, 0) \in L(\Pi, G)$ with a given by

$$(27.11) \quad a(x) = \phi \partial[x \mid_3 x] + 2G, \quad x \in \Pi.$$

Now apply the map β' of the universal coefficient theorem. To the extension e corresponding to the map ϕ there is associated the cohomology class $\beta'e$ given by the cocycle $w = \phi \partial$. An application of θ^6 to this cocycle yields the element

$(a, 0) \in L(\Pi, G)$ with

$$a(x) = w[x \mid_3 x] + 2G = \phi \partial[x \mid_3 x] + 2G$$

exactly as above. This proves the commutativity of the upper square.

THEOREM 27.2. *There is a natural isomorphism*

$$(27.12) \quad \theta^7: H^7(\Pi, 4; G) \cong \text{Hom}(\Pi, G/2G) + \text{Hom}({}_2\Pi, G) \quad (\text{stable})$$

obtained by assigning to each cocycle $w \in Z^7(\Pi, 4; G)$ the pair of homomorphisms $a: \Pi \rightarrow G/2G$ and $c: {}_2\Pi \rightarrow G$ determined by

$$(27.13) \quad a(x) = w[x \mid_3 x] + 2G,$$

$$(27.14) \quad c(s) = w(\zeta(s, 2)), \quad s \in {}_2\Pi.$$

This can be proved either by repeating the previous argument, with the obvious simplification, or by comparing the map θ^7 with the previous map θ^6 , using suspension and the monomorphisms $\kappa: \text{Hom}(\Pi, G/2G) \rightarrow L(\Pi, G)$.

THEOREM 27.3. *There is a natural isomorphism*

$$(27.15) \quad \theta^8: H^8(\Pi, 4; G) \\ \cong \text{Hom}({}_2\Pi, G/2G) + \text{Hom}(\Gamma_4(\Pi), G) + \text{Hom}(\Pi/3\Pi, G),$$

obtained by assigning to each cocycle $w \in Z^8(\Pi, 4; G)$ the triple of homomorphisms $a: {}_2\Pi \rightarrow G/2G$, $b: \Gamma_4(\Pi) \rightarrow G$, $c: \Pi/3\Pi \rightarrow G$ determined by

$$(27.16) \quad a(x) = w\gamma_6(x, 2) = w([x \parallel x \parallel x]),$$

$$(27.17) \quad b(x) = w\gamma_8(x, 4) = w([x \mid_4 x]),$$

$$(27.18) \quad c(s) = w([s, s \parallel s, s]) + 2G, \quad s \in {}_2\Pi.$$

PROOF. We first observe that θ^8 is a homomorphism. In view of Theorem 24.2, it is clear that a and b as defined are homomorphisms. As to c , we observe that $\nu(s) = [s \mid s \parallel s, s]$ has $\partial\nu(s) = 2\zeta(s, 2)$ by the definition of ζ . Hence $\nu(s)$ is a cycle, modulo 2. The process of splitting variables applied to the cycle $\nu(s + t)$ then yields the homology

$$\nu(s, t) \sim \nu(s) + \nu(t) + [s, s]_*[t, t],$$

modulo 2. But $\partial[s, s] = 2[s] \equiv 0 \pmod{2}$, and hence

$$\partial[s, s \mid_3 t, t] \equiv [s, s]_*[t, t] \pmod{2}.$$

There results the homology $\nu(s + t) \sim \nu(s) + \nu(t)$, mod 2, which proves that c as defined by (27.18) is indeed a homomorphism.

Using the isomorphism

$$\rho: \text{Extabel}({}_2\Pi, G) = \text{Hom}({}_2\Pi, G/2G)$$

obtained from Theorem 26.5, we now set up a diagram involving the universal coefficient theorem parallel to the diagram of the preceding section. As before, the proposition reduces to the commutativity of the upper square of this diagram. As before, consider the exact sequence

$$0 \rightarrow B_7(\Pi, 4) \rightarrow Z_7(\Pi, 4) \rightarrow {}_2\Pi \rightarrow 0.$$

The factor set f_0 then satisfies

$$f_0(s, s) = 2\zeta(s, 2) = \partial[s, s \parallel s, s], \quad s \in {}_2\Pi.$$

The remainder of the proof is just as before.

By suspension of this result we obtain

THEOREM 27.4. *There is a natural isomorphism*

$$(27.19) \quad \theta^9: H^9(\Pi, 5; G) \\ \cong \text{Hom}({}_2\Pi, G/2G) + \text{Hom}(\Pi/2\Pi, G) + \text{Hom}(\Pi/3\Pi, G) \quad (\text{stable}),$$

obtained by assigning to each cocycle $w \in Z^9(\Pi, 5; G)$ the triple (a, b, c) of homomorphisms defined by

$$(27.20) \quad a(x) = w\gamma_6(x, 2) = w([x \parallel x \parallel x]),$$

$$(27.21) \quad b(x + 2\Pi) = w\gamma_8(x, 4) = w([x \mid_4 x]),$$

$$(27.22) \quad c(s) = w([s, s \parallel s, s]) + 2G, \quad s \in {}_2\Pi.$$

THEOREM 27.5. *There is a natural isomorphism*

$$(27.23) \quad \theta^{10}: H^{10}(\Pi, 5; G) \cong L(\Pi, G) + \text{Hom}(\Pi, G/3G) + \text{Hom}({}_2\Pi + {}_3\Pi, G),$$

where $L(\Pi, G)$ is the functor introduced in (27.2) and (27.3). Here θ^{10} assigns to each cocycle $w \in Z^{10}(\Pi, 5; G)$ the element (a, b) in $L(\Pi, G)$ defined by setting

$$a(x) = w[x \mid_5 x] + 2G, \quad b(x, y) = w([x]_*[y])$$

and the homomorphisms

$$a': \Pi \rightarrow G/3G, \quad c: {}_2\Pi \rightarrow G, \quad d: {}_3\Pi \rightarrow G$$

defined, for $x \in \Pi$, $s \in {}_2\Pi$, and $u \in {}_3\Pi$ by

$$a'(x) = -w[x \mid_3 x \parallel x] + 3G, \quad c(s) = w(\zeta(s, 4)), \quad d(u) = w(\sigma(u)).$$

The proof is parallel to that of Theorem 27.1 and involves the functor $L(\Pi, G)$ in much the same way. To show that θ^{10} is indeed a homomorphism, we use the homology modulo 2,

$$\gamma_{10}(x \tau y, 5) \sim [x]_*[y], \quad \gamma_{10}(x, 5) = [x \mid_5 x].$$

This shows that $a(x \tau y) = b(x, y) + 2G$, and hence that θ^{10} is a homomorphism as of $L(\Pi, G)$. We next observe the boundary formula

$$-\partial[x \mid_3 x \parallel x] = 3[x \parallel x \parallel x] = 3\gamma_6(x, 2).$$

Hence $\nu(x) = [x \mid_3 x \parallel x]$ is a cycle modulo 3 in $A(\Pi, 3)$. Splitting variables in $A(\Pi, 5)$, where $\nu(x)$ is a suspended cycle, gives the homology $\nu(x + y) \sim \nu(x) + \nu(y)$, mod 3. This proves that a' as defined is a homomorphism of Π into $G/3G$. A similar but simpler argument proves that c and d are homomorphisms.

We now construct a diagram like that in the proof of Theorem 27.1, the top row being

$$E(\Pi/2\Pi + \Pi/3\Pi, G) \cong E(\Pi/2\Pi, G) + E(\Pi/3\Pi, G)$$

$$\xrightarrow{\rho'} \text{Hom}(\Pi, G/2G) + \text{Hom}(\Pi, G/3G),$$

where the isomorphism ρ' is the sum of two isomorphisms ρ_2 and ρ_3 given by Theorem 26.5 for the primes $p = 2$ and $p = 3$. Now choose representatives $v(r) \in \Pi$ for $r \in \Pi/2\Pi$ and $w(t) \in \Pi$ for $t \in \Pi/3\Pi$, with $v(0) = 0 = w(0)$. In the exact sequence

$$0 \rightarrow B_9(\Pi, 5) \rightarrow Z_9(\Pi, 5) \rightarrow \Pi/2\Pi + \Pi/3\Pi \rightarrow 0$$

we then choose cycles $[v(r) \mid_4 v(r)]$ and $[w(t) \parallel w(t) \parallel w(t)]$ representing the elements r and t . With these representatives the factor set f_0 for this exact sequence has

$$f_0(r, r) = 2[v(r) \mid_4 v(r)] = -\partial[v(r) \mid_5 v(r)],$$

$$f_0(t, t) + f_0(t, t^2) = 3[w(t) \parallel w(t) \parallel w(t)] = -\partial[w(t) \mid_3 w(t) \parallel w(t)].$$

The expressions on the left are exactly those needed for the application of the isomorphisms ρ_2 and ρ_3 ; the boundaries on the right are applied exactly to the cells used to define a and a' in the theorem. The rest of the proof proceeds as before.

By suspension of this result we obtain

THEOREM 27.6. *There is a natural isomorphism*

$$(27.24) \quad \theta^{11}: H^{11}(\Pi, 6; G) \cong \text{Hom}(\Pi, G/2G) + \text{Hom}(\Pi, G/3G) \\ + \text{Hom}({}_2\Pi, G) + \text{Hom}({}_3\Pi, G) \quad (\text{stable})$$

obtained by assigning to each cocycle $w \in Z^{11}(\Pi, 6; G)$ the homomorphisms

$$a: \Pi \rightarrow G/2G, \quad a': \Pi \rightarrow G/3G, \quad c: {}_2\Pi \rightarrow G, \quad d: {}_3\Pi \rightarrow G$$

defined for $x \in \Pi$, $s \in {}_2\Pi$, and $u \in {}_3\Pi$ by setting

$$a(x) = w[x \mid_5 x] + 2G, \quad a'(x) = -w[x \mid_3 x \parallel x] + 3G, \\ c(s) = w(\zeta(s, 4)), \quad d(u) = w(\sigma(u)).$$

The stable cohomology results formulated in Theorems 27.2, 27.4, and 27.6 above are implicitly contained in some of the geometrical results of Nakaoka [15, 16].

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